# Oil-Price Volatility and the Asymmetric Transmission of Energy Shocks

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**Abstract**

How important are oil shocks for the macroeconomy and recessions? We investigate the asymmetric nature of oil shocks which means that price increases reduce output and investment by more than price declines raise these variables. We argue that the asymmetry comes from an increase in price volatility that accompanies large price shocks. Using identified supply shocks, we document that in response to world oil price shocks, investment in the EU exhibits this asymmetry. At the same time, the absolute value of price shocks raises future price volatility. We build a two-sector, heterogeneous agent model with adjustment costs in production, firm liquidation, and portfolio diversification costs to quantitatively reproduce the pattern. Joint movements in the price and volatility of future prices reproduce the asymmetry in aggregate European investment as well as capturing changes in expected returns. We can also replicate investment reallocation between sectors. To solve the model, we propose a fast non-linear solution method which maintains the non-linear responses crucial to our theory. The model has implications for why recent oil price increases have not led to a significant world recession.

**Keywords:** oil prices, investment, uncertainty, heterogeneous agents, computational methods.

**JEL codes:** E21, E22, E32, G11, G32, Q43.

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# 1 Introduction

Oil price movements are widely regarded as an important source of business cycle fluctuations and as the cause of several severe recessions, particularly those of the 1970s and 1980s. Oil and other forms of energy are important inputs into production, yet the energy-intensive sector accounts for a fairly small share of aggregate output. This raises two questions: are oil price shocks an important cause of recessions, and has their importance changed over time? In this paper, we study the role of oil shocks in the macroeconomy and highlight an asymmetry that helps explain why oil price increases matter so much for recessions.

Most standard macroeconomic models treat oil as a flexible input into production. As a result, they predict a close to linear response to oil price changes: a decline in the oil price raises output by as much as an equal-sized increase lowers it. This prediction is inconsistent with the data. Oil price increases are associated with larger contractions than oil price declines are with expansions. Moreover, there is evidence that this nonlinearity grows with the size of the shock (Caravello and Martinez Bruera, 2024). In this paper, we propose and quantify a new mechanism for this asymmetry: an increase in volatility that accompanies oil price changes. We document evidence for this mechanism in the data and then build a quantitative heterogeneous-agent model, calibrated to the data, that reproduces these patterns. Our main contribution is to isolate the interaction between oil price changes and oil price volatility and to show that it accounts for a meaningful share of the asymmetry in the data.

We begin by documenting the nonlinearity in the data. The asymmetry itself is not new (see Mork, 1989; Hamilton, 1996, 2003). [1] Our empirical contribution is to estimate it using structurally identified supply shocks and to confirm it in European data. The central empirical challenge is to separate oil supply shocks from oil demand shocks: a large oil price change may itself reflect a shift in demand, in which case the comovement of oil prices and output need not reflect a causal effect of prices on output. To address this, we use the oil supply shocks identified by Baumeister and Hamilton (2019), who estimate a Bayesian structural VAR of the global oil market. They identify supply shocks using sign restrictions together with informative priors on the short-run price elasticities of oil supply and demand.

We then estimate nonlinear local projections in which the response depends on the

[1] Kilian and Vigfusson (2011) question this evidence. We discuss this debate in our review of the related literature.

sign of the shock. Using quarterly data,[2] we find that one year after a supply shock of median absolute size, aggregate European investment is 1.79 log points lower following a shock that raises oil prices, compared with 0.79 log points higher following a shock that lowers them. The decline is statistically significant, while the increase is not.

Next, we show that oil price shocks are followed by higher oil price volatility. We find this in two related settings. Again using the identified supply shocks, we show that both the cross-sectional dispersion of energy prices across EU countries and the aggregate time-series volatility of oil prices rise after a shock. Across EU countries doubling the absolute size of the shock raises annualized oil price volatility in the following month by 40 log points, and the effect persists at the 6- and 12-month horizons.

We interpret this evidence as indicating that oil supply shocks affect both the mean and the variance of future oil prices. A change in the price today not only shifts the expected oil price tomorrow but also widens the distribution of prices that producers will face. Firms making investment decisions therefore face more volatile oil prices after a price change. Because volatility can depress investment, this channel can generate a nonlinear aggregate response to oil price shocks.

To illustrate this channel, we first set up a simple single-sector model. The firm produces with a Leontief technology in capital and oil that has decreasing returns to scale, plans its inputs before the oil price is realized, and is owned by an agent with CARA utility. In this setting, the owner's objective has a certainty-equivalent form, and we show that the firm lowers its capital investment when the standard deviation of future oil prices rises. Quantitatively, an increase in the standard deviation of the oil price in line with the data, reduces output by 8 percent and capital and oil use by 11 percent.

We next show that with a more conventional CES production technology and a firm owned by a household with CRRA utility, the response to volatility is significantly attenuated.[3] Although the mechanism holds qualitatively, its magnitude is economically negligible: symmetric oil price volatility can not generate asymmetric movements in investment of the size observed in the data. Finally, we show that adding a revenue shock that can trigger liquidation restores the mechanism. A firm that has already committed to purchasing oil and may be liquidated if revenue is low behaves as if it were more risk averse, because the threat of liquidation introduces concavity into its problem. Greater oil price volatility interacts with the revenue shock to raise the probability that revenue

[2]The Baumeister and Hamilton (2019) supply shocks are monthly; we aggregate them to quarterly frequency.

[3]Because inputs are still planned before the oil price is realized, the oil bill is linear in the price, so the firm responds to volatility only through the curvature of its owner's utility.

falls short of the oil bill and the firm is liquidated. In the simple model, adding the revenue shock restores the decline in output and capital, which fall by 13 and 14 percent, respectively.

The simple model illustrates how a rise in oil price volatility can reduce production, but it lacks several features needed to quantify the mechanism. It has a single firm and a household owning it, so it misses the interaction between the firms that demand capital and the households that supply it in the capital market. It also has no general-equilibrium prices. We therefore develop a heterogeneous-agent model with two sectors, input adjustment costs, firm liquidation, incomplete markets, and portfolio diversification costs. To introduce idiosyncratic oil price volatility while retaining general-equilibrium prices, we introduce "regions" that receive region-specific oil price shocks. The variance of these region-specific shocks rises after a change in the world oil price. From the firm's ex ante perspective, an increase in regional dispersion and volatility are equal. The regional dispersion also maps directly into our empirical evidence on price dispersion across European countries. Household heterogeneity allows the model to capture the aggregate marginal propensity to save and its realistic dependence on volatility. With sticky prices, it would also generate a realistic household demand response, as in a heterogeneous-agent New Keynesian (HANK) model.

To capture the role of oil as an input into production, the economy has two sectors: one uses oil intensively, and the other uses substantially less. Firms in both sectors choose capital and plan oil and labor inputs before the regional oil price shock is realized. They can adjust oil and labor after the shock, subject to an adjustment cost. Firms pay for operating inputs before revenue arrives, using a within-period working capital loan. If, following a revenue shock, revenue cannot cover the cost of inputs, the firm is insolvent and must liquidate part of its capital. A claim on a single firm is therefore riskier than a diversified claim on its sector. Households, in turn, hold portfolios with idiosyncratic exposure: the cheapest portfolio to hold is a claim on the firm and sector in which they work, and diversifying away from it is costly. This makes household saving more sensitive to regional dispersion in input prices, and it makes the dispersion of firm-level returns on capital relevant for aggregate dynamics.

We calibrate the model to match salient features of the role of oil in aggregate production. The oil-intensive sector is a fifth of final demand and has an oil weight of 0.175, compared with 0.025 in the other sector. Oil expenditure is 8 percent of GDP, labor compensation is 60 percent, and about 1.5 percent of firms liquidate in a quarter, with liquidation destroying 30 percent of surviving capital. Discounting, risk aversion,

and depreciation take standard quarterly values. The remaining parameters come from external estimates or from these steady-state targets, as described in Section 6. As in the simple model, the asymmetry in the quantitative model comes from the interaction between the price shock and the volatility shock, which occur together. The price change moves production costs. When the oil price rises, higher costs and a greater chance of liquidation both discourage investment, so the price change and the rise in volatility reinforce each other. When the oil price falls, lower costs encourage investment, but the rise in volatility offsets part of the expansion. Because larger supply shocks are followed by larger increases in volatility, this interaction can strengthen with the size of the shock. Changes in the level of the oil price alone produce responses that are close to symmetric.

To quantify the mechanism, we study the model's nonlinear responses to MIT shocks. When the price and volatility move together, output falls by about 6.1 percent at the peak after an adverse shock but rises by only about 3.5 percent after a favorable one. Total capital and investment display the same asymmetry. The volatility shock deepens the contraction and offsets part of the expansion, because own-sector exposure translates the additional liquidation risk faced by individual firms into a lower supply of capital.To decompose this effect, we study price and volatility changes separately. Price changes alone produce nearly symmetric responses: output rises by about 4.3 percent at the peak after a price decline and falls by about 4.1 percent after a price increase, so the small remaining gap goes in the opposite direction from the data. A pure rise in regional dispersion lowers output by about 1.7 percent at the peak and reallocates capital toward the sector that uses less oil. These separate responses do not add up to the joint ones. Adding the same dispersion increase to a price decline reduces the peak rise in output from about 4.3 percent to about 3.5 percent. Adding it to a price increase deepens the peak contraction from about 4.1 percent to about 6.1 percent. The dispersion shock is more contractionary when the oil price has risen, because more firms are already near the liquidation threshold.

The two sectors respond to the increase in volatility differently. In the oil-intensive sector, cash flow moves more closely with the oil price, and a rise in regional price dispersion pushes more firms across the liquidation threshold and widens the distribution of firm-level returns. Households who work in the oil-intensive sector hold more of their portfolios in these firms, so their supply of capital falls as liquidation risk rises. In the other sector, direct oil exposure is smaller, so a pure price change moves input costs and liquidation risk by less. That sector is nonetheless affected: it uses some oil, and wages transmit the shock through the labor market. Moreover, lower household wealth reduces

the capacity to bear risk in both sectors. A rise in volatility or regional dispersion can therefore reduce capital in both sectors while also reallocating capital between them. This reallocation is stronger than the one that follows a price change alone, because a price change mainly shifts relative costs, whereas a regional dispersion shock mainly shifts relative liquidation risk.

The sectoral responses in the model provide a test of the mechanism. In the data, investment and returns are measured separately for sectors that use oil and energy intensively and for sectors that use little of either. The two groups do not move as one aggregate. After a shock that lowers the oil price, investment in the energy-intensive group can have smaller rise than the other group. After a shock that raises the oil price, the relative response of the two groups changes, and the same change in comovement appears in returns. The correlation between the groups therefore depends on the sign of the shock. The model produces this sign dependence in sectoral capital. A price change alone leaves the correlation of the two sectors' capital near $-0.3$ for either sign. Adding the rise in dispersion raises that correlation to 0.88 when the oil price falls, so capital in the two sectors moves together, and lowers it to $-0.54$ when the oil price rises, so the sectors move apart.

Comparing large favorable and adverse shocks requires a solution method that keeps the nonlinearities behind the asymmetry. Liquidation probabilities, portfolio shares, and borrowing constraints depend on the level and the sign of the oil price, so a first-order approximation around the steady state would remove the mechanism. We compute the transitions with an almost-nonlinear level-$k$ algorithm, set out in Section 8. Households and firms solve nonlinear policies conditional on believed paths of wages, returns, and oil prices. Within each date, markets are cleared with a local correction, which avoids resolving both dynamic programs after every small price change. The algorithm retains exact liquidation risk, portfolio corners, and the dependence of household wealth on returns, at a cost that makes it feasible to compare large shocks of opposite sign.

The same model can be used to think about why oil price increases seem to have become less important for recessions. Three changes weaken the mapping from a higher oil price into lower aggregate investment. Lower diversification costs let households move away from the firm and sector in which they work, so liquidation risk is spread beyond the households who supply that sector's capital and the supply of capital falls by less. A lower liquidation cost reduces the capital destroyed when revenue cannot cover the oil bill, which shrinks the downside that own-sector exposure passes into saving. Growth of the energy-intensive sector, as in an energy-demand boom of the kind associated with

data centers, changes the composition of the capital stock. More production sits in the sector that is directly exposed to energy, but the expansion of that sector raises the return to investing there and can offset part of the drop in investment that a higher oil price would otherwise cause. These are comparative-static questions the model is built to answer.

Section 2 reviews the related literature. Section 3 presents the empirical observations in detail. Section 4 shows why oil-price uncertainty has only a small effect under a conventional technology, and a large effect once a liquidation threshold is added. Section 5 sets out the two sectors, liquidation, and own-sector exposure. Section 6 discusses how the model is disciplined. Section 7 reports the experiments that separate the price shock from the volatility shock. Section 8 explains the solution method, and Section 9 concludes.

## 2 Related literature

The debate on oil price transmission has moved forward in stages. Hamilton (1983, 1996, 2003), Mork (1989), and the work that followed argued that large oil price increases have repeatedly preceded contractions and that sufficiently disruptive oil events need not have a mirror image in price declines.[4] A later literature, built on newer VAR and structural VAR methods, reached a more linear conclusion. Kilian (2009) stressed that an oil price movement is not a primitive shock: supply, global demand, and precautionary demand disturbances can move the same price while producing different macroeconomic responses. Kilian and Vigfusson (2011) showed that censored oil price specifications can exaggerate asymmetry and that a linear, approximately symmetric response is a useful benchmark for ordinary U.S. innovations.

Baumeister and Hamilton (2019) challenged the identifying assumptions of those linear VARs with a Bayesian structural VAR that combines sign restrictions with explicit priors on the elasticities of oil supply and oil demand. Although their decomposition is based on a linear model, it restores an important role for oil supply shocks and finds that these shocks lower economic activity with a delay.[5] The associated dataset is a

[4]Proposed explanations for this asymmetry include costly reallocation of labor and capital across sectors (Hamilton, 1988; Davis and Haltiwanger, 2001) and the option value of delaying irreversible investment under uncertainty (Bernanke, 1983; Pindyck, 1991).

[5]Their prior on the short-run supply elasticity is contested: Kilian and Zhou (2018) argue that a prior closer to microeconomic estimates restores a limited role for supply shocks. The asymmetry is not confined to this series. The next paragraph reports the same pattern for the oil-supply news shocks of

monthly time series of world crude oil production, global real activity, the real price of oil, and petroleum inventories. Each price movement is decomposed into an oil supply shock, an economic activity shock, an oil consumption demand shock, and an inventory demand shock. The empirical section uses their posterior-median oil supply shock, which is available over a long monthly sample, can be aggregated to the quarterly frequency of investment, and isolates the supply component of the price. Because Kilian (2009) shows that the macroeconomic response depends on which component moved the price, the supply series is the empirical counterpart of the disturbance in the model.

A more recent group of papers returns to sign and size nonlinearities using the oil supply news shocks of Kanzig (2021), which are constructed from high-frequency movements in oil futures prices around OPEC announcements. Forni, Franconi, Gambetti, and Sala (2025) estimate a nonlinear proxy structural VAR for the United States: news that raises the oil price produces a large and persistent decline in real activity, whereas news that lowers the price produces a smaller expansion, and uncertainty rises after a supply shock of either sign. They also report that the asymmetry is robust when the shock is identified as in Baumeister and Hamilton (2019). Caravello and Martinez Bruera (2024) separate sign nonlinearities from size nonlinearities and, applying the same news series to the United States, find that the macroeconomic effect grows with the magnitude of the shock. Mareels (2026) uses externally identified oil supply shocks in panel local projections for European countries and finds that temporary oil price increases leave real GDP lower for more than a decade, while temporary price decreases do not generate comparable lasting gains. Our evidence on sign asymmetry concerns aggregate European investment, and the rise in oil price volatility after larger supply shocks links our mechanism to size nonlinearities. Relative to Forni, Franconi, Gambetti, and Sala (2025), who interpret the asymmetry in terms of increased returns uncertainty, we provide a quantitative general-equilibrium model of this channel.

Our mechanism also builds on work that studies oil price uncertainty directly. Oil price volatility has been found to depress output and investment (Lee, Ni, and Ratti, 1995; Ferderer, 1996; Elder and Serletis, 2010; Bredin, Elder, and Fountas, 2011; Jo, 2014), and oil producers reduce drilling when expected volatility rises (Kellogg, 2014).[6] Volatility itself depends on the short-run elasticities of oil supply and demand (Baumeister and

Kanzig (2021) and for other externally identified supply shocks, and Forni, Franconi, Gambetti, and Sala (2025) find that their estimates remain when the shock is identified as in Baumeister and Hamilton (2019).

[6] Jo (2014) shows that GARCH-based measures cannot separate uncertainty from past price changes; using stochastic volatility, she finds a smaller but still negative effect on world industrial production.

Peersman, 2013; Van Robays, 2016).[7] These papers treat oil price volatility as a separate disturbance or study its determinants; we document how it moves with identified supply shocks and model the price and volatility responses jointly.

The mechanism is related to the real-options and uncertainty-shock logic of Bloom (2009) and to models in which volatility interacts with financial frictions and default (Christiano, Motto, and Rostagno, 2014; Gilchrist, Sim, and Zakrajsek, 2014; Arellano, Bai, and Kehoe, 2019), but it operates through household risk bearing and capital ownership.[8] Under-diversification can already distort aggregate investment in general-equilibrium production economies (Bhamra and Uppal, 2019), and uninsured idiosyncratic risk lowers saving and investment (Angeletos, 2007; Panousi and Papanikolaou, 2012). Own-sector exposure gives under-diversification a concrete form: households are persistently invested in the firm and sector in which they work, in the spirit of employer stock, local information, and concentrated portfolios (Huberman, 2001; Benartzi, 2001; Moskowitz and Vissing-Jorgensen, 2002; Van Nieuwerburgh and Veldkamp, 2009), and moving away from that position is costly. The relevant firm-level risk is endogenous: it comes from oil costs, an idiosyncratic revenue shock, and liquidation.

Recent heterogeneous-agent New Keynesian models of energy shocks emphasize the household demand channel (Auclert et al., 2023; Pieroni, 2023) but abstract from the volatility that accompanies supply shocks. A first-order transition (Boppart, Krusell, and Mitman, 2018; Auclert et al., 2021) would suppress the dependence of liquidation, portfolios, and the wealth distribution on the sign and size of the oil shock, which is why the experiments are computed with a nonlinear transition and an almost-nonlinear level-$k$ algorithm, in the spirit of Farhi and Werning (2019), that preserves those nonlinearities.

# 3 Empirical Motivation

This section examines how oil-supply disturbances relate to price risk, aggregate investment, and differences across energy-intensive and other industries. We combine European, Swedish, and U.S. evidence to motivate the model's transmission channels. The results document several relevant patterns, while the financial responses remain less precisely estimated.

[7] Whether volatility responds symmetrically to price increases and decreases is less clear: in GARCH models, crude oil, unlike most commodities, exhibits higher volatility after negative price shocks (Chen and Mu, 2021).

[8] Liquidation in our model plays a role similar to default in these models, but its effect on investment operates through the capital supplied by under-diversified households.

## 3.1 Data

We use monthly posterior-median oil-supply shocks from Baumeister and Hamilton (2019), denoted by $s_t$, retaining their original signs. Quarterly shocks are sums of the three monthly observations. European data comprise domestic energy producer prices for ten EU economies in 2005–2019 and aggregate investment and GDP in 1996–2019. Oil volatility uses daily Brent prices converted into real Swedish kronor using SEK/USD and Swedish CPI, covering 1987–2025. Swedish industry investment begins in 1993; U.S. monthly data provide dividend dispersion through 2024 and an industry return premium through August 2026. Shock availability and lag requirements determine each estimation sample. Appendix Figures 21, 26–28 describe the European price series and energy-intensity groups.

## 3.2 Supply shocks, price dispersion, and volatility

We distinguish cross-country price dispersion from time-series oil volatility. Let $D_t$ be the cross-country standard deviation of log energy-price residuals after country and month fixed effects and country-specific trends. Let $e_d$ be the residual from an AR(5) for daily real-oil-price log returns, expressed in log points. For the next $K \in \{1, 3, 6, 12\}$ calendar months, define

$$F^D_{t,K} = \left( \frac{1}{K} \sum_{j=1}^{K} D^2_{t+j} \right)^{1/2},$$
$$F^V_{t,K} = \left( \frac{252}{N_{t,K}} \sum_{j=1}^{K} \sum_{d \in t+j} e^2_d \right)^{1/2}, \tag{1}$$

where $N_{t,K}$ counts the daily innovations in the future window. Both measures exclude the shock month. For each measure $m \in \{D, V\}$, we estimate

$$F^m_{t,K} = \alpha^m_K + \beta^m_K |s_t| + \gamma^m_K U^m_{t-1} + u^m_{t,K}, \tag{2}$$

where $U^m_{t-1}$ is the corresponding previous-month dispersion or realized volatility. Regression inference uses $K$-lag Newey–West standard errors; the binned-mean bars are descriptive intervals.

Figures 1 and 2 plot next-month binned means and $5\widehat{\beta}^m_K$. The dispersion association is strongest at the shortest horizon and fades thereafter. Oil-volatility estimates remain

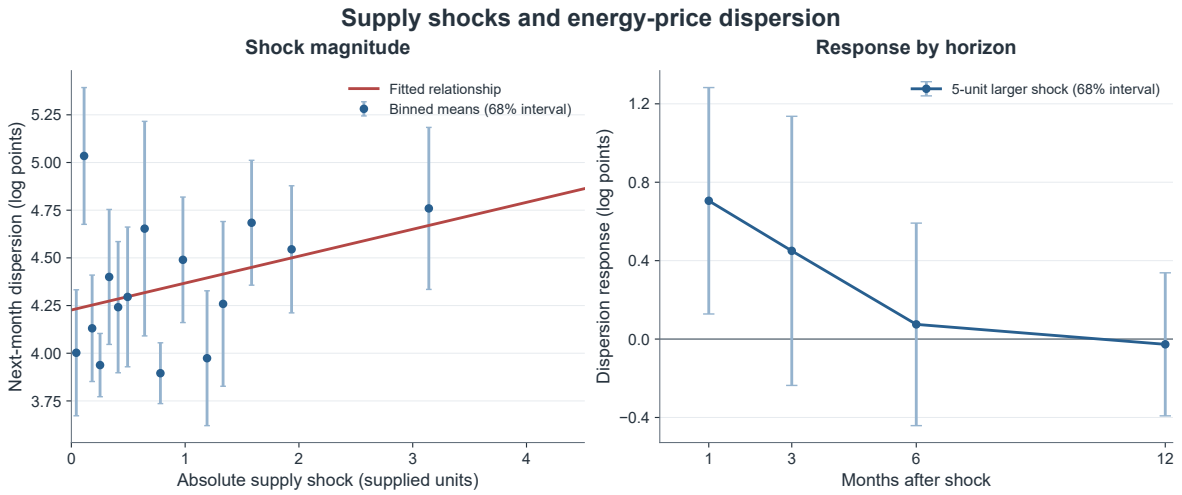



Figure 1: Supply-shock magnitude and EU10 energy-price dispersion. Left: next-month binned means, controlling for lagged dispersion. Right: effects of a five-unit larger absolute shock. Bars show 68% intervals.

positive at longer horizons: a five-unit larger absolute shock is associated with approximately 14.5 and 11.3 additional log points of annualized volatility over six and twelve months. The corresponding 68% intervals exclude zero. Sign-specific estimates appear in Appendix Figures 22 and 23.

## 3.3 Aggregate asymmetry

We estimate quarterly local projections for aggregate EU10 investment. Writing $S_t$ for the quarterly supply shock, define $S_t^+ = \max\{S_t, 0\}$ and $S_t^- = \max\{-S_t, 0\}$. With $i_t = 100 \log I_t$, the specification is

$$i_{t+h} - i_{t-1} = \alpha_h + \theta_h^+ S_t^+ + \theta_h^- S_t^- + \Gamma_h' Z_t + u_{t+h}, \tag{3}$$

where $Z_t$ contains two lags of investment growth, GDP growth, real oil-price growth, inflation, the short rate, and both shock components, together with quarter indicators. Local-projection inference uses pointwise Newey–West intervals with $h + 1$ lags.

Figure 3 scales both responses to the common median absolute quarterly shock, approximately 1.43 shock units. At twelve quarters, the point estimates are 0.79 log points after a positive shock and $-1.79$ after a negative shock; only the latter 90% interval excludes zero. The paths suggest sign dependence, although most individual responses are imprecise. Shock-size extensions, the exclusion of 2009 shock origins, and the GDP

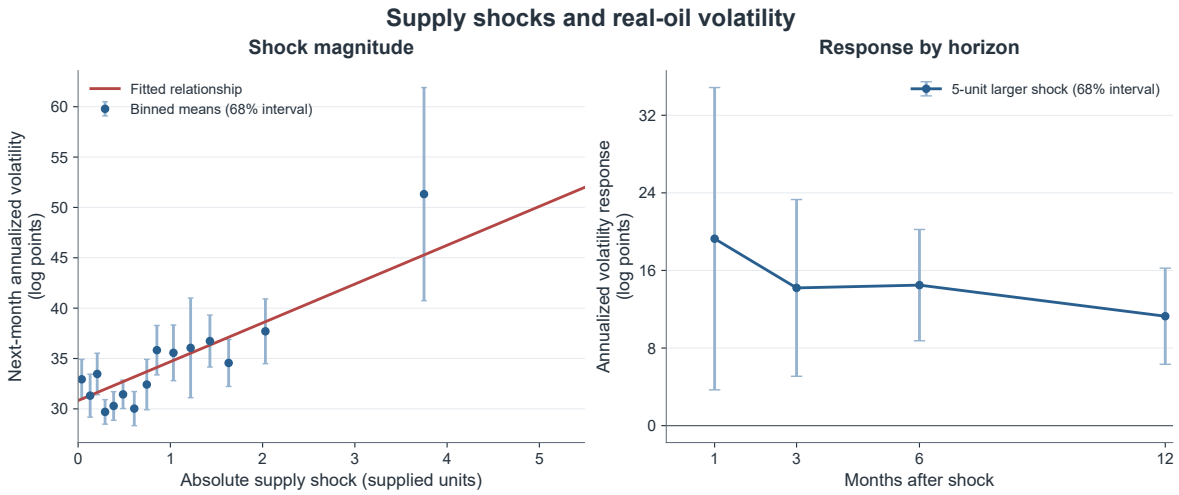



Figure 2: Supply-shock magnitude and annualized real-oil volatility. Left: next-month binned means, controlling for lagged volatility. Right: effects of a five-unit larger absolute shock. Bars show 68% intervals.

counterparts are reported in Appendix Figures 24, 25, 30, and 31.

## 3.4 Cross-sectional facts

**Dividend dispersion.** For the United States, we use monthly dividend dispersion within high- and low-energy-intensity groups. For $g \in \{H, L\}$, we estimate

$$\text{Disp}_{g,t+h} = \alpha_{g,h} + \beta_{g,h} s_t + \Gamma'_{g,h} W_t + u_{g,t+h}, \qquad h = 0, \ldots, 12, \tag{4}$$

where $W_t$ includes six lags of both groups' dispersion, the shock, U.S. industrial-production growth, inflation, the three-month Treasury rate, and real oil-price growth, together with month indicators. Figure 4 excludes the flagged 1998–2001 and 2020 episodes, including lag and response windows crossing them. The estimates provide limited evidence of a persistent increase in dispersion or a systematically larger high-intensity response. The full-sample counterpart is in Appendix Figure 32.

**Return premia.** We next examine the weighted U.S. return on energy-intensive industries minus the weighted return on other industries, denoted by $P_t$. This realized equity-return spread differs from the excess bond premium studied by Forni, Franconi, Gambetti, and Sala (2025). We project $P_{t+h}$ jointly on $s_t^+ = \max\{s_t, 0\}$ and $s_t^- = \max\{-s_t, 0\}$, using six lags of the premium, both shock components, the same monthly

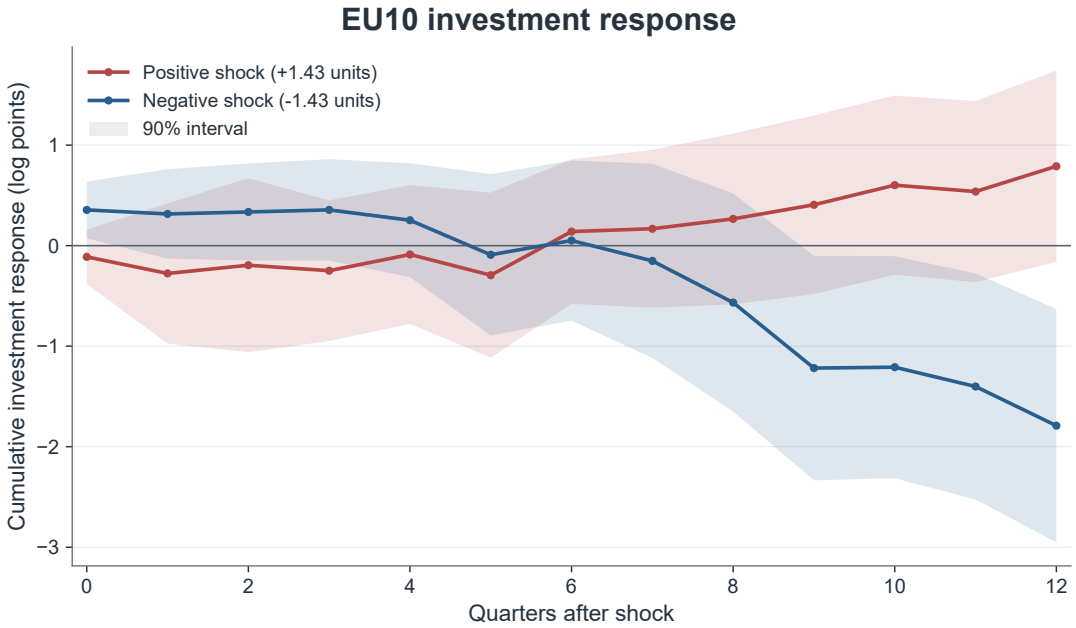



Figure 3: Cumulative EU10 investment responses to positive and negative supply shocks of the common median absolute size. Shading shows pointwise 90% intervals.

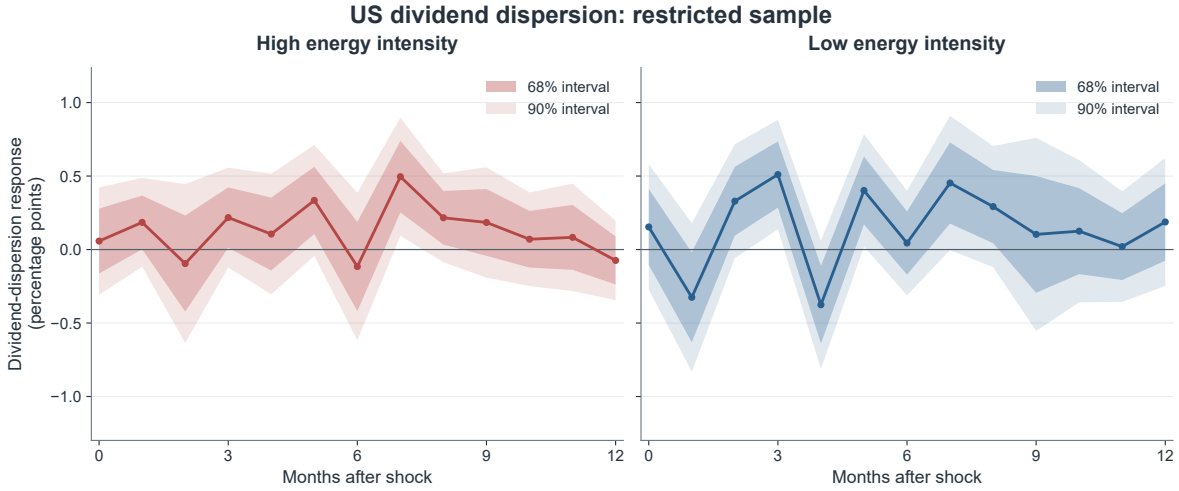



Figure 4: U.S. dividend-dispersion responses to a one-unit signed supply shock, by energy intensity. Dark and light shading show pointwise 68% and 90% intervals.

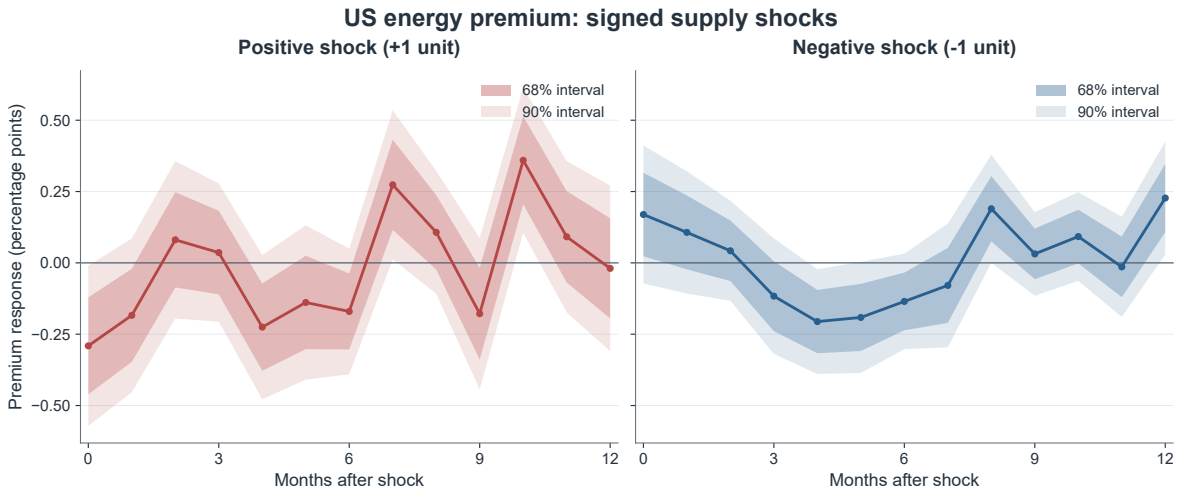



Figure 5: U.S. energy-intensity return-premium responses to positive and negative one-unit supply shocks, estimated jointly. Dark and light shading show pointwise 68% and 90% intervals.

U.S. macro controls, and month indicators.

Figure 5 reports impact estimates of −0.29 and 0.17 percentage points after positive and negative shocks, respectively; most subsequent responses are imprecise. Appendix Figure 33 instead projects the premium on $|s_t|$, imposing a common response to shock magnitude across signs.

**Sectoral investment.** Finally, we estimate the sign-specific investment projection separately for Swedish high- and low-energy-intensity industries over 1993–2019. Groups are fixed using 2015–2019 direct energy purchases divided by gross output: high intensity exceeds 5%, giving eight high- and 24 low-intensity industries. The controls follow (3), using each group's investment growth.

Figure 6 uses the same 1.43-unit reference magnitude as the aggregate exercise. Six quarters after a positive shock, the point estimates are −2.56 log points for high-intensity investment and 0.62 for low-intensity investment. Negative-shock responses are less stable. Appendix Figure 29 extends the sample through 2026Q2.

# 4 A Qualitative Model

This section puts the idea in the introduction into one-firm problems. Each step uses only capital and oil. The firm sells output at a given price $P = 1$, rents capital at

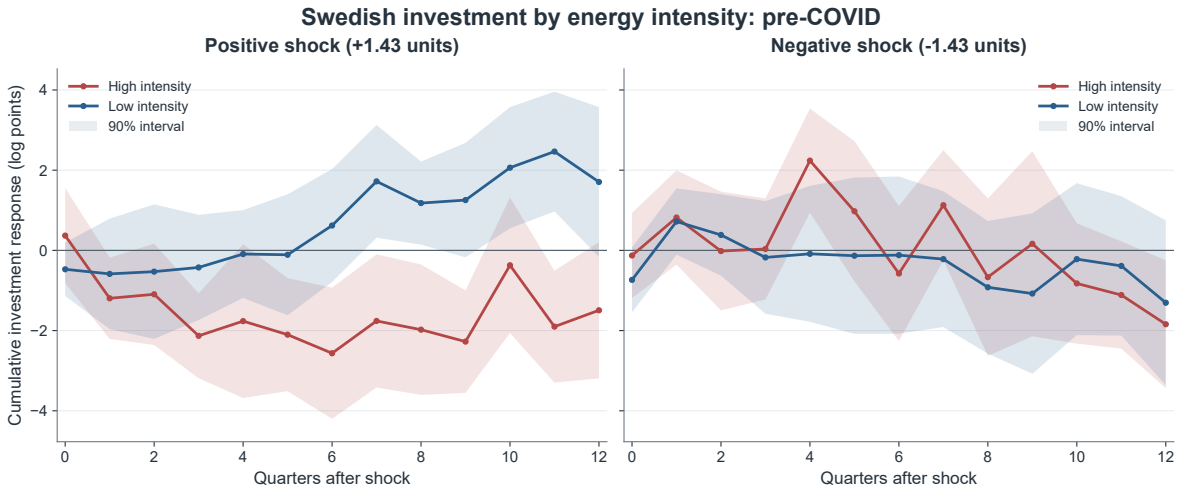



Figure 6: Cumulative Swedish investment responses by energy intensity, 1993–2019, at the common reference shock magnitude. Shading shows pointwise 90% intervals.

$r = 0.05$, and faces a normal oil price with mean $\bar{p}^o = 1$. The experiment raises the standard deviation of that price from 0.16 to 0.24 and holds the mean fixed. That is the same increase in dispersion used on impact in the quantitative experiments, where cross-regional dispersion goes from 0.16 to 0.24. Nothing in the comparison depends on whether the mean price has gone up or down. The first step locks the inputs in fixed proportions and gives the firm a strong distaste for the variance of the oil bill. Output and capital fall by several percent. The second step replaces that technology and those preferences with a conventional CES technology and CRRA utility. The sign survives and the magnitude does not. The third step has the firm pay for oil before a revenue shock, and liquidate when revenue cannot cover that bill. The chance of liquidation depends on how much oil the firm bought, and the same increase in dispersion cuts scale by much more. The quantitative model in Section 5 is that last structure, disciplined by external parameters and by internal targets.

## 4.1 Fixed proportions

Capital and oil are used in fixed proportions inside a decreasing-returns technology,

$$Y = \min\left(K, \frac{O}{\theta}\right)^{\nu}, \tag{5}$$

with $\nu = 0.70$ and $\theta = 0.10$. The rental rate on capital is $r = 0.05$. Both inputs are chosen before the oil price is drawn. The oil price is normal, with mean $\bar{p}^o = 1$ and standard deviation $\sigma_p$. Profit risk comes only from the oil bill. The firm has constant absolute risk aversion $\lambda = 0.15$.

Inputs chosen off the fixed-proportions ray are wasteful. Idle capital costs rent and does not reduce risk. Oil beyond $\theta K$ raises both the mean and the variance of the oil bill. The firm therefore sets $K = X$ and $O = \theta X$, which gives $Y = X^\nu$. Profit is normal. Its certainty equivalent is revenue minus expected input cost minus a penalty for the variance of the oil bill,

$$X^\nu - (r + \bar{p}^o\theta)\, X - \frac{\lambda}{2}\left(\theta\sigma_p X\right)^2. \tag{6}$$

The first-order condition equates marginal revenue to expected marginal cost plus a risk premium,

$$\nu X^{\nu-1} = r + \bar{p}^o\theta + \lambda\theta^2\sigma_p^2 X. \tag{7}$$

The premium rises with the scale of the firm, with $\theta^2$, and with the variance of the oil price. The objective is strictly concave, so the root is unique. When $\sigma_p = 0$ the scale is explicit,

$$X = \left(\frac{\nu}{r + \bar{p}^o\theta}\right)^{1/(1-\nu)}. \tag{8}$$

When $\sigma_p > 0$, (7) is solved for $X$. Differentiating that condition gives

$$\frac{dX}{d\sigma_p^2} = -\frac{\lambda\theta^2 X}{(1-\nu)\,(r + \bar{p}^o\theta)\, X^{-1} + (2-\nu)\lambda\theta^2\sigma_p^2} < 0. \tag{9}$$

A higher oil-price variance reduces capital, oil, and output. Raising $\sigma_p$ from 0.16 to 0.24, with the mean oil price held at 1, lowers output by 8.1 percent. Capital and oil fall together, by 11.3 percent, because the firm stays on the ray $O = \theta K$. Nothing in this calculation depends on whether the mean oil price has risen or fallen. Uncertainty that arrives with either shock reduces the scale of the firm.

## 4.2 A conventional technology and CRRA utility

The fixed-proportions example is useful because the response is visible, and it is not a conventional firm. The technology forces oil and capital to move together, the oil requirement is large relative to the capital rental, and absolute risk aversion penalizes the variance of profit one for one. This step replaces all three ingredients.

The firm has a CES technology in capital and oil. The conventional benchmark is the Cobb–Douglas case, with elasticity of substitution equal to one,

$$Y = AK^{0.30}O^{0.05}. \tag{10}$$

The capital elasticity 0.30 and the oil elasticity 0.05 are standard output elasticities: oil is a small input, and the residual 0.65 is decreasing returns. The owner has CRRA utility with relative risk aversion $\gamma = 2$,

$$\max_{K,O} \mathbb{E}\left[\frac{\pi^{1-\gamma}}{1-\gamma}\right], \qquad \pi = Y - rK - p^oO, \tag{11}$$

and chooses both inputs before the oil price is drawn.

A risk-neutral owner of this firm does not respond to $\sigma_p$ at all. The oil bill is linear in the oil price, so a mean-preserving spread leaves expected profit unchanged and the optimal inputs depend only on $\bar{p}^o$. Risk aversion breaks that invariance, but only by a little. The variance of profit is $(O\sigma_p)^2$, and under CRRA the penalty for that variance is divided by the level of profit. With oil's output elasticity at 0.05, the gamble is small relative to profit. Raising $\sigma_p$ from 0.16 to 0.24 lowers oil by about 0.5 percent. Output and capital barely move. The qualitative result is the same as in the fixed-proportions firm: higher oil-price uncertainty reduces investment. The movement is not a significant one.

## 4.3 Paying for oil before the revenue shock

The conventional firm barely moves because a high oil price only raises a linear bill, and the utility cost of that bill is smooth. This step turns that smooth penalty off and uses the liquidation rule of the quantitative model. The firm still chooses capital and oil before the oil price is drawn. When the price is realized it pays the oil bill $p^oO$ in advance. It then draws a revenue shock $\zeta$, with $\mathbb{E}[\zeta] = 1$, and sells $\zeta Y$. It liquidates when it cannot cover the bill,

$$\zeta Y < p^oO, \tag{12}$$

and liquidation destroys a fraction $\psi = 0.30$ of installed capital, the same loss used in the quantitative model. The revenue shock is uniform on $[0.75, 1.25]$, again as in that model. The threshold

$$\zeta^*(K, O, p^o) = \frac{p^oO}{Y(K, O)} \tag{13}$$

is the firm's own decision. Buying more oil raises the bill. Producing more output lowers the bill relative to sales. A higher realized oil price pushes $\zeta^*$ up through the revenue-shock distribution, so more draws fail to cover the bill.

Under the elasticities of the previous subsection this rule never binds. The oil bill is about 5 percent of revenue, and every revenue draw is at least 75 percent of mean revenue, so the firm always covers the bill. Dispersion in the oil price then has nothing to tilt, and the firm does not change scale. The quantitative model avoids that irrelevance because the prepaid bill includes labor as well as oil, and labor is most of revenue. This example has no labor, so oil has to stand in for that operating bill. Oil's output elasticity is set to 0.70 and capital's to 0.10,

$$Y = AK^{0.10}O^{0.70}, \tag{14}$$

which puts the mean oil bill at about two thirds of revenue, below the bottom of the revenue-shock support. At the mean oil price the firm covers the bill for every revenue draw. A high oil-price draw pushes the bill into the support.

The firm maximizes expected profit. The only reason dispersion matters is that it changes the probability of (12). Raising $\sigma_p$ from 0.16 to 0.24 raises that probability from 2.0 percent to 3.5 percent. Output falls by 13 percent, capital by 14 percent, and oil by 16 percent. The firm cuts the oil purchase because that purchase is what moves $\zeta^*$ into the revenue-shock support. The same increase in dispersion that left the conventional firm almost unchanged now cuts its scale by more than a tenth.

## 4.4 From the example to the quantitative model

The quantitative model is this last firm, with the rest of the operating bill put back in and the magnitudes disciplined rather than set for illustration. Firms use a CES technology and households have CRRA utility. Inputs are chosen before the oil-price state is realized, the oil bill and the other operating costs are paid before revenue, and a fraction of capital is destroyed when the revenue shock leaves revenue short of that bill, as in (12) and (22). External parameters pin down preferences, elasticities, and depreciation. Internal targets pin down the expenditure shares, the liquidation rate, and the variance of idiosyncratic returns. Sections 5 and 6 set out that structure. The experiments in Section 7 then separate the oil-price level from the dispersion around it.

# 5 Model

## 5.1 Environment and timing

Time is discrete. There are two intermediate sectors, $s \in \{H, L\}$, that differ in oil intensity. A competitive final-good producer combines their output. Households differ in financial wealth $a$, labor productivity $z$, and own-sector exposure $e \in \{H, L\}$. Aggregate oil conditions determine the oil price and the distribution of sectoral returns.

Within a period the aggregate state is known before firms plan, the regional oil price is realized before spot markets clear, and the idiosyncratic revenue shock determines returns before households choose next period's capital. Figure 7 records the order. A region is a local oil-price island: a country inside the European Union, or a state inside the United States. Because the revenue shock is i.i.d., expected and realized aggregate goods demand and labor supply coincide, so the spot markets in the third event clear at the aggregates that households deliver after returns are realized.

## 5.2 Oil prices and volatility

The oil price in region $j$ is

$$p_j^o = \bar{p}^o(1 + \nu_j), \qquad \mathbb{E}[\nu_j] = 0, \qquad \mathrm{Var}(\nu_j) = \sigma_\nu^2. \tag{15}$$

MIT experiments allow both $\bar{p}_t^o$ and $\sigma_{\nu,t}$ to vary over time. This separation permits experiments involving a change in the expected oil-price level, a pure volatility shock, or a joint shock in which a positive or negative price innovation occurs under elevated volatility.

## 5.3 Production

Define the capital–labor composite

$$M_{js} = \left[ \alpha_{L,s} L_{js}^{\frac{\epsilon_f - 1}{\epsilon_f}} + \alpha_{K,s} K_s^{\frac{\epsilon_f - 1}{\epsilon_f}} \right]^{\frac{\epsilon_f}{\epsilon_f - 1}}, \tag{16}$$

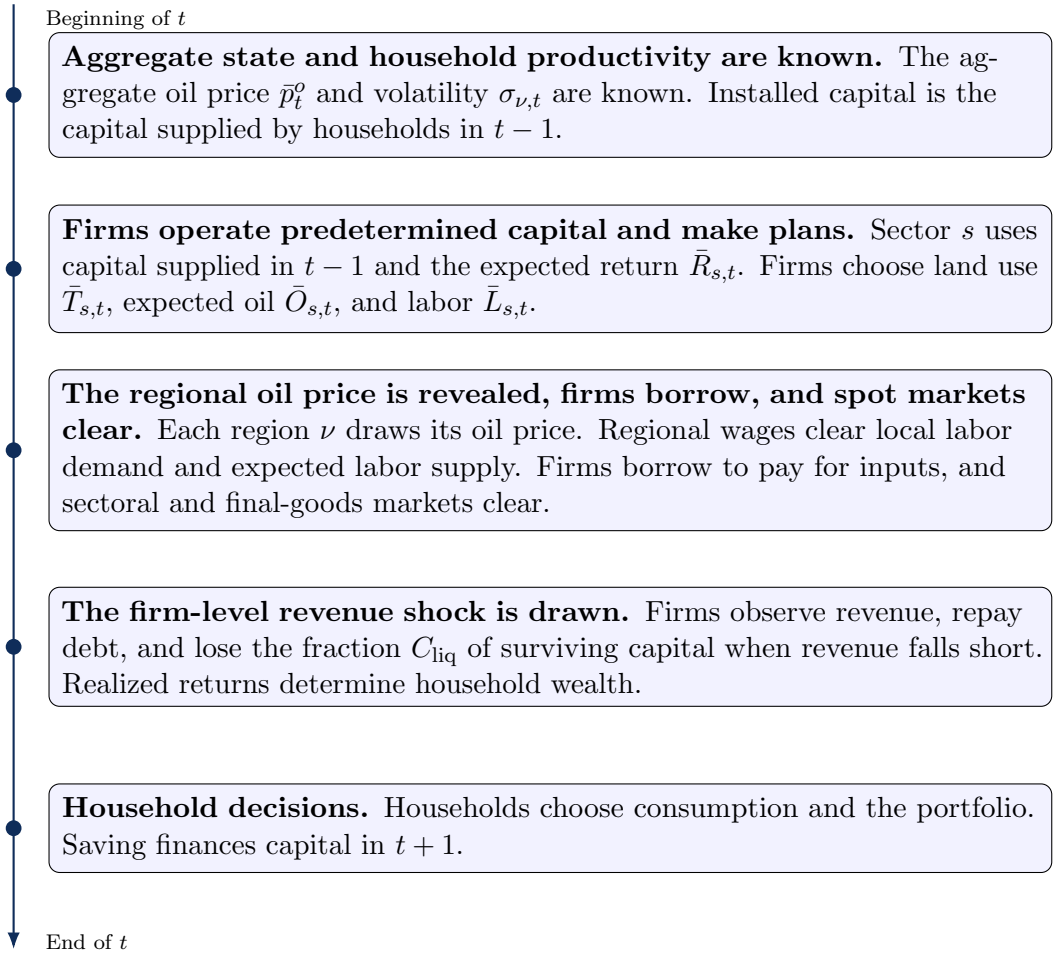



Figure 7: Timing inside a period.

Gross output combines oil, the capital–labor composite, and the fixed factor directly in one CES nest:

$$Y_{js}^{g} = \left[(1-\phi_s)\alpha_{0,s} O_{js}^{\frac{\rho_f-1}{\rho_f}} + (1-\phi_s)\alpha_{LK,s} M_{js}^{\frac{\rho_f-1}{\rho_f}} + \phi_s T_s^{\frac{\rho_f-1}{\rho_f}}\right]^{\frac{\rho_f}{\rho_f-1}}, \qquad \sum_s T_s = \bar{T}. \tag{17}$$

An exactly zero $\alpha_{0,s}$ omits oil from that sector's technology. The fixed aggregate supply $\bar{T}$ gives decreasing returns in the variable inputs and anchors scale. Sectoral land is allocated with a softmax transformation so that the land market clears exactly, and its competitive rent is rebated equally to households.

Firms choose planned oil and labor inputs, $\bar{O}_s$ and $\bar{L}_s$, before the aggregate oil realization. Deviations incur resource costs

$$\Phi_{js}^{I} = \kappa_O \frac{(O_{js} - \bar{O}_s)^2}{\bar{O}_s} + \kappa_L \frac{(L_{js} - \bar{L}_s)^2}{\bar{L}_s}. \tag{18}$$

Net sectoral output is $Y_{js} = Y_{js}^{g} - \Phi_{js}^{I}$.

## 5.4 Final goods

A competitive producer combines sectoral goods:

$$Y_j = A_Y \left[\omega Y_{jH}^{\frac{\eta-1}{\eta}} + (1-\omega) Y_{jL}^{\frac{\eta-1}{\eta}}\right]^{\frac{\eta}{\eta-1}}. \tag{19}$$

The final good is the numeraire. Sectoral prices satisfy the dual price-index condition and sectoral demand equations associated with (19).

## 5.5 Firm finance, liquidation, and individual-firm returns

Variable inputs, adjustment expenditures, and land rent are paid before an idiosyncratic revenue shock $\zeta_i$ is realized. Firm $i$ in sector $s$ and aggregate state $j$ owes

$$D_{js} = p_j^o O_{js} + w_j L_{js} + p_{js} \Phi_{js}^{I} + q_T T_s. \tag{20}$$

Revenue is $\zeta_i p_{js} Y_{js}^{g}$, with $\mathbb{E}[\zeta_i] = 1$. The firm liquidates whenever

$$\zeta_i p_{js} Y_{js}^{g} < D_{js}. \tag{21}$$

Liquidation destroys the fraction $C_{\text{liq}}$ of surviving capital. The expected real loss is therefore

$$\mathcal{L}_{js} = C_{\text{liq}}(1-\delta)K_s \Pr\left(\zeta_i < \frac{D_{js}}{p_{js}Y_{js}^g}\right). \tag{22}$$

The benchmark revenue shock is uniform. Numerically, narrow linear ramps are placed at the two support edges to make the density and liquidation derivatives continuous; this approximation is arbitrarily close to the uniform benchmark away from the endpoints.

Individual-firm returns retain the discrete downside exposure generated by liquidation. Diversified sectoral returns integrate over $\zeta_i$. Oil volatility matters disproportionately for concentrated investors because

$$\text{Var}(R_{js}^i) = \text{Var}\big(\mathbb{E}[R_{js}^i \mid j]\big) + \mathbb{E}\big[\text{Var}(R_{js}^i \mid j)\big], \tag{23}$$

and the second term can rise sharply when liquidation moves into the support of the idiosyncratic shock.

## 5.6 Households

Household preferences are

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \psi \frac{n_t^{1+\gamma}}{1+\gamma}\right], \tag{24}$$

where $\gamma$ is the inverse Frisch elasticity. The consumption composite permits nonhomothetic oil demand:

$$C = \left[\chi_o^{1/\epsilon_o} c^{\frac{\epsilon_o-1}{\epsilon_o}} + (1-\chi_o)^{1/\epsilon_o}(o-\underline{o})^{\frac{\epsilon_o-1}{\epsilon_o}}\right]^{\frac{\epsilon_o}{\epsilon_o-1}}. \tag{25}$$

The subsistence term $\underline{o}$ is set to zero at the baseline calibrations, so oil demand is homothetic.

Own-sector exposure means that the cheapest portfolio is a claim on the firm and sector in which the household works. The exposure state follows

$$\Pi_e = \rho_e I + (1-\rho_e)\mathbf{1}\bar{\pi}_e', \tag{26}$$

where $\bar{\pi}_e$ is the invariant distribution of own-sector exposure. In the baseline, this distribution is anchored to average sectoral labor and moves with it in the MIT transitions.

## 5.7 Portfolio choice and diversification

Let $d \in [0,1]$ denote the share of savings placed in a diversified account, and let $\theta \in [0,1]$ denote the low-oil-sector share within that account. The remaining fraction $1-d$ is invested in an individual firm from the household's own sector. The gross portfolio return is

$$R'_p = (1-d)R'_{i,e} + d\left[(1-\theta)R'_H + \theta R'_L\right]. \tag{27}$$

Let $\theta_e^0 = 0$ when the household's own sector is the high-oil sector and $\theta_e^0 = 1$ when it is the low-oil sector. Diversification uses resources according to

$$\Phi(W, d, \theta; e) = W\left[\kappa_d d^2 + \kappa_s d^2(\theta - \theta_e^0)^2\right]. \tag{28}$$

The household budget constraint is

$$x + \left[1 + \kappa_d d^2 + \kappa_s d^2(\theta - \theta_e^0)^2\right] W = a + w_j z n + \tau_T, \tag{29}$$

where $x = c + p_j^o o$ and $\tau_T = q_T \bar{T}$ is the equal land-rent transfer. Next period's financial wealth is

$$a' = R'_p W. \tag{30}$$

The savings Euler equation is

$$\lambda\left[1 + \kappa_d d^2 + \kappa_s d^2(\theta - \theta_e^0)^2\right] = \beta \mathbb{E}[\lambda' R'_p]. \tag{31}$$

The portfolio conditions compare marginal-utility-weighted returns. Thus an asset with a high mean return can remain unattractive if it performs poorly in states with high marginal utility or if it exposes the household to sufficiently large idiosyncratic downside risk.

## 5.8 The volatility and own-sector-exposure mechanism

Consider an increase in $\sigma_\nu$ around a given mean oil price. In sector $H$, cash flows become more sensitive to the aggregate oil state. The liquidation threshold in (22) moves through the firm-shock distribution, increasing both liquidation probability and the variance of individual returns. For a household with own-sector exposure, reducing this exposure requires a higher $d$. Doing so is costly, and reallocating the diversified account toward the other sector is additionally costly when it moves away from $\theta_e^0$.

The household therefore responds along both the intensive and extensive portfolio margins. Risk-adjusted capital supply to sector $H$ falls directly. Sector $L$ also has a direct risk channel: its positive oil weight makes its cash flow and liquidation threshold vary with oil-price volatility, while state-contingent wages transmit oil risk to both sectors according to their labor intensities. Capital supply to sector $L$ can fall further because lower wealth and higher marginal utility reduce aggregate risk-bearing capacity. These direct and general-equilibrium effects distinguish the model from a pure portfolio-reallocation economy in which capital moves one-for-one from $H$ to $L$.

The asymmetry follows because volatility interacts differently with the direct cost effect of the price innovation. For a price increase, both channels reduce investment and activity. For a price decline, the direct cost effect raises desired production, while the volatility–portfolio channel lowers capital supply. The latter offsets the former. Full liquidation, borrowing constraints, and nonlinear portfolio costs make the offset state dependent and stronger for larger shocks.

# 6 Calibration

The model period is one quarter. Parameters are divided into external assignments, normalizations fixed at one, and objects adjusted to six steady-state moments. The final-good productivity $A_Y$, the labor-disutility coefficient, baseline labor productivity, and the aggregate land endowment are normalizations and are set to one. They are not calibrated. The only parameters left free of both an external source and a moment are the input-plan adjustment costs $\kappa_L$ and $\kappa_O$, with the restriction $\kappa_L = 2\kappa_O$.

## 6.1 Sectoral technology

Sector $H$ is mapped to oil-intensive production and sector $L$ to less oil-intensive activity. Their oil weights preserve a sizable but not exclusive difference in oil exposure. The final-good weight $\omega = 0.20$ is the high-energy sector's share in our European industry classification for 2015–2019, and $\eta$ governs substitution following relative-price movements. Capital–labor weights are not fixed in advance. With $\alpha_{K,s} = 1 - \alpha_{L,s}$, the two labor weights are adjusted jointly to an aggregate labor-compensation share of 0.60 and a sector-$H$ employment share of 0.30. Labor compensation, the residual capital share, and land rent are 0.60, 0.35, and 0.05 of GDP. Oil expenditure is 0.08 of GDP. Oil is an intermediate input, so it is not counted in GDP; the implied total of factor and oil bills is

Table 1: Working parameterization

| Parameter | Description | Value | Source/target |
|---|---|---|---|
| $\beta$ | Discount factor | 0.99 | Kydland and Prescott (1982) |
| $\sigma$ | Relative risk aversion | 3 | Aiyagari (1994) |
| $\gamma$ | Inverse Frisch elasticity | 2 | Chetty (2012); Chetty et al. (2011) |
| $\rho_z$ | Earnings persistence | 0.95 | Cooley and Prescott (1995) |
| $\sigma_z$ | Earnings innovation s.d. | 0.08 | Iacoviello and Pavan (2013) |
| $\delta$ | Depreciation | 0.015 | Cooley and Prescott (1995) |
| $\chi_o$ | Normal-good CES weight | 0.90 | Audzei and Sutoris (2024) |
| $\epsilon_o$ | Household oil substitution | 0.30 | Natal (2012) |
| $\rho_f$ | Oil–$(KL)$–land elasticity | 0.30 | Natal (2012) |
| $\epsilon_f$ | Capital–labor elasticity | 0.50 | Chirinko (2008); Knoblach, Roessler, and Zwerschke (2020) |
| $\eta$ | Cross-sector elasticity | 0.75 | Atalay (2017) |
| $\omega$ | Weight on sector $H$ | 0.20 | Own EU industry classification |
| $\alpha_{0,H}, \alpha_{0,L}$ | Oil weights | 0.175, 0.025 | High/low oil-intensity distinction |
| $\kappa_O$ | Oil plan-adjustment cost | 1.0 | Free; $\kappa_L = 2\kappa_O$ |
| $\kappa_L$ | Labor plan-adjustment cost | 2.0 | set on $[0.5, 1, 1.5]$ for robustness |
| $\rho_e$ | Own-sector exposure persistence | 0.90 | Employment persistence |
| $\kappa_d$ | Firm-diversification cost | 0.001 | Calvet, Campbell, and Sodini (2007); Betermier, Calvet, and Sodini (2017) |
| $\kappa_s$ | Cross-sector portfolio cost | 0.005 | Sectoral return premium |

Notes: Values are the current benchmark. The Stone–Geary oil floor is zero and is omitted. The six internally calibrated objects—$\phi_s$, $\bar{p}^o$, $\alpha_{L,s}$, $\alpha_{K,s}$, $\mathrm{Var}(\zeta)$, and $C_{\mathrm{liq}}$—are reported with their targets in Table 2. $\kappa_d$ and $\kappa_s$ remain at these values until they are matched to the concentrated-portfolio share and the high-energy return premium. The oil weights are reported before the common technology scaling used in the code. Normalizations set to one are omitted.

1.08 of GDP. The elasticity of substitution between capital and labor is set to $\epsilon_f = 0.50$. Chirinko (2008) places the long-run aggregate elasticity in a range of roughly 0.4 to 0.6, and the meta-analysis of Knoblach, Roessler, and Zwerschke (2020) centers on a value near one half.

The costs of missing the input plan, $\kappa_O$ and $\kappa_L$, are the remaining free parameters. A proportional miss of size $e = (x - \bar{x})/\bar{x}$ costs $\kappa e^2$ units of the planned input, so $\kappa \in [0.1, 2]$ means that a 10 percent deviation costs between 0.1 and 2 percent of the planned oil bill or wage bill. That interval is the order of magnitude of the convex factor-adjustment costs surveyed by Hamermesh and Pfann (1996) and estimated by Hall (2004). The benchmark sets $\kappa_O = 1.0$ and $\kappa_L = 2.0$. Labor receives the larger cost because hours are harder to reset within the quarter than spot energy purchases (Hamermesh and Pfann, 1996).

## 6.2 Households and portfolios

The productivity process uses standard quarterly persistence and dispersion. Relative risk aversion is $\sigma = 3$, the intermediate value in the incomplete-markets calibration of Aiyagari (1994). The inverse Frisch elasticity is $\gamma = 2$, so the Frisch elasticity is one half. Chetty (2012) finds that an intensive-margin Frisch elasticity of one half is consistent with the micro evidence once optimization frictions are allowed for, and Chetty et al. (2011) report the same central intensive-margin value. The labor-disutility coefficient is a normalization equal to one.

The portfolio costs are reserved for two moments that are not yet imposed. $\kappa_d$ governs the share of saving kept in the firm from the household's own sector. The source for that share is the direct-stockholding evidence in Calvet, Campbell, and Sodini (2007) and the finding in Betermier, Calvet, and Sodini (2017) that professionally close stocks account for about 16 percent of direct stockholdings. $\kappa_s$ governs how costly it is to tilt the diversified account away from the household's own sector, and it will be used to match the return premium of the high-energy sector over the low-energy sector in our industry portfolios. Until those two targets are imposed, the parameters stay at $\kappa_d = 0.001$ and $\kappa_s = 0.005$.

## 6.3 Firm risk and liquidation

The idiosyncratic revenue shock is approximately uniform, with narrow endpoint ramps used only to smooth derivatives. Its support is the instrument for the liquidation rate.

Liquidation destroys the fraction $C_{\text{liq}}$ of surviving capital and is the instrument for the idiosyncratic return variance. The two objects are solved jointly, because a wider revenue-shock support changes both the frequency of liquidation and the dispersion of returns among surviving firms, while a larger liquidation loss widens the gap between liquidation and repayment returns.

The sixth moment is the capital-weighted variance of idiosyncratic firm returns. Inside each sector $s$ and region $j$, firm returns are measured as deviations from the mean return in that region. Let $\omega_\zeta$ be the probability weights on the revenue shock and $\pi_j$ the probability of region $j$. The sectoral variance and the aggregate moment are

$$v_s = \sum_j \pi_j \sum_i \omega_{\zeta,i} \left(R_{ijs} - \bar{R}_{js}\right)^2, \qquad v = \sum_s \frac{K_s}{K_H + K_L} v_s. \tag{32}$$

Subtracting $\bar{R}_{js}$ removes the variation that comes from the regional oil price itself, so $v$ is within-region dispersion. The weights $K_s$ are installed capital. The target is $v = 0.0038$ per quarter. That number is the square of the pooled cross-sectional standard deviation of the mild-loss payout that replaces the adjusted dividend with the quarterly return when that return lies in $(-20\%, 0)$, with zero-dividend years included and 1998–2001 and 2020 left out. The corresponding standard deviation is 6.16 percent. The same capital weights and revenue-shock probabilities define the liquidation rate, whose target is 0.015 per quarter, about 5–6 percent at an annual frequency. The variance moment uses the returns themselves, including the liquidation return on the shortfall side of that threshold.

For the household calculation, the full firm-return distribution is reduced to four weighted groups: liquidation, low surviving, middle surviving, and high surviving. This is a numerical quadrature rather than an additional calibration target. It preserves group probabilities and conditional means and is checked against diversified sectoral mean returns.

# 7 Nonlinear MIT Experiments

An oil-price shock in the model is a change in the expected oil price together with a rise in cross-regional price dispersion. The dispersion increase that accompanies the price movements raises cross-regional dispersion on impact from its steady-state value of 0.16 to 0.24, the same increase used in Section 4, and then returns toward steady state. The

Table 2: Moments used to discipline the working calibration

| Moment | Target | Primary adjusted object | Calibrated value |
|---|---|---|---|
| Land rent / model GDP | 0.05 | $\phi_s$ | 0.03 |
| Firm and household oil expenditure / model GDP | 0.08 | $\bar{p}^o$ | 0.175 |
| Aggregate labor compensation / model GDP | 0.60 | $\alpha_{L,s}$ | 0.60, 0.70 |
| Sector-$H$ employment share | 0.30 | $\alpha_{K,s} = 1 - \alpha_{L,s}$ | 0.40, 0.30 |
| Average quarterly liquidation probability | 0.015 | $\mathrm{Var}(\zeta)$ | 0.0208 |
| Idiosyncratic return variance, (32) | 0.0038 | $C_{\mathrm{liq}}$ | 0.30 |

Notes: Wealth-distribution moments and average hours are not calibration targets. Calibrated values are those of the current working parameterization. Sectoral pairs are reported as sector $H$, then sector $L$. The residual capital share is 0.35 of GDP. Oil is an intermediate input, so labor, capital, and land add to 1 of GDP and the oil bill of 0.08 of GDP brings total expenditure to 1.08 of GDP. The return-variance target is the square of the pooled mild-loss dispersion that uses quarterly returns in $(-20\%, 0)$. $\mathrm{Var}(\zeta)$ is the variance of the approximately uniform revenue shock with support $[0.75, 1.25]$. $C_{\mathrm{liq}}$ is the share of surviving capital destroyed in liquidation. The final-good weight $\omega$ is set externally to 0.20. The portfolio costs $\kappa_d$ and $\kappa_s$ are reserved for the concentrated-portfolio share and the high-energy return premium and are not yet adjusted to those moments. The liquidation rate and $v$ are capital-weighted and are solved jointly.

paths are scaled to the empirical magnitudes in Section 3. Each path is unexpected on impact. After that date, agents have perfect foresight over the aggregate paths, while firms still do not know the regional price they will draw. The experiments compare an oil-price increase with an oil-price decline, and they shut down either the price change or the dispersion increase. The contribution is the interaction: the same dispersion shock does not have the same effect after the two price movements.

## 7.1 Joint price and dispersion shocks

The joint experiment is the one suggested by the data. A supply shock raises both the volatility of the real oil price and the dispersion of European energy prices.

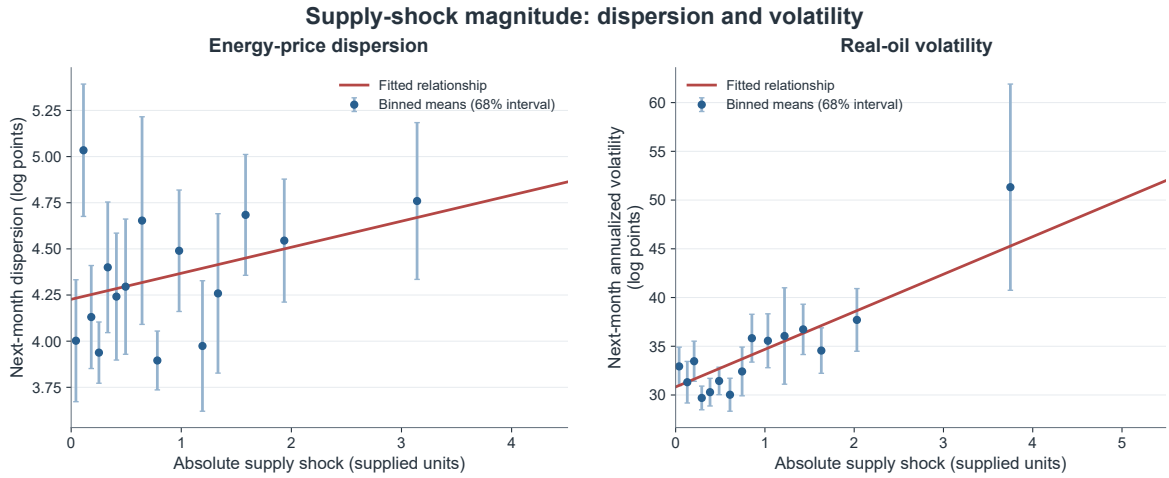



Figure 8: A supply shock and subsequent energy-price dispersion and oil-price volatility. The left panel is cross-country dispersion. The right panel is time-series volatility.

Figure 9 shows the price and dispersion paths used to compare a favorable joint shock with an adverse one. Both experiments include the rise in dispersion. Figure 10 plots the resulting GDP and total-capital responses. The adverse joint shock has the larger absolute effect.

The asymmetry between the two joint responses is

$$\mathcal{A}_t = r_t(p^o \uparrow, \sigma_\nu \uparrow) + r_t(p^o \downarrow, \sigma_\nu \uparrow). \tag{33}$$

A negative value means that the contraction after the adverse shock exceeds the expansion after the favorable shock in absolute value. Figure 11 plots (33) for GDP and total capital.

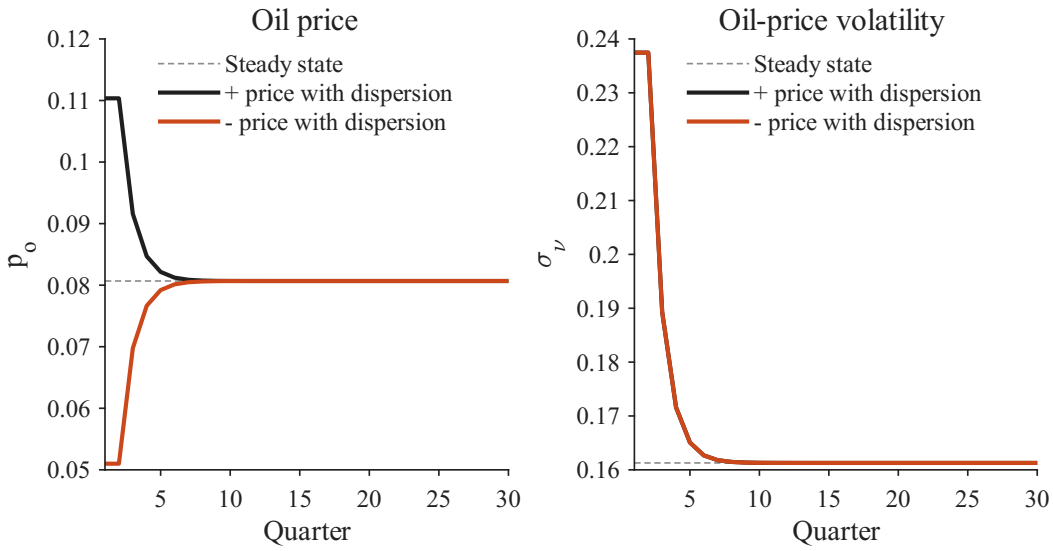
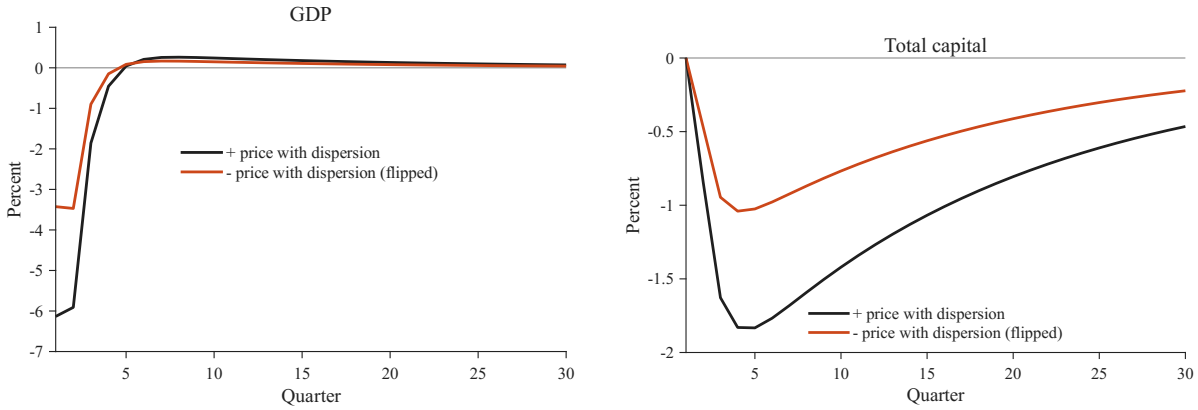
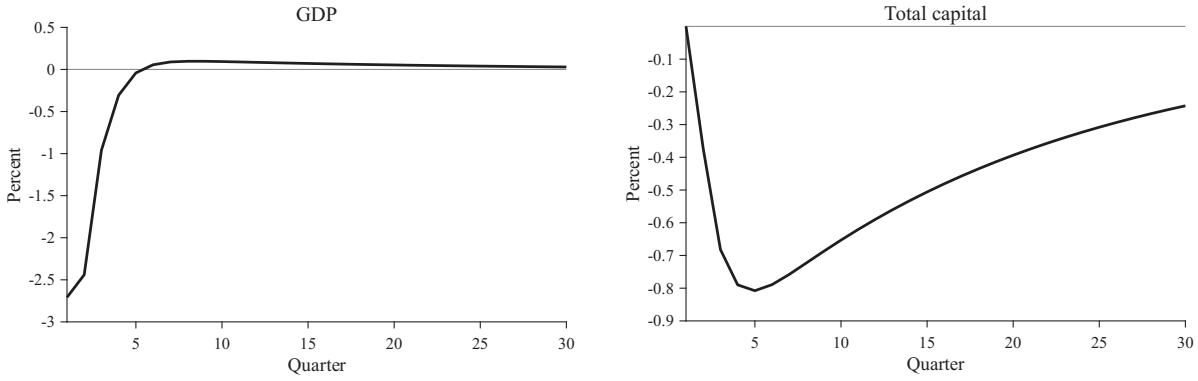



Figure 9: Price and dispersion paths for the joint comparison of an oil-price increase and an oil-price decline.




(a) GDP (b) Total capital

Figure 10: GDP and total capital after joint price and dispersion shocks. Both experiments include the rise in dispersion.




(a) GDP (b) Total capital

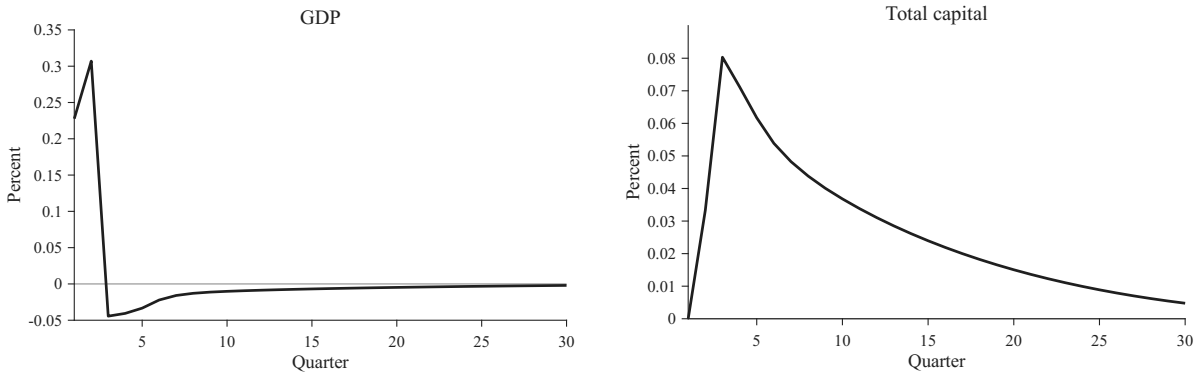
Figure 11: Asymmetry between the adverse and favorable joint shocks. A negative path means the contraction is larger in absolute value.

## 7.2 Price changes alone

Holding dispersion at its steady-state value, the same comparison is

$$\mathcal{A}_t = r_t(p^o \uparrow) + r_t(p^o \downarrow). \tag{34}$$

A positive value means that the expansion after the price decline exceeds the contraction after the price increase. Figure 12 shows that price shocks alone are close to symmetric. The small gap that remains has the opposite sign from the joint experiment. The asymmetry in the joint experiment is therefore not a property of the price change by itself.

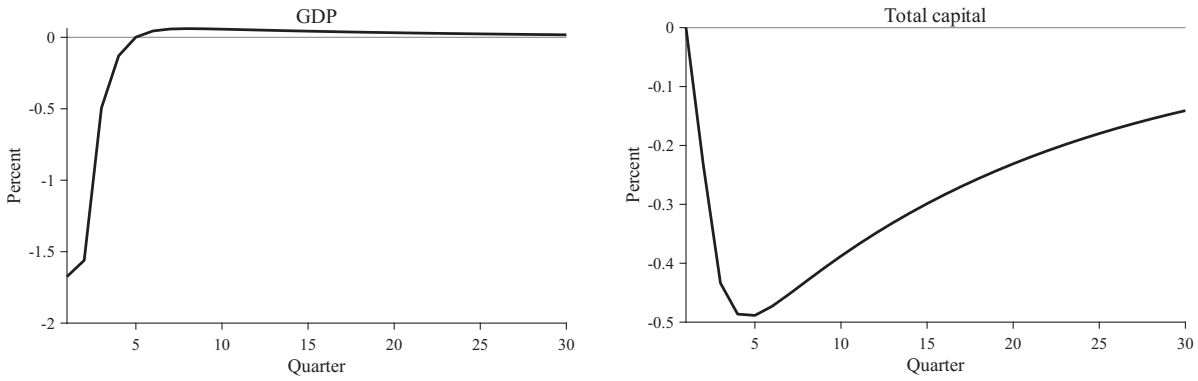



(a) GDP (b) Total capital

Figure 12: Asymmetry between an oil-price increase and an oil-price decline, with dispersion held at steady state.

## 7.3 Dispersion by itself, and its interaction with the price

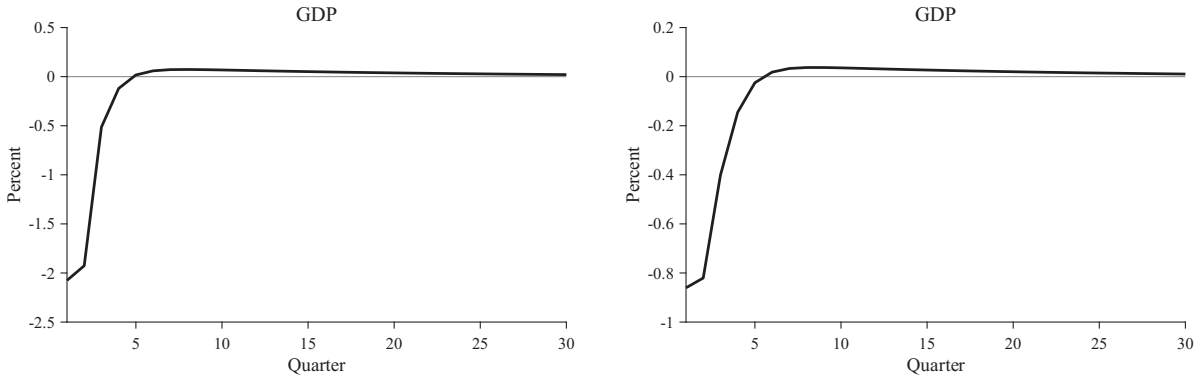
A pure dispersion shock leaves the mean oil price at steady state and raises cross-regional dispersion. Output and total capital both fall (Figure 13).




(a) GDP (b) Total capital

Figure 13: Aggregate responses to a pure increase in price dispersion.

The interaction is the incremental effect of adding that same dispersion shock to a price movement. Figure 14 shows the two cases. The dispersion shock is more contractionary after an oil-price increase than after an oil-price decline.

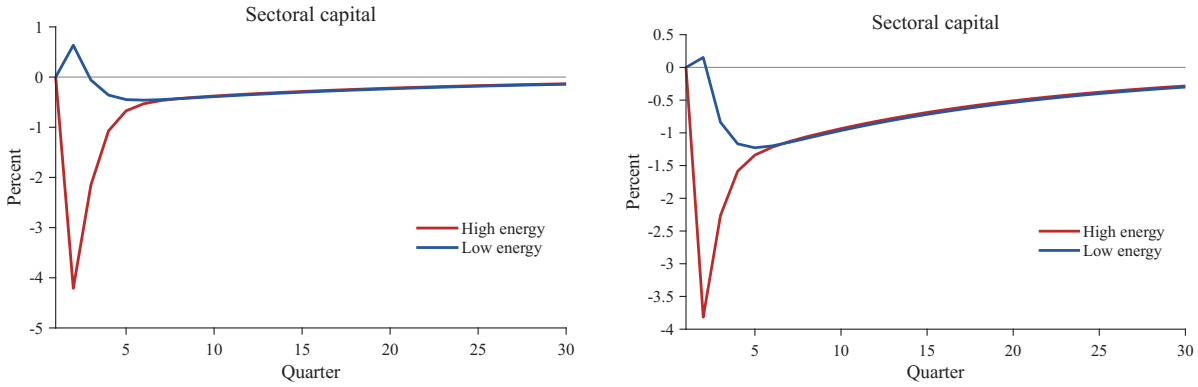



(a) With an oil-price increase (b) With an oil-price decline

Figure 14: Incremental GDP effect of the same dispersion shock added to each price movement.

Dispersion also reallocates capital across sectors more strongly than a pure price increase does (Figure 15).

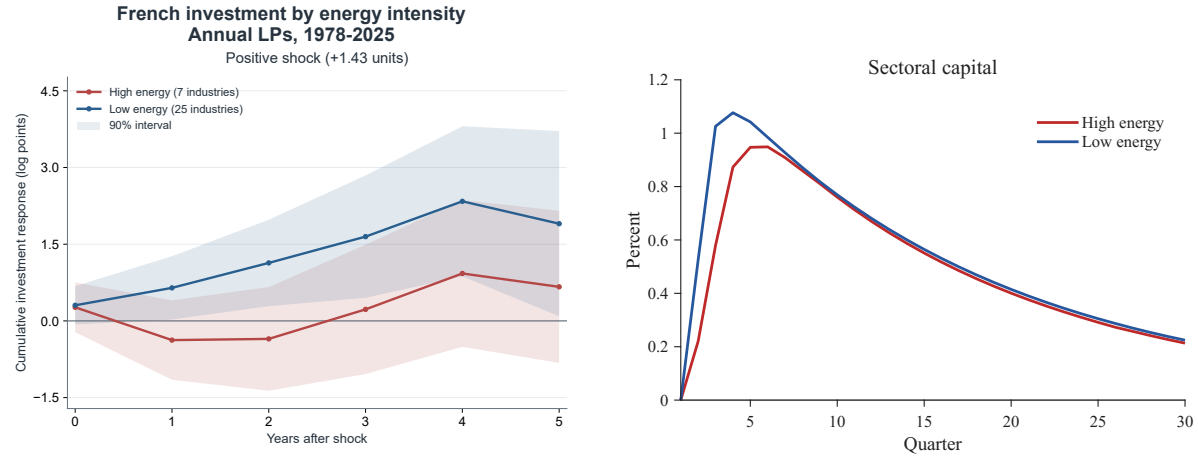
The sectoral pattern lines up with investment by energy intensity. Figure 16 places cumulative French investment responses, by energy intensity, next to the model's sectoral




(a) Dispersion increase (b) Oil-price increase

Figure 15: Sectoral capital after a dispersion increase and after an oil-price increase.

capital response when a price decline is combined with the dispersion shock. In the French panel a positive shock is a positive supply shock, and therefore an oil-price decline.




(a) French investment by energy intensity (b) Model, price decline plus dispersion

Figure 16: Sectoral investment in the data and sectoral capital in the joint favorable experiment.

Table 3 collects the peak aggregate responses. Joint shocks are asymmetric. Price shocks alone are close to symmetric. The dispersion shock lowers output and investment, and the adverse joint movement is the large contraction.

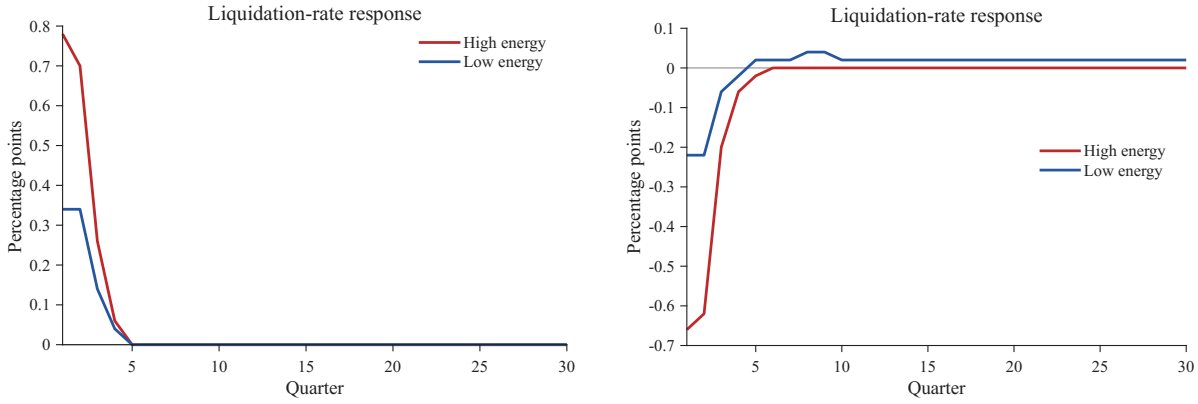
Table 3: Peak aggregate responses

| | Single shock | | | With a dispersion increase | |
|---|---|---|---|---|---|
| Outcome | $p^o \downarrow$ | $p^o \uparrow$ | $\sigma_\nu \uparrow$ | $\sigma_\nu \uparrow$, $p^o \downarrow$ | $\sigma_\nu \uparrow$, $p^o \uparrow$ |
| Peak output (%) | 4.29 | −4.06 | −1.67 | 3.46 | −6.13 |
| Peak total investment (%) | 39.4 | −37.2 | −15.5 | 32.1 | −56.3 |
| Sectoral capital correlation | −0.27 | −0.31 | −0.84 | 0.88 | −0.54 |

$p^o \downarrow$ is a favorable price decline and $p^o \uparrow$ is an adverse price increase. The correlation is calculated through the last quarter with a nonzero shock path.

## 7.4 Liquidation and sectoral returns

The aggregate asymmetry shows up in liquidation and in sectoral returns. A price increase raises liquidation in both sectors, and a price decline lowers it (Figure 17). The same price shocks move sectoral returns in opposite directions (Figure 18). A price decline raises the relative return of the oil-intensive sector, and a price increase lowers it.



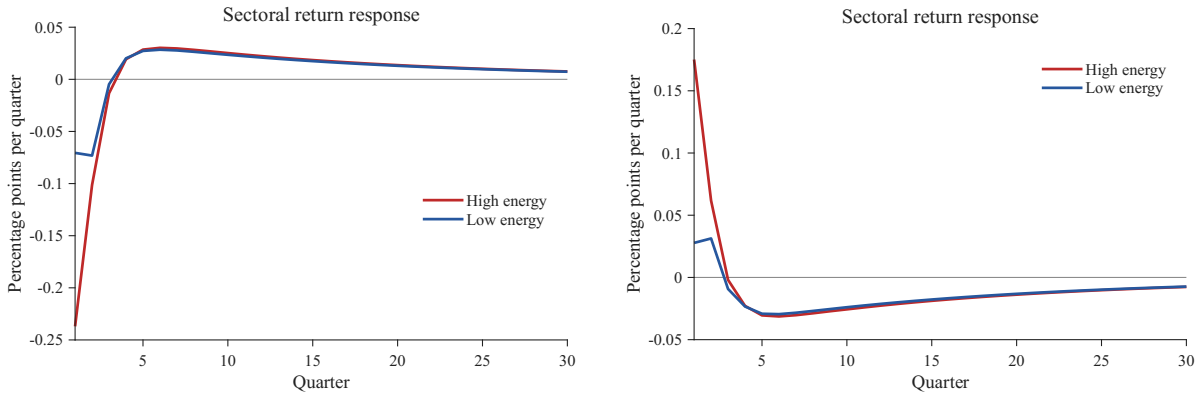

(a) Oil-price increase

(b) Oil-price decline

Figure 17: Liquidation-rate responses to price shocks alone.

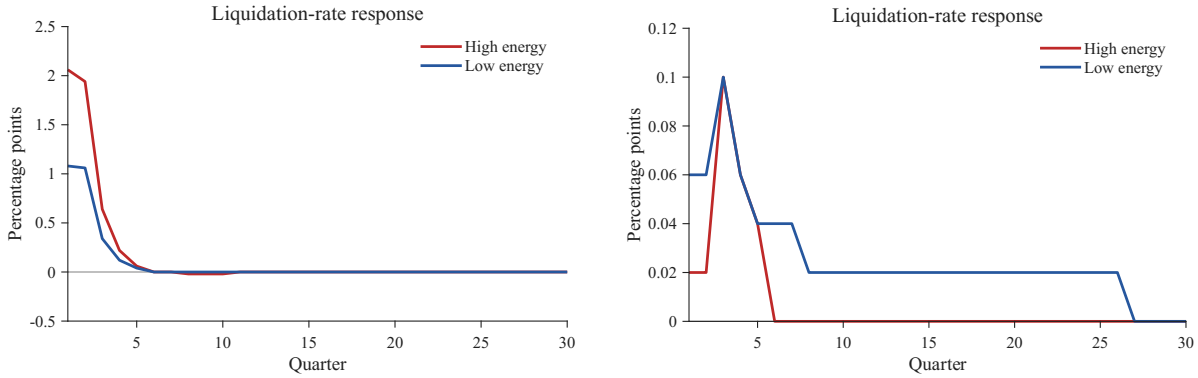
Adding the same dispersion increase raises liquidation after either price movement, and the larger response follows the price increase (Figure 19). The joint shocks also widen the sectoral return response (Figure 20). The volatility shock cancels the relative-return gain of the oil-intensive sector after a price decline and deepens its loss after a price increase. Liquidation is the main channel, so the variance of idiosyncratic returns rises with it.

These responses are the interaction the paper is about. Price movements alone push liquidation and relative returns in opposite directions and by similar amounts. The dispersion shock raises liquidation after either price movement, by more after the price




(a) Oil-price increase (b) Oil-price decline

Figure 18: Sectoral return responses to price shocks alone, in percentage points from each sector's steady state.

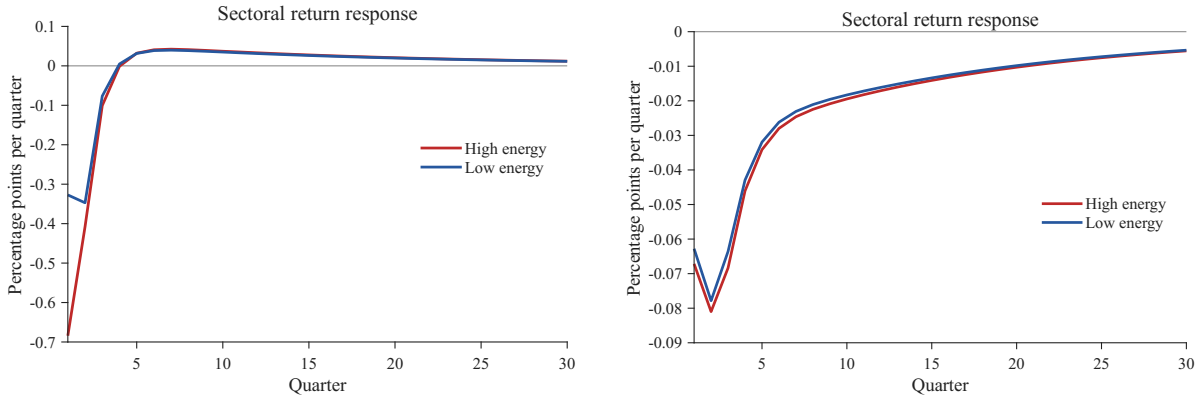



(a) Price increase plus dispersion (b) Price decline plus dispersion

Figure 19: Liquidation-rate responses when the dispersion shock is added to each price movement.

Table 4: Peak liquidation and return responses

| | Single shock | | | With a dispersion increase | |
|---|---|---|---|---|---|
| Outcome | $p^o \downarrow$ | $p^o \uparrow$ | $\sigma_\nu \uparrow$ | $\sigma_\nu \uparrow$, $p^o \downarrow$ | $\sigma_\nu \uparrow$, $p^o \uparrow$ |
| Peak relative return (p.p.) | 0.14 | −0.16 | −0.19 | −0.00 | −0.34 |
| Peak liquidation (p.p.) | −0.29 | 0.41 | 0.71 | 0.09 | 1.25 |
| Peak variance ($\times 10^4$) | −4.8 | 6.6 | 10.1 | 1.4 | 19.2 |




(a) Price increase plus dispersion

(b) Price decline plus dispersion

Figure 20: Sectoral return responses when the dispersion shock is added to each price movement.

increase, and that is when the contraction in output and capital becomes the larger one.

Uncertainty rises after supply news of either sign (Forni, Franconi, Gambetti, and Sala, 2025). Section 3 finds only limited evidence that U.S. dividend dispersion and the energy-intensity return premium respond persistently to these shocks. In the model, the same dispersion shock raises liquidation and the variance of idiosyncratic returns by more after a price increase, and by more where oil use is higher. That cross-section is the sectoral counterpart of the aggregate interaction.

Separating the capital-demand channel from the household-exposure channel is left to counterfactuals that are not yet computed: removing the cost of diversifying within the sector, removing the cost of diversifying across sectors, removing the liquidation loss, removing the persistence of own-sector exposure, and giving the two sectors the same oil intensity.

# 8 An Almost-Nonlinear Level-$k$ Algorithm for MIT Transitions

The MIT experiments generate nonlinear responses because liquidation probabilities, household portfolio shares, borrowing constraints, and firm input choices move with the level and sign of the shock. We therefore use an almost-nonlinear algorithm that combines level-$k$ learning with local on-impact corrections. Dynamic household and firm policies remain nonlinear conditional on believed paths; linearity is used only to accelerate small contemporaneous market-clearing adjustments.

## 8.1 Belief levels and the equilibrium fixed point

Let $B^k$ denote the collection of paths believed at level $k$, including state-contingent wages, sectoral returns, oil prices and dispersion, firm demand scales, and market-clearing prices. Let $\mathcal{S}(B)$ be the nonlinear policy operator and $\mathcal{C}$ the forward simulation and clearing operator. One level of the algorithm is

$$P^k = \mathcal{S}(B^{k-1}), \qquad X^k = \mathcal{C}(P^k, B^{k-1}), \qquad B^k = \mathcal{M}(B^{k-1}, X^k), \tag{35}$$

where $\mathcal{M}$ applies damping and, in later iterations, Gaussian weights over recent levels. Level zero consists of steady-state endogenous paths together with the announced oil-price and volatility paths. At convergence,

$$B^\star = X^\star,$$

so beliefs agree with realized equilibrium outcomes. The iteration follows the level-$k$ procedure of Farhi and Werning (2019). Here it is used as a computational fixed-point iteration, not as an assumption that households or firms permanently stop at a finite reasoning level.

## 8.2 Nonlinear anticipated policies

At each level, households solve their complete policy path backward given the believed paths of wages, sectoral and individual-firm returns, oil prices, and transition probabilities. Consumption, labor, saving, the diversified share, and the sectoral portfolio share are therefore nonlinear policy functions. The household distribution is subsequently propagated forward using realized returns and the nonlinear policy-induced transition matrix.

Firms solve backward policy functions over lagged sectoral capital. For every on-grid current capital choice $K_{s,t}$, the capital first-order condition is inverted to recover the endogenous lagged state $K^{\text{endo}}_{s,t-1}$ that rationalizes that choice. These endogenous points are interpolated back to a fixed lagged-capital grid. Post-horizon policies use a cached steady-state continuation. The baseline MIT exercises set

$$\boxed{\kappa_K = 0}.$$

Positive capital-adjustment costs are reserved for robustness exercises; oil and labor

planning costs remain active.

## 8.3 On-impact response banks

Re-solving both dynamic programs after every small within-period price update would be prohibitively expensive. Before the level iteration, we therefore place standardized innovations at an interior date, $T_{\text{shock}} = 50$, and compute normalized one-step response banks:

$$J_{\text{oi}}^{HH} = \frac{\partial(x, W', d, \theta, n)}{\partial(w_j, R_j^H, R_j^L, \text{wealth})}, \qquad J_{\text{oi}}^{F} = \frac{\partial(K, R, w, L, O, Y)}{\partial(\text{wage ladder}, \bar{R}, \text{firm scale})}. \tag{36}$$

If current clearing inputs differ only slightly from the believed inputs, a policy object $g$ is corrected locally:

$$g^{\text{current}} \approx g^{\text{belief}} + J_{\text{oi}}\Delta. \tag{37}$$

The firm policy is first interpolated at realized lagged capital and then corrected. If return-induced wealth revaluation exceeds the admissible local range, the household distribution transition is rebuilt using realized returns instead of relying on the local correction.

## 8.4 Forward clearing and belief updating

Within each date, the algorithm iterates on the wage ladder to close state-contingent labor gaps, on sectoral claim prices to close the two capital markets, and on the firm demand scale to close the goods market. It then advances sectoral capital and the household distribution. Expected sectoral returns are recovered from claim prices and next-period payoffs. The resulting cleared paths become the input to the next belief level.

## 8.5 Algorithm summary

**Step 1.** Construct level-zero beliefs from steady-state endogenous paths and the announced paths for $\bar{p}_t^o$ and $\sigma_{\nu,t}$.

**Step 2.** Precompute household and firm on-impact response banks at the interior shock date.

**Step 3.** Given level-$(k-1)$ beliefs, solve nonlinear household policies backward and nonlinear firm capital policies by backward inversion.

**Step 4.** Starting from the initial distribution and capital stocks, clear wages, claim markets, and the firm demand scale date by date using local on-impact corrections.

**Step 5.** Propagate the household distribution and sectoral capital forward, rebuilding transitions whenever wealth revaluation leaves the local range.

**Step 6.** Convert cleared outcomes into level-$k$ belief paths and update them with damping and recent-level weighting.

**Step 7.** Repeat until the maximum normalized belief distance and all market residuals satisfy their tolerances.

The procedure preserves nonlinear anticipated policy responses, exact liquidation probabilities, portfolio corners, and the dependence of realized wealth on returns. Its approximation is confined to local contemporaneous corrections inside each date's clearing loop. This compromise is substantially cheaper than repeatedly resolving both dynamic programs after every price update while retaining the nonlinearities needed to compare large positive and negative oil shocks.

# 9 Conclusion

This paper proposes a portfolio-based explanation for the asymmetric transmission of oil-price shocks. Oil-price volatility raises idiosyncratic firm-return risk, especially in oil-intensive production. Own-sector exposure and costly diversification prevent households from costlessly neutralizing this risk. Investment consequently falls across sectors, productive capacity contracts, and the favorable effect of an oil-price decline is attenuated while the adverse effect of a price increase is amplified.

The framework connects three objects typically studied separately: oil-price uncertainty, firm-level distress, and household portfolio concentration. Its nonlinear solution is important because liquidation, borrowing constraints, and portfolio corners cannot be summarized adequately by first-order dynamics. Future revisions will discipline the empirical shock process and portfolio parameters more tightly and use microdata to test the predicted interaction between oil intensity, ownership concentration, and firm investment.

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# A Figures

The figures below provide the descriptive context and additional specifications referenced in Section 3. Shock signs and normalizations match the corresponding main-text exercises. Monthly regressions use complete observations without interpolation; inference retains the calendar spacing across missing or excluded observations.

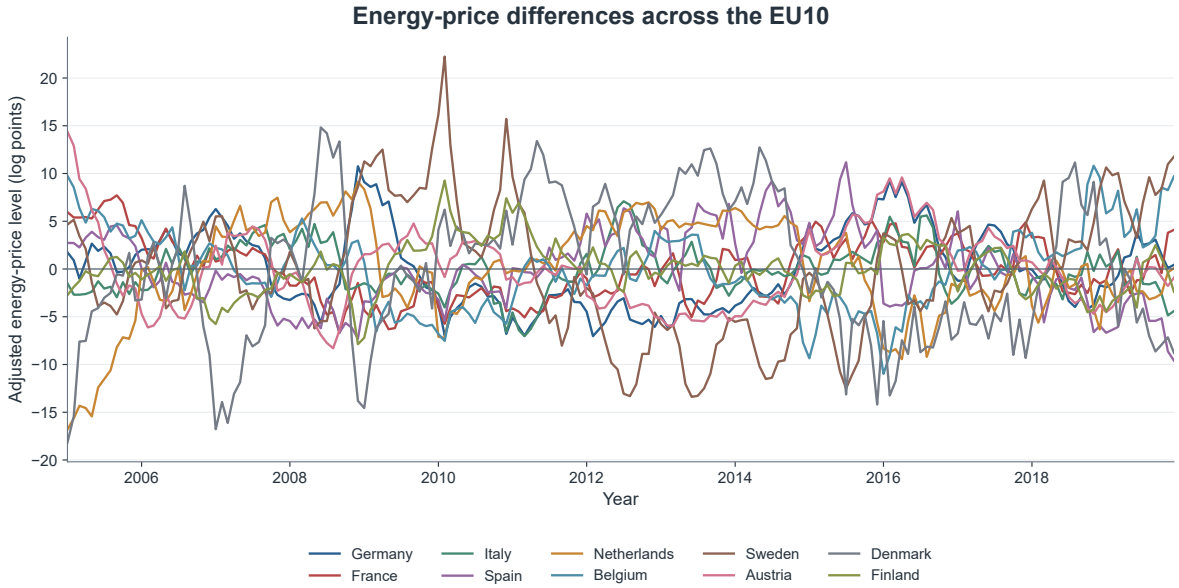



Figure 21: EU10 domestic energy-price residuals, 2005–2019, after country and month fixed effects and country-specific linear trends.

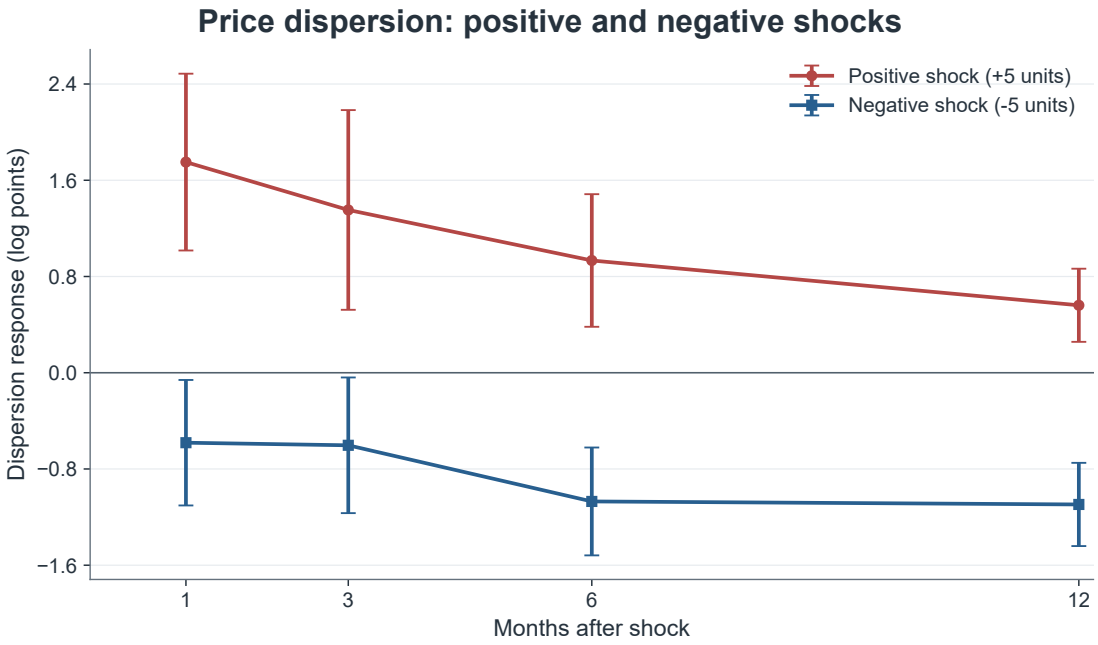



Figure 22: Future EU10 energy-price dispersion after positive and negative five-unit supply shocks, estimated jointly. Bars show pointwise 68% intervals.

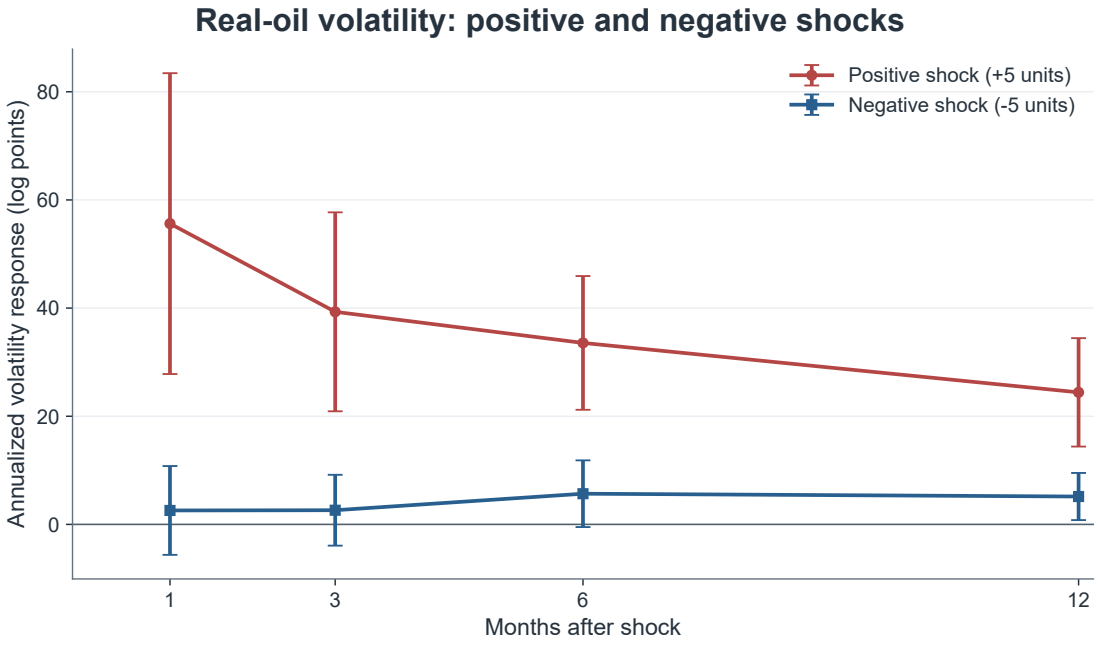



Figure 23: Future annualized real-oil volatility after positive and negative five-unit supply shocks, estimated jointly. Bars show pointwise 68% intervals.

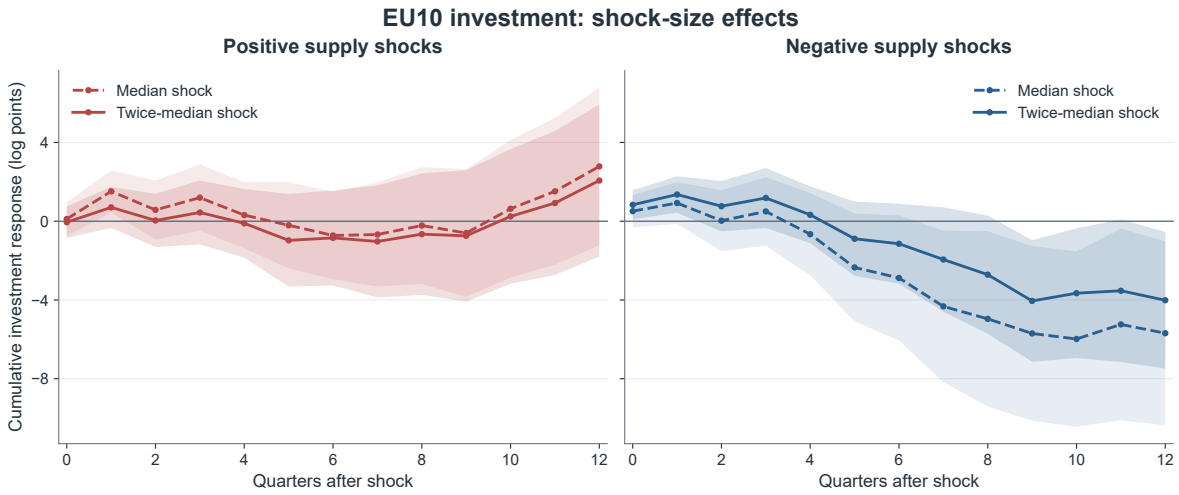



Figure 24: EU10 investment responses at the median and twice-median shock magnitudes in a continuous threshold specification. Shading shows pointwise 90% intervals.

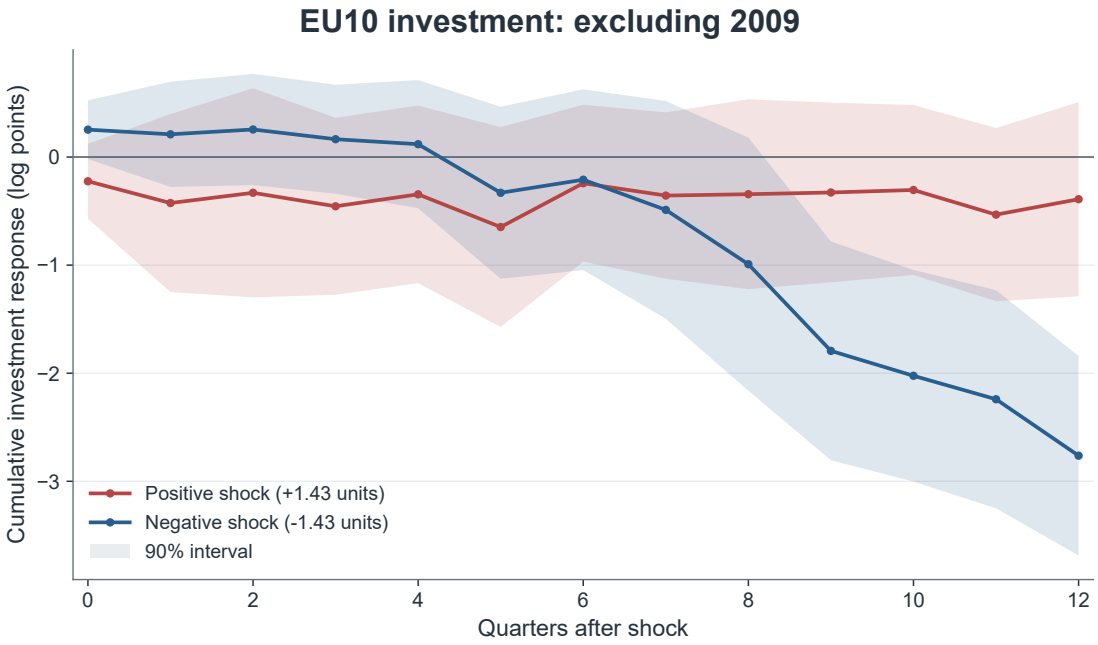



Figure 25: EU10 investment responses excluding 2009 shock-origin quarters, with the full-sample reference magnitude held fixed. Shading shows pointwise 90% intervals.

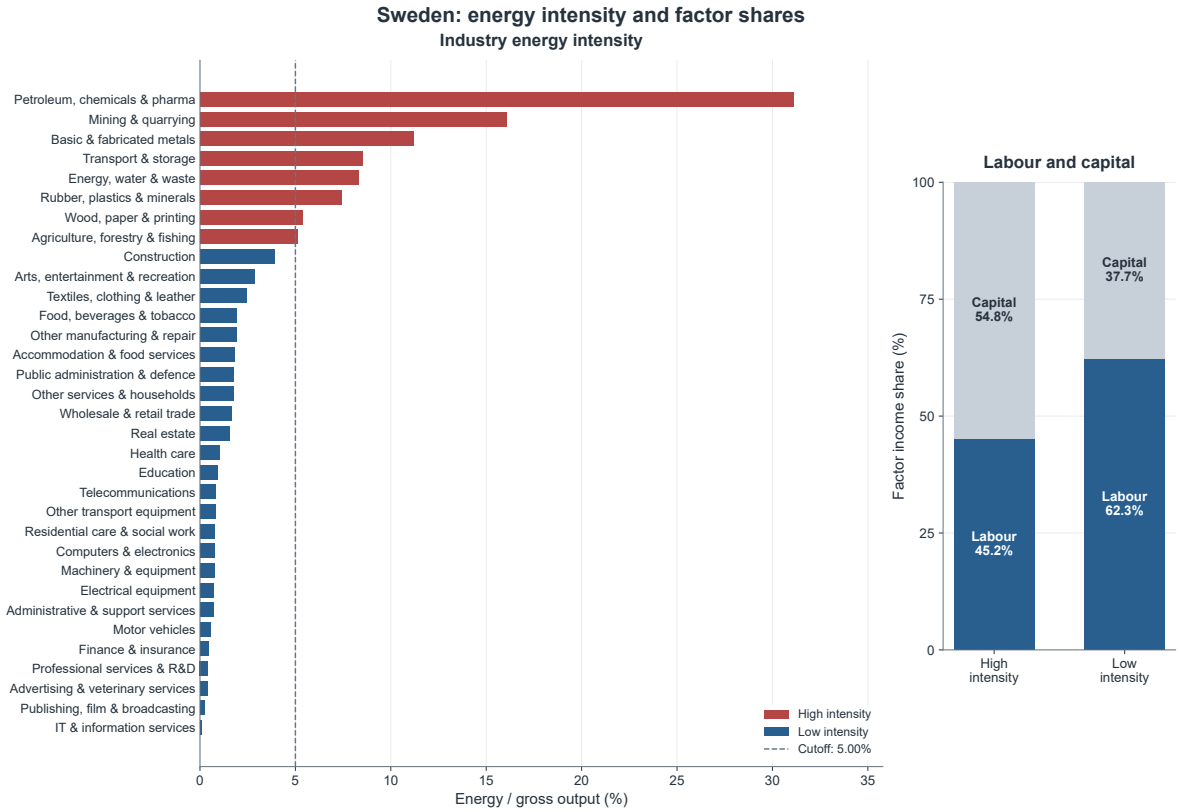



Figure 26: Swedish industry energy intensity and normalized labour and capital income shares. Classification uses 2015–2019 energy purchases relative to gross output and a 5% cutoff.

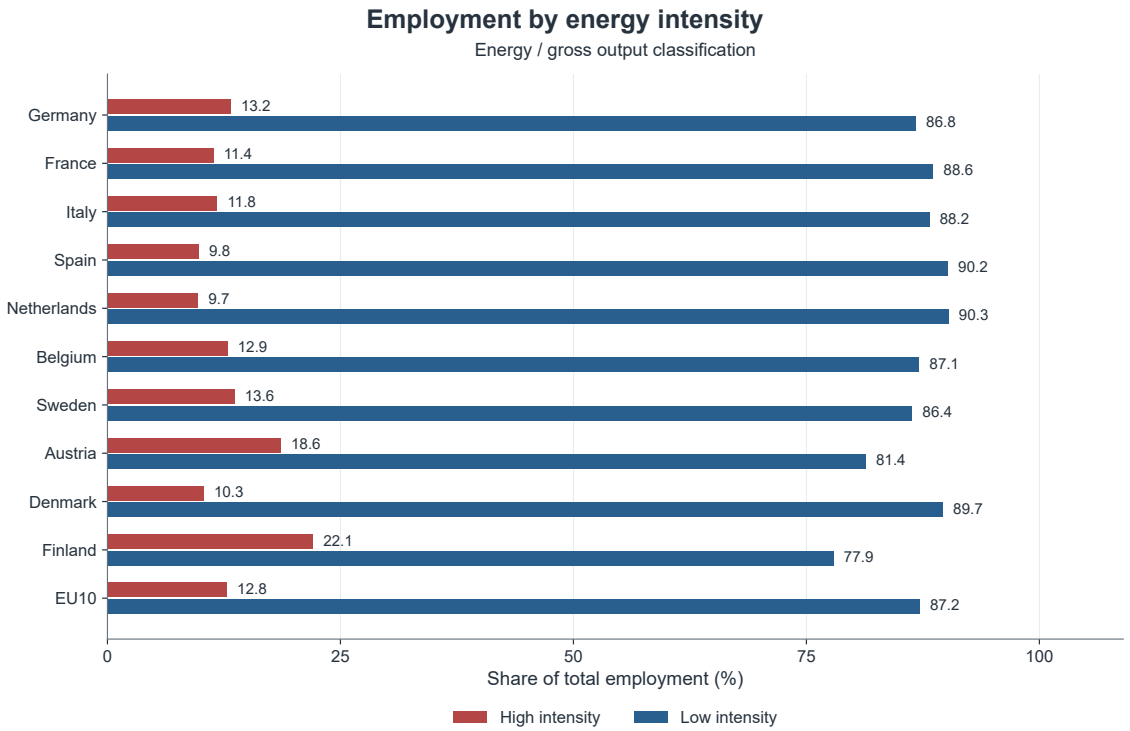



Figure 27: Employment shares of high- and low-energy-intensity groups, using the energy-purchases-to-gross-output classification.

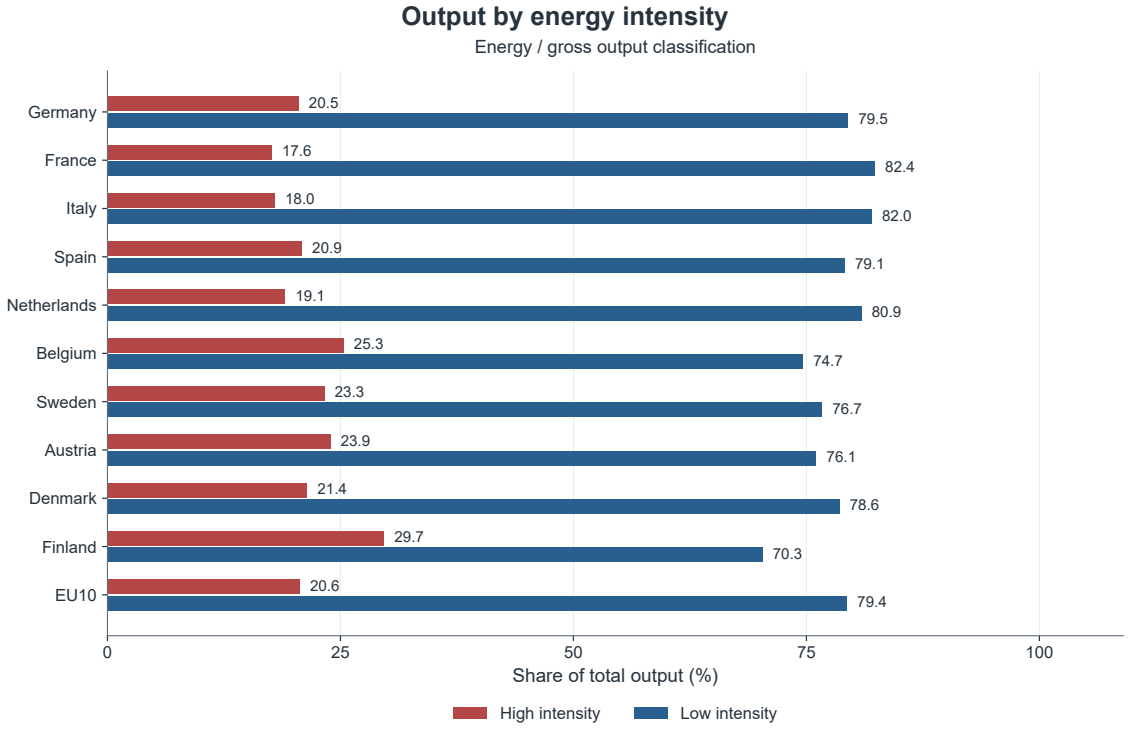



Figure 28: Gross-output shares of high- and low-energy-intensity groups, using the energy-purchases-to-gross-output classification.

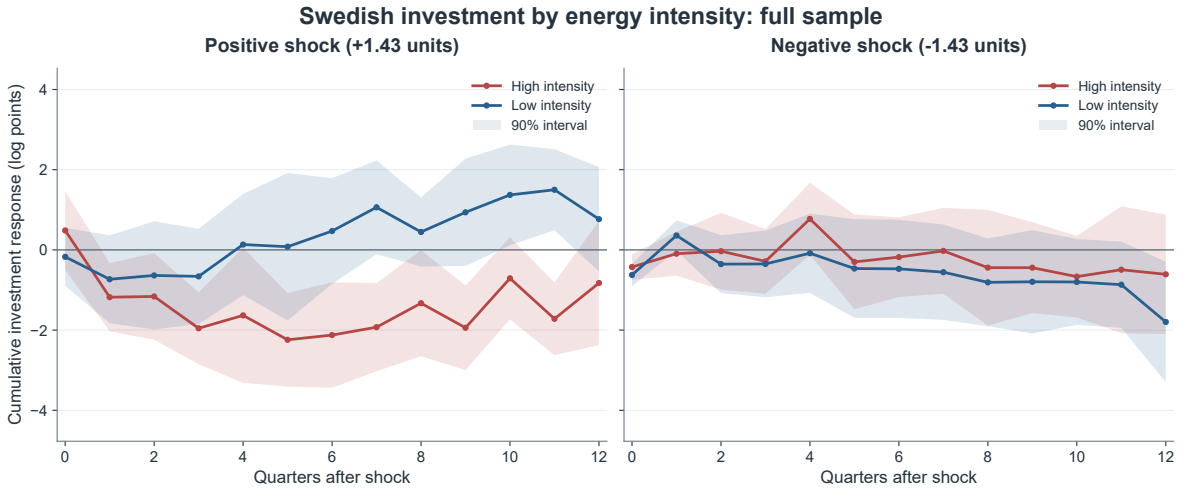



Figure 29: Swedish investment responses by energy intensity with data through 2026Q2. Groups and the reference shock magnitude are held fixed. Shading shows pointwise 90% intervals.

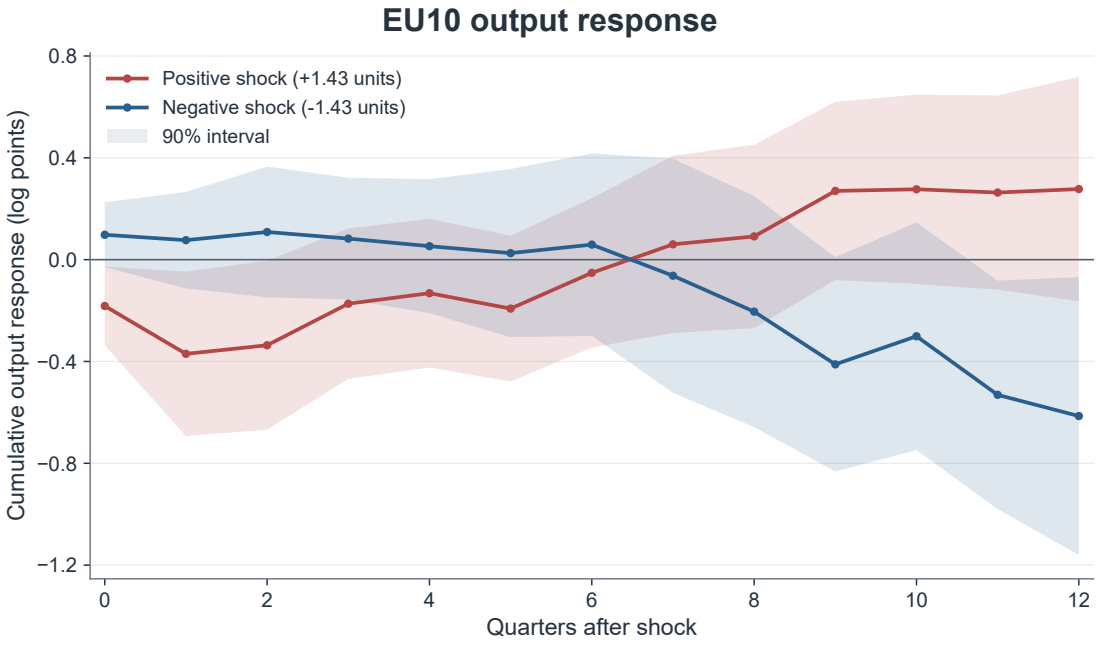



Figure 30: Cumulative EU10 real-GDP responses using the aggregate investment specification's controls and shock magnitude. Shading shows pointwise 90% intervals.

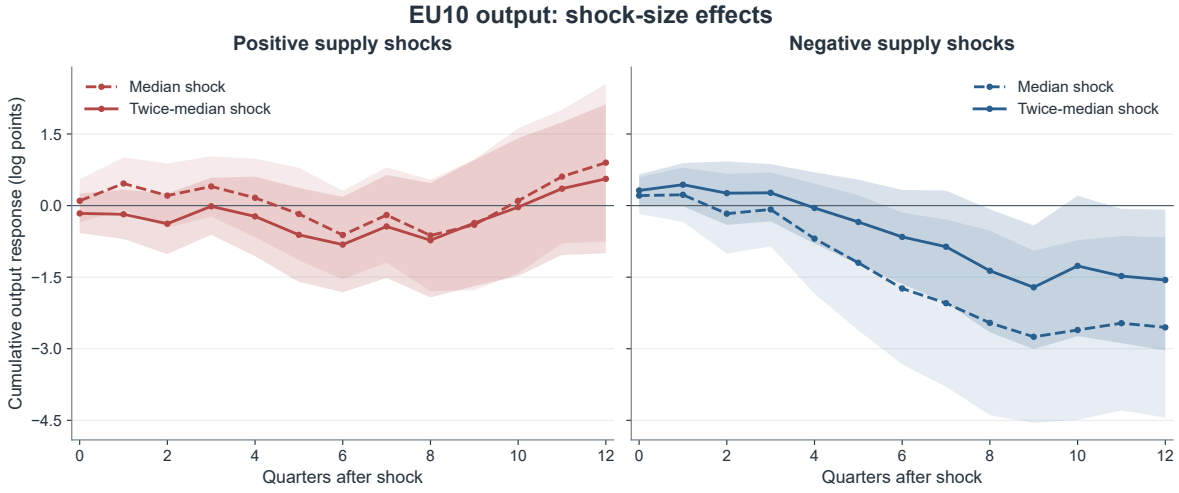



Figure 31: EU10 real-GDP responses at the median and twice-median shock magnitudes in a continuous threshold specification. Shading shows pointwise 90% intervals.

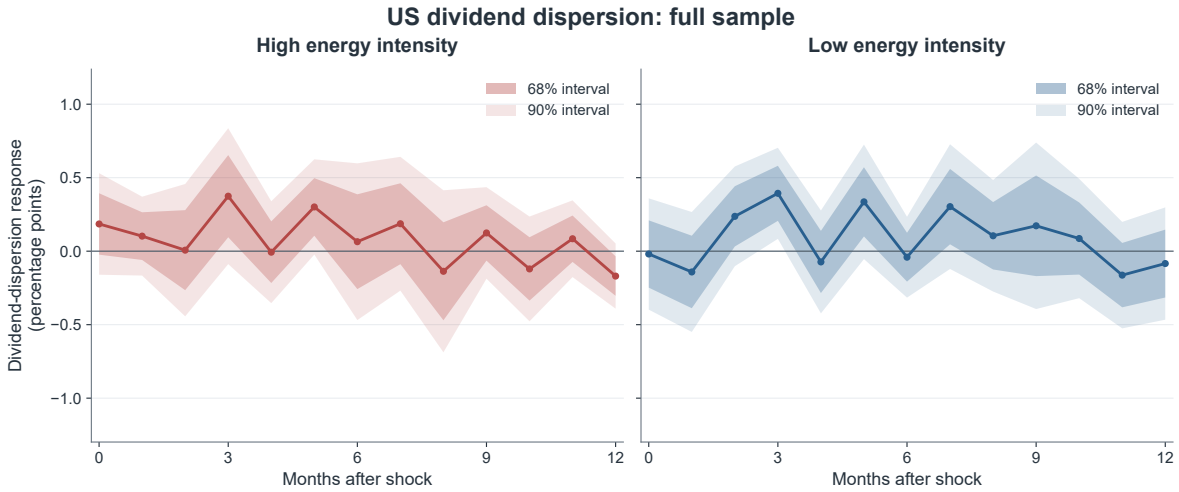



Figure 32: U.S. dividend-dispersion responses using the full sample, including 1998–2001 and 2020. Dark and light shading show pointwise 68% and 90% intervals.

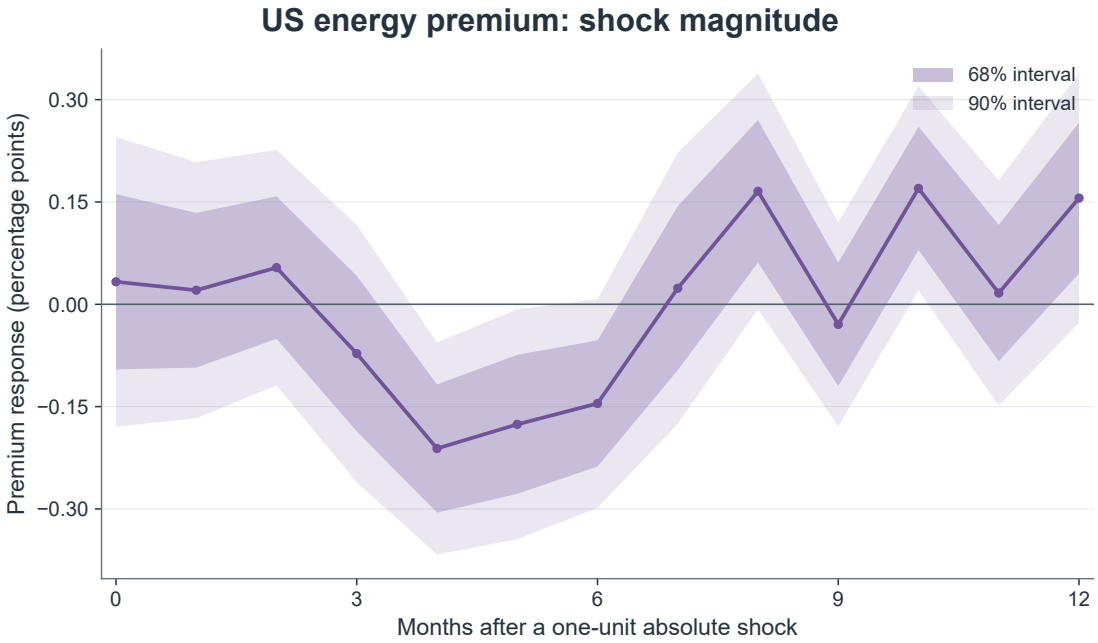



Figure 33: U.S. energy-intensity return-premium responses to a one-unit increase in absolute supply-shock magnitude. Dark and light shading show pointwise 68% and 90% intervals.

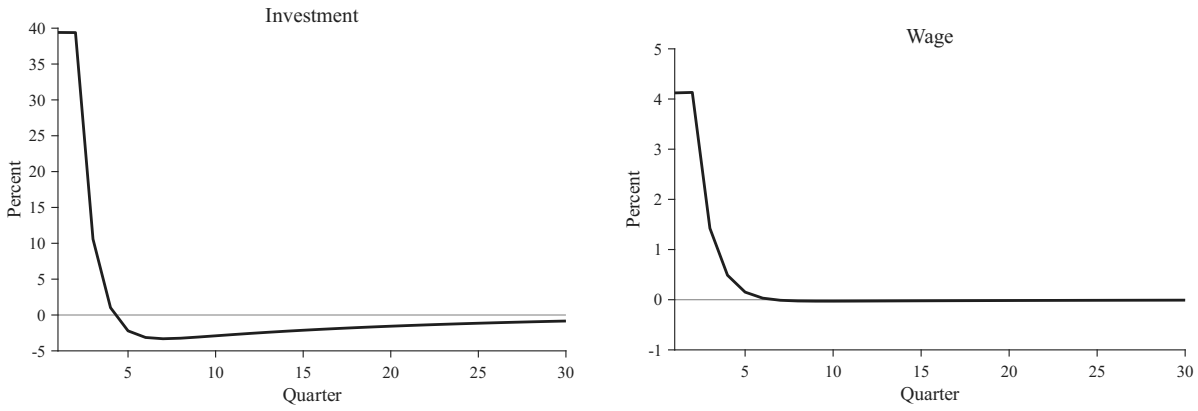
# B Additional MIT Impulse Responses

Section 7 reports a subset of the series written by the MIT plotters. The figures below use the same folders and file names as `plot_HANC_MIT_results.m` and `plot_HANC_MIT_compare.m`: `oil_shock`, `gdp`, `total_capital`, `sectoral_capital`, `investment`, `wage`, `labor`, `returns`, `liquidation_rate`, and `idio_variance`. Paths remain `faculty macro mock/images/<folder>/`.

## B.1 Remaining series for the main-text experiments

The main text shows the joint, price-only, and dispersion experiments used to isolate the interaction. The remaining series for each experiment appear here.

### B.1.1 MIT1: Decrease in average prices




(a) Investment response

(b) Wage response

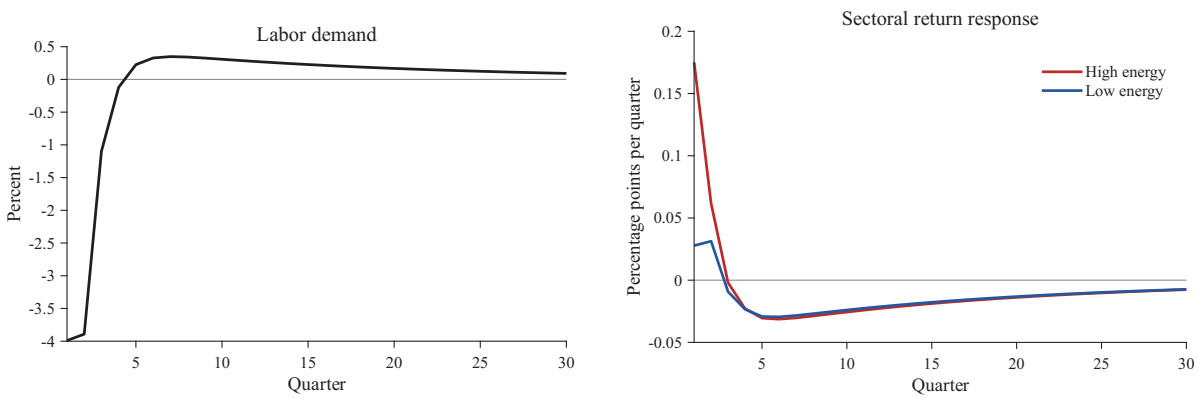
Figure 34: Investment and wage responses to a decrease in average oil price

### B.1.2 MIT2: Increase in average prices

### B.1.3 MIT4: Increase in volatility

### B.1.4 MIT1 versus MIT9: Volatility in a price decline

### B.1.5 MIT2 versus MIT10: Volatility in a price increase




(a) Labor response (b) Return response

Figure 35: Labor and return responses to a decrease in average oil price

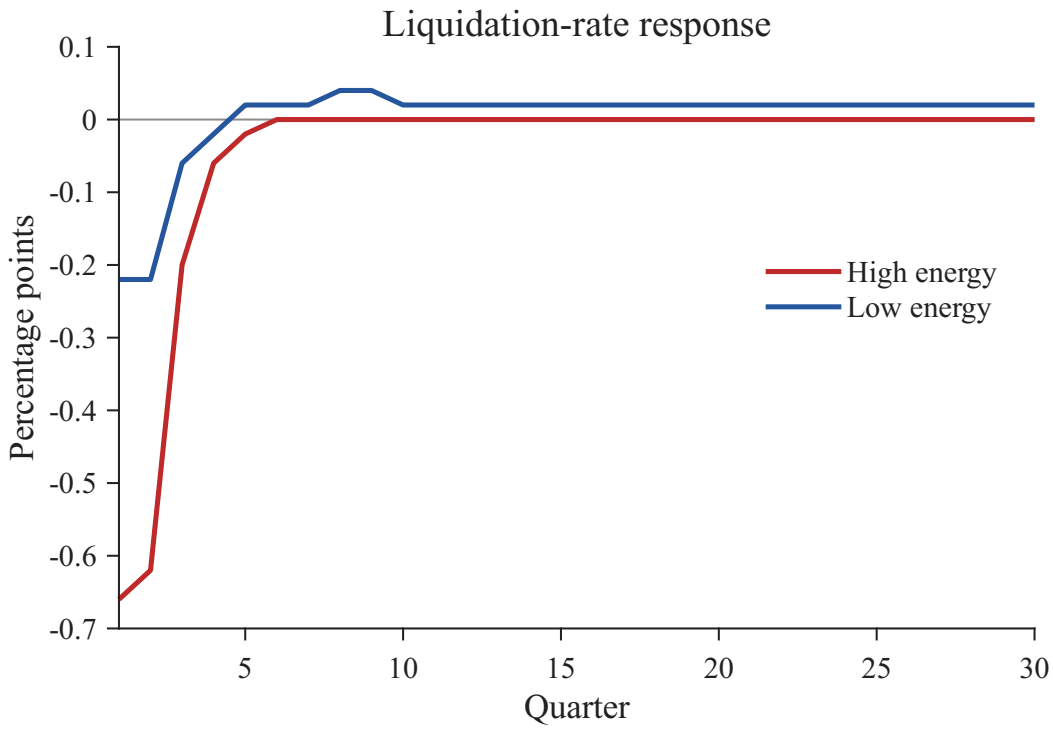
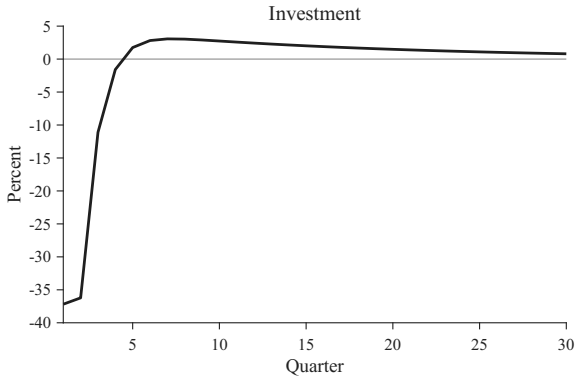
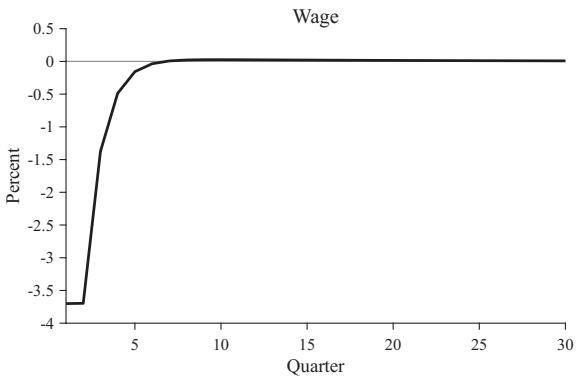



Figure 36: Liquidation-rate response to a decrease in average oil price




(a) Investment response

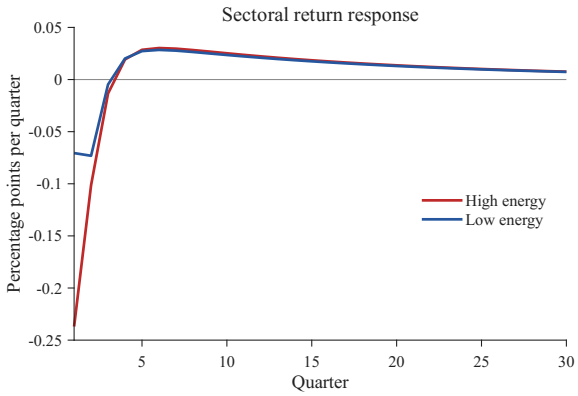



(b) Wage response

Figure 37: Investment and wage responses to an increase in average oil price




(a) Labor response




(b) Return response

Figure 38: Labor and return responses to an increase in average oil price

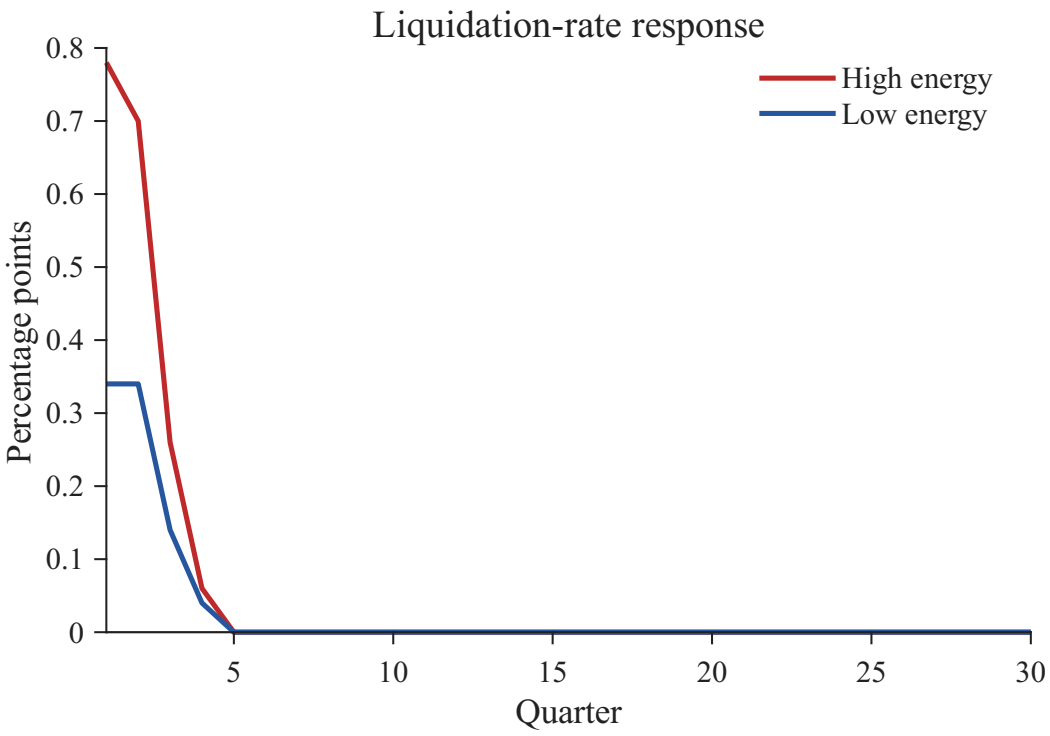
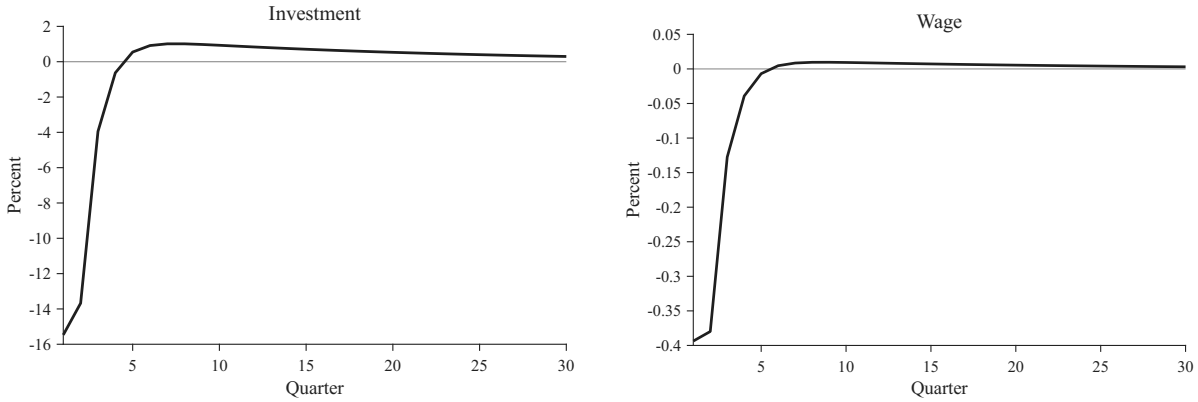


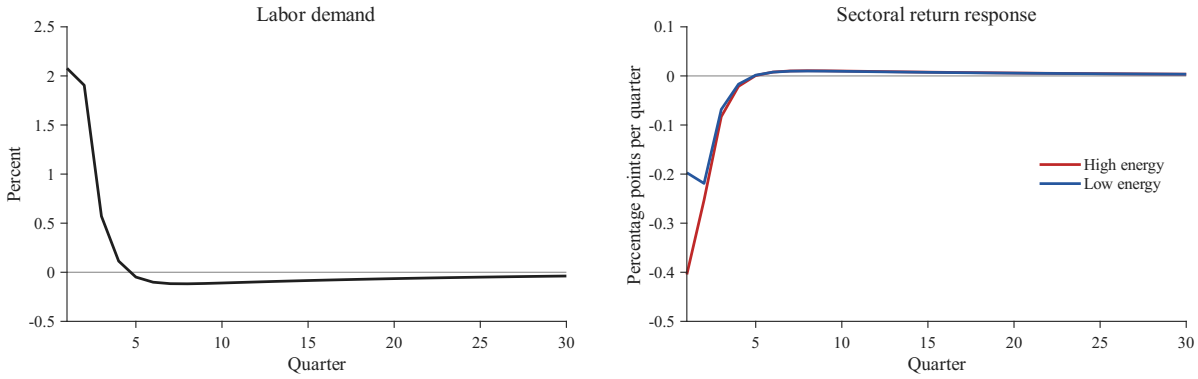

Figure 39: Liquidation-rate response to an increase in average oil price




(a) Investment response

(b) Wage response

Figure 40: Investment and wage responses to an increase in volatility




(a) Labor response (b) Return response

Figure 41: Labor and return responses to an increase in volatility

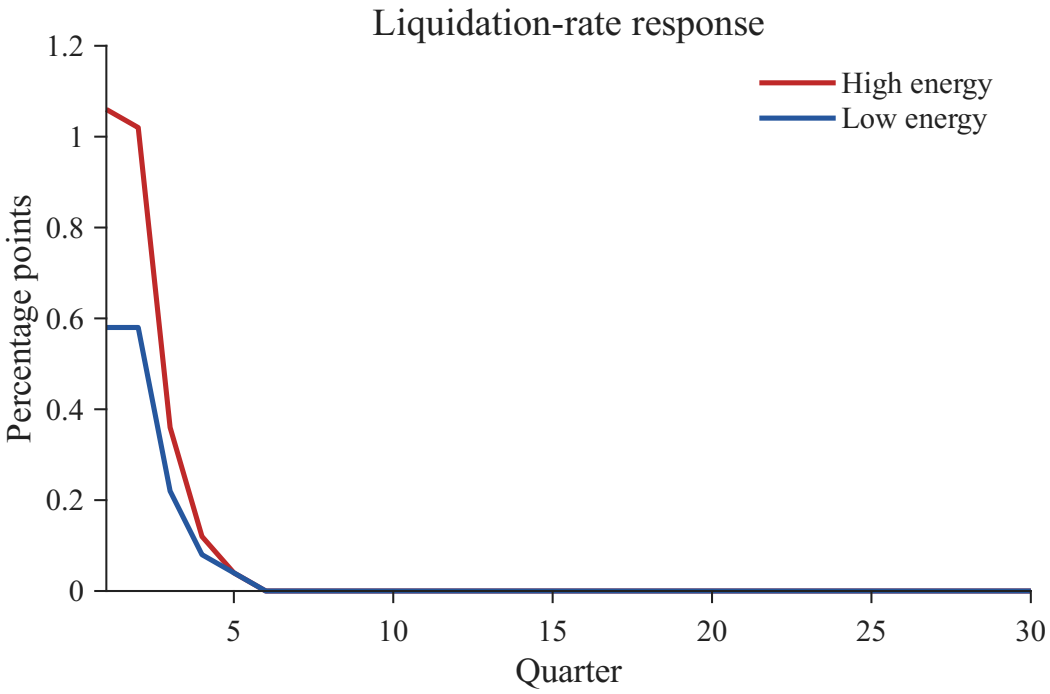
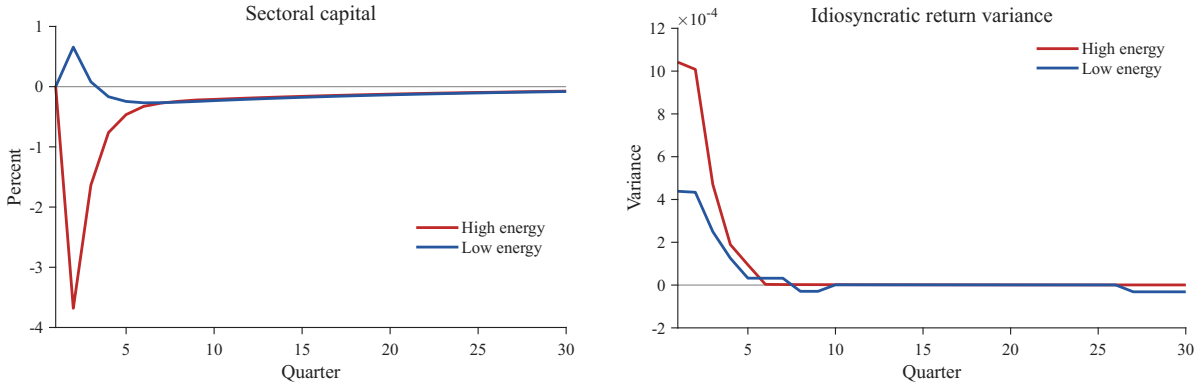
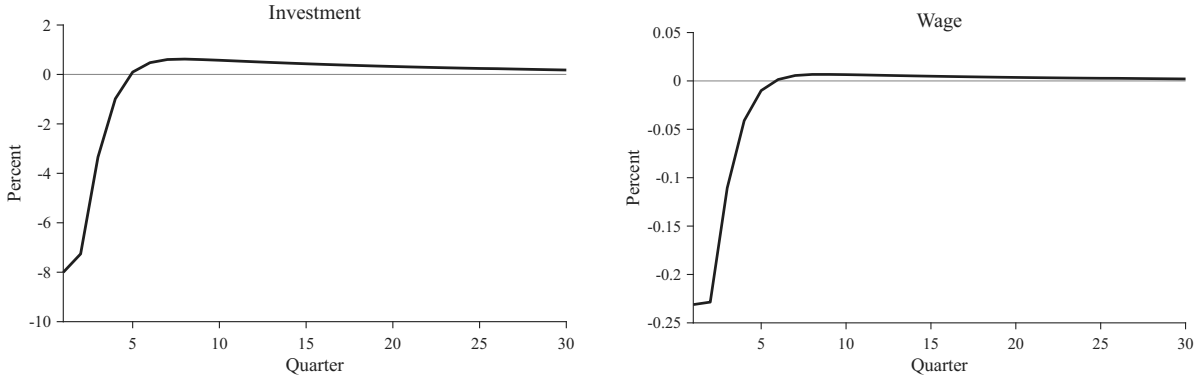



Figure 42: Liquidation-rate response to an increase in volatility



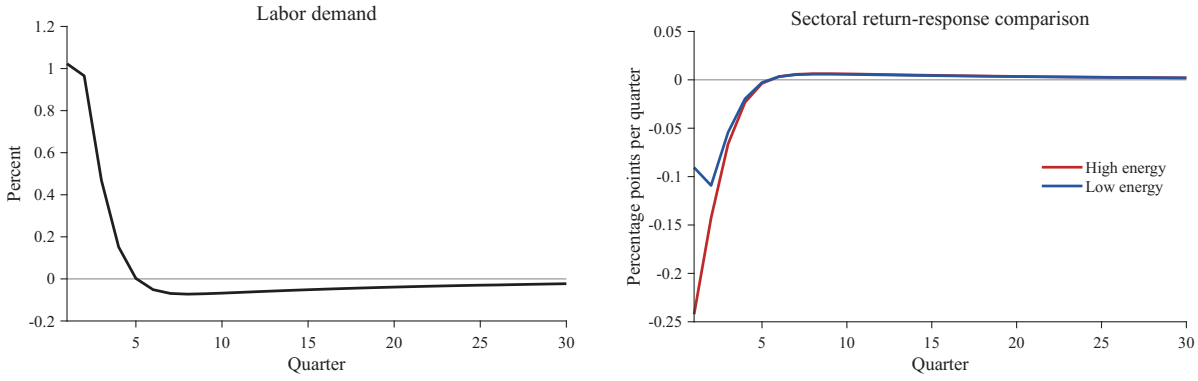

(a) Sectoral capital response (b) Variance response

Figure 43: Difference in sectoral responses to a price decline with and without a volatility shock




(a) Investment response (b) Wage response

Figure 44: Difference in investment and wage responses to a price decline with and without a volatility shock




(a) Labor response (b) Return response

Figure 45: Difference in labor and return responses to a price decline with and without a volatility shock

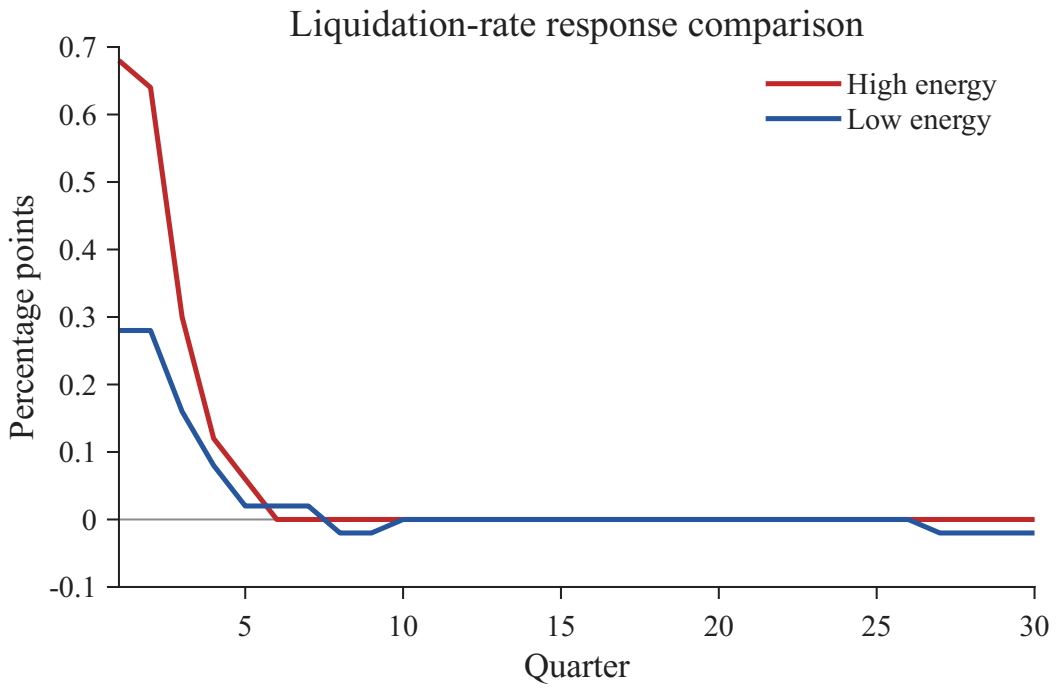
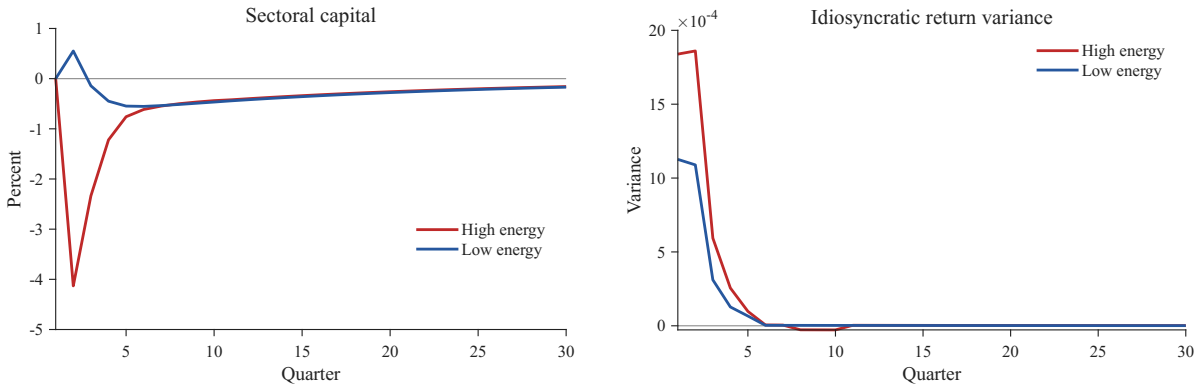


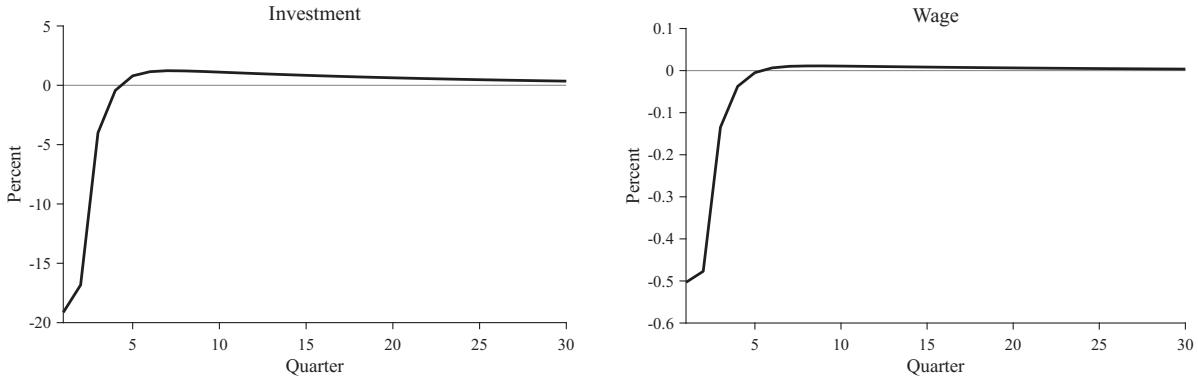

Figure 46: Difference in the liquidation-rate response to a price decline with and without a volatility shock




(a) Sectoral capital response

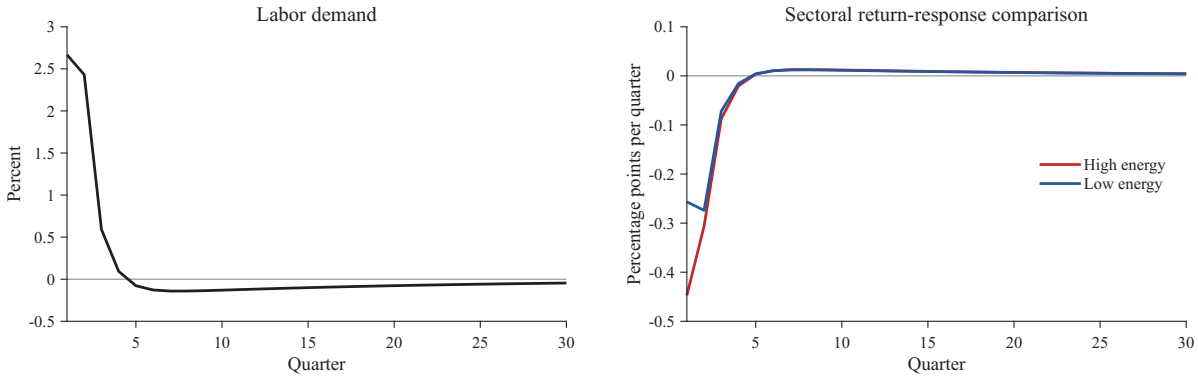
(b) Variance response

Figure 47: Difference in sectoral responses to a price increase with and without a volatility shock




(a) Investment response (b) Wage response

Figure 48: Difference in investment and wage responses to a price increase with and without a volatility shock




(a) Labor response (b) Return response

Figure 49: Difference in labor and return responses to a price increase with and without a volatility shock

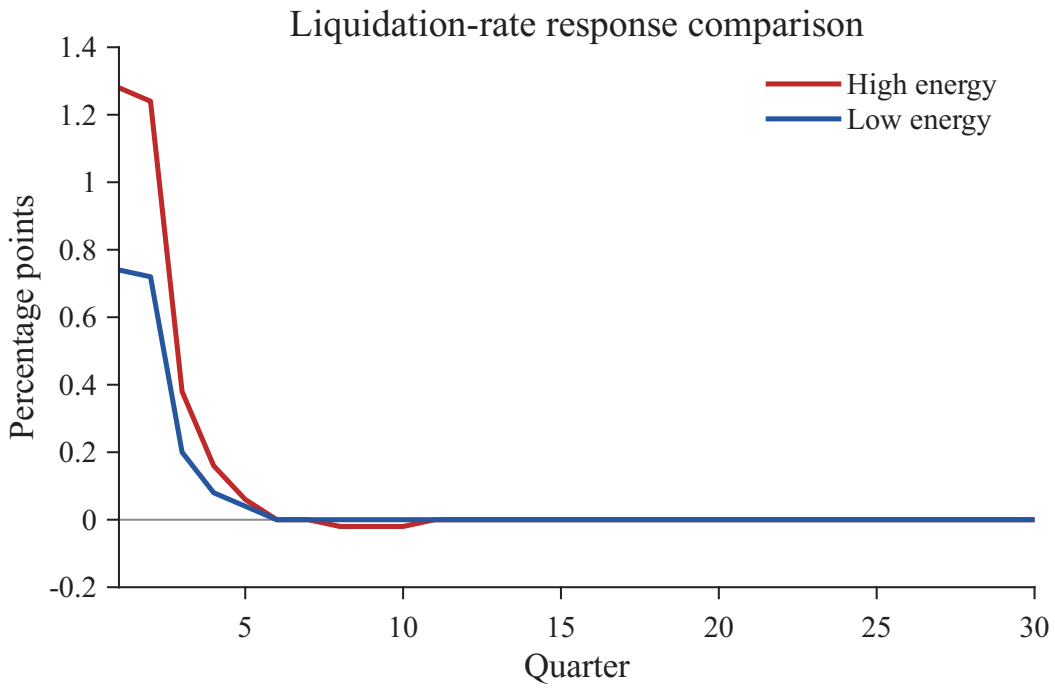



Figure 50: Difference in the liquidation-rate response to a price increase with and without a volatility shock

## B.2 Additional MIT scenarios

The remaining MIT files are reported in full, using the scenario names and numbers from the experiment list.

## B.3 MIT3: Large variance shock

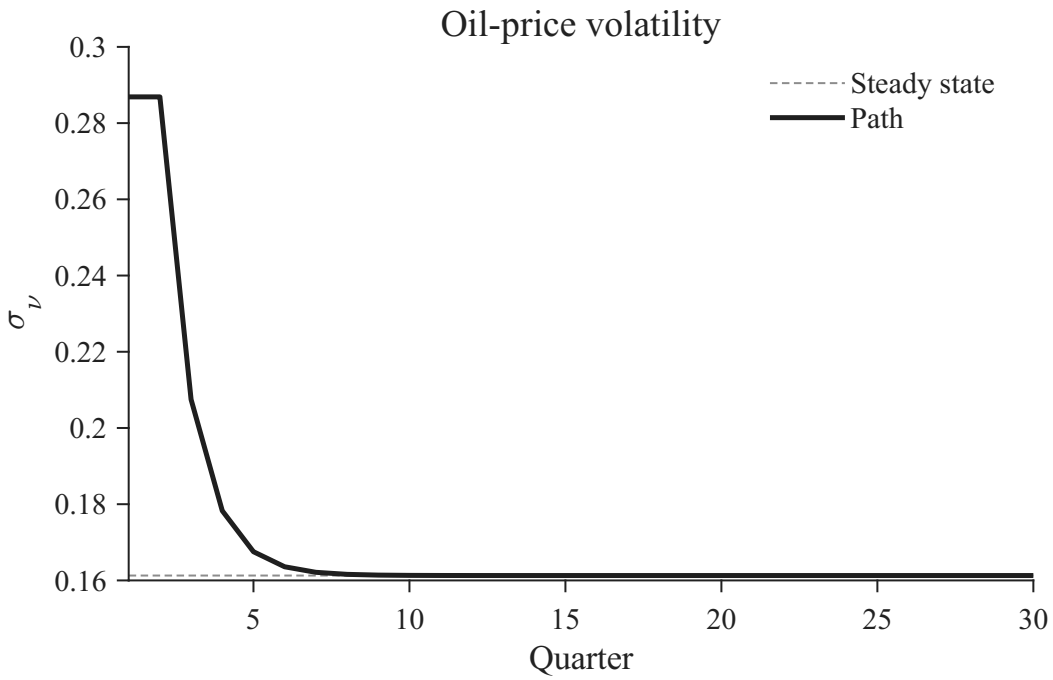
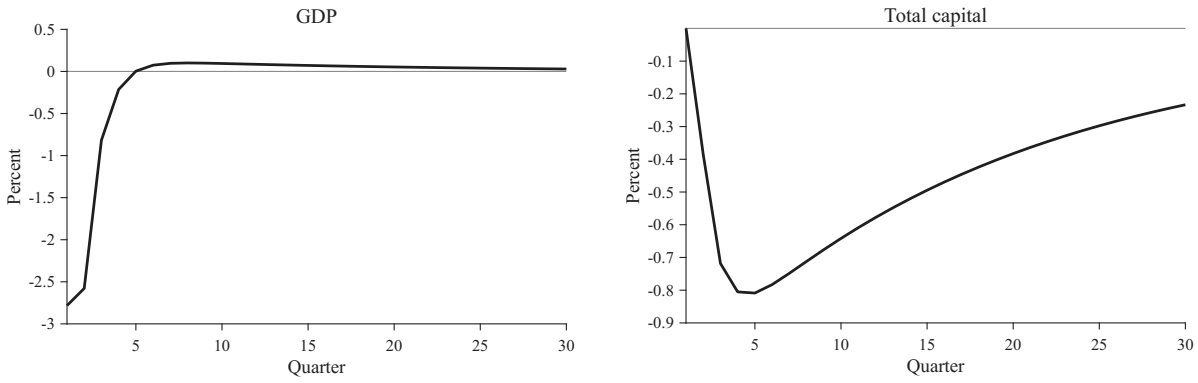


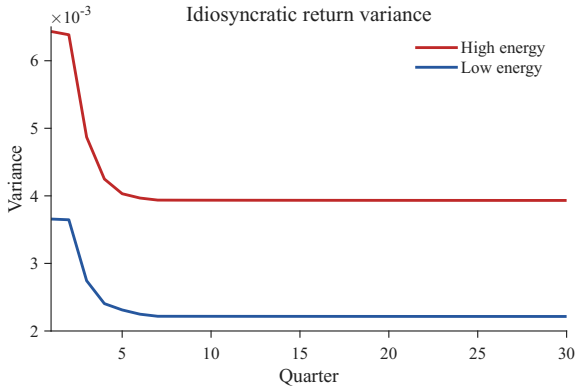

Figure 51: Shock path



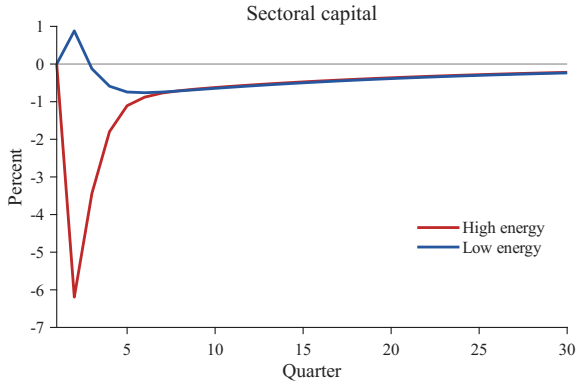

(a) GDP response

(b) Capital response

Figure 52: Aggregate responses

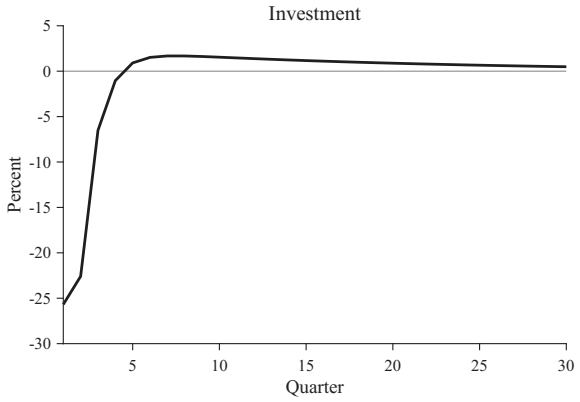



(a) Variance response

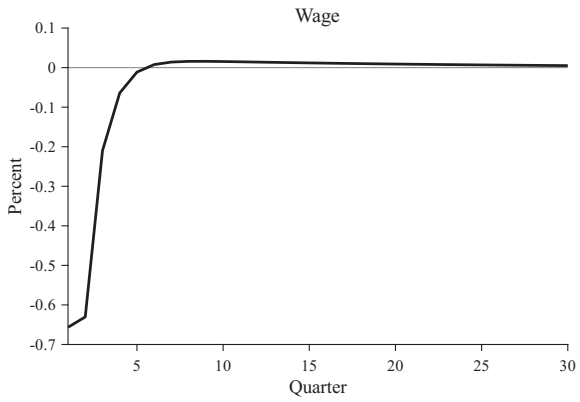



(b) Sectoral capital response

Figure 53: Sectoral responses

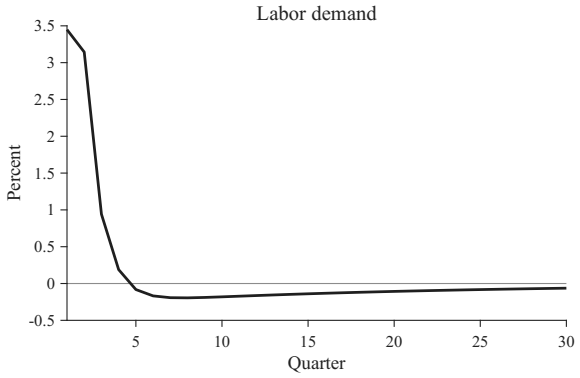



(a) Investment response

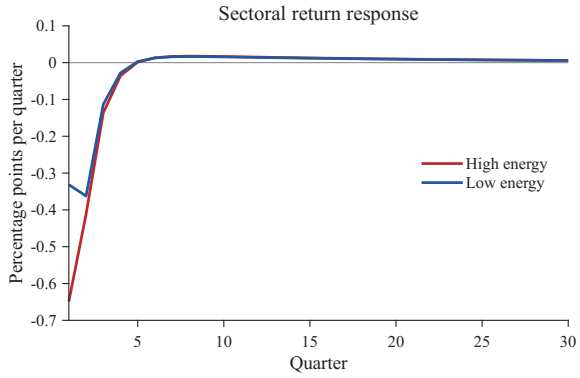



(b) Wage response

Figure 54: Investment and wage




(a) Labor response




(b) Return response

Figure 55: Labor and returns

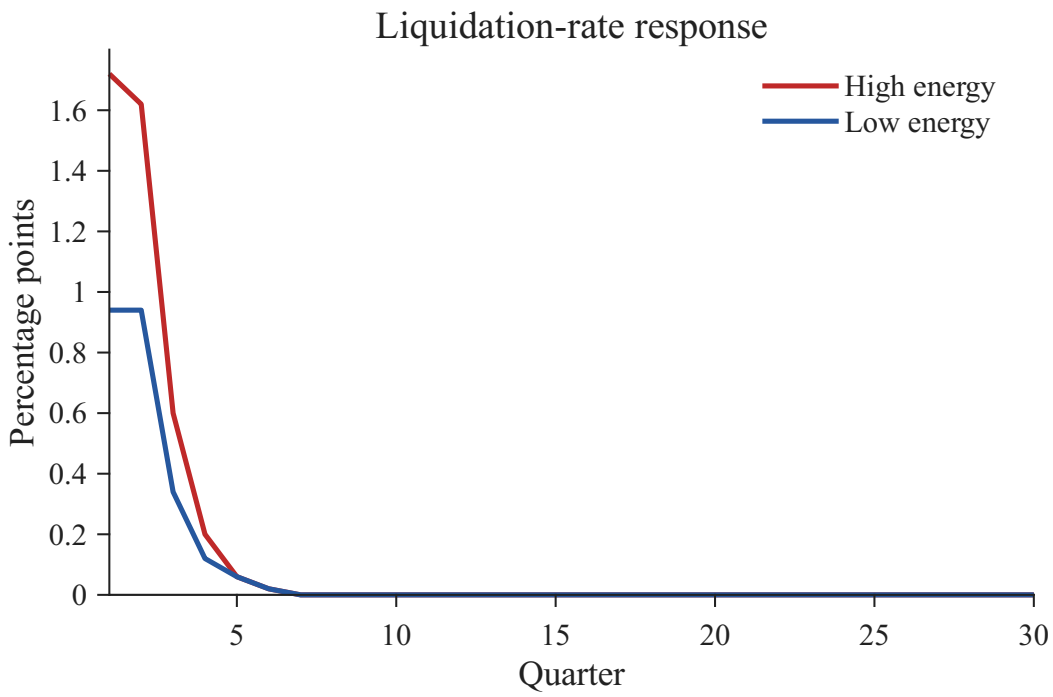



Figure 56: Liquidation-rate response

## B.4 MIT5: Variance shock with medium duration

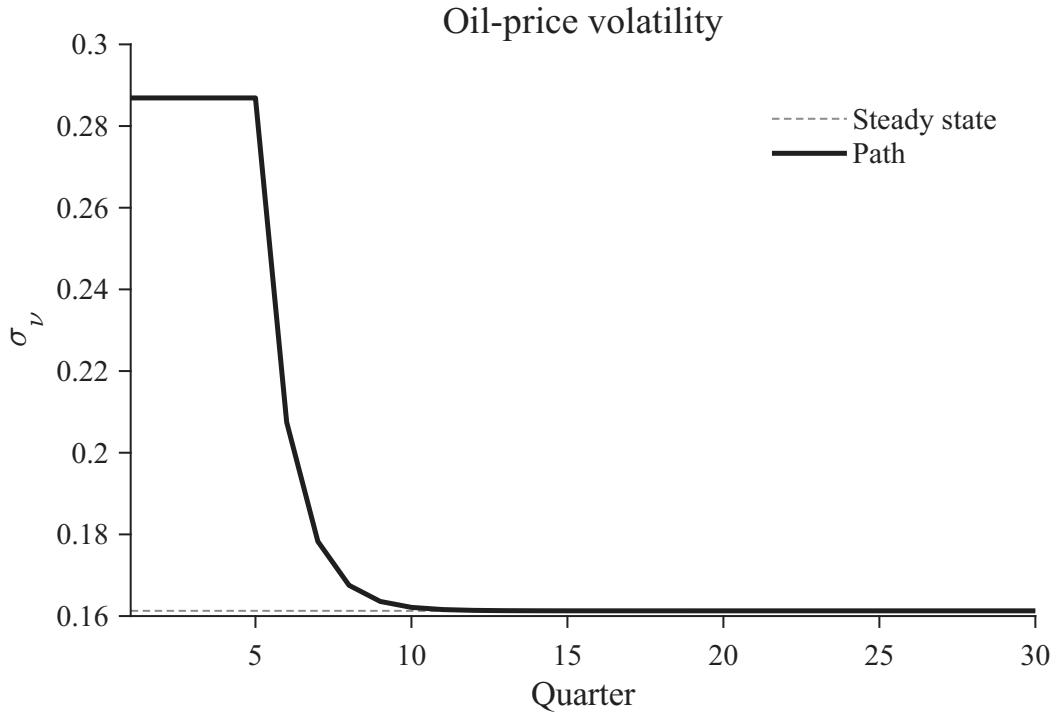
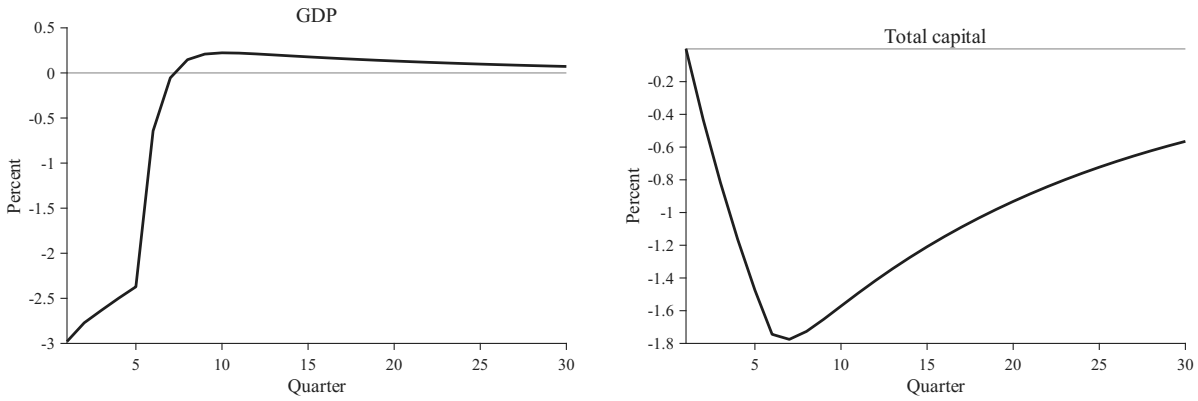


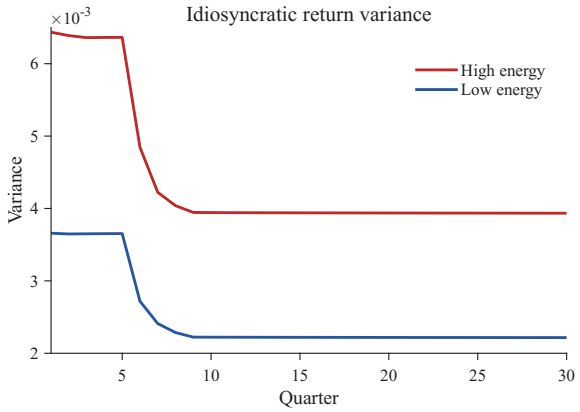

Figure 57: Shock path



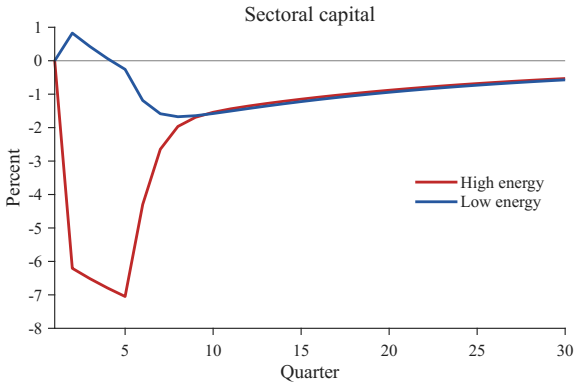

(a) GDP response

(b) Capital response

Figure 58: Aggregate responses

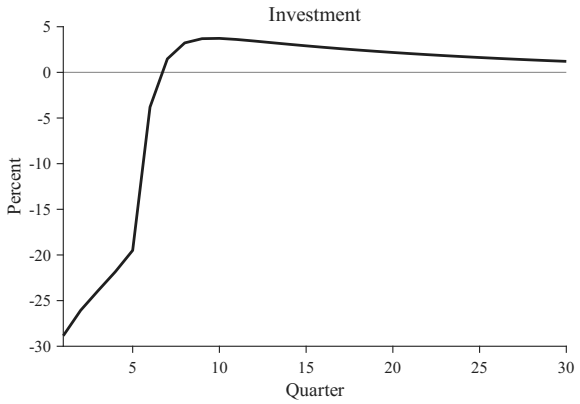



(a) Variance response

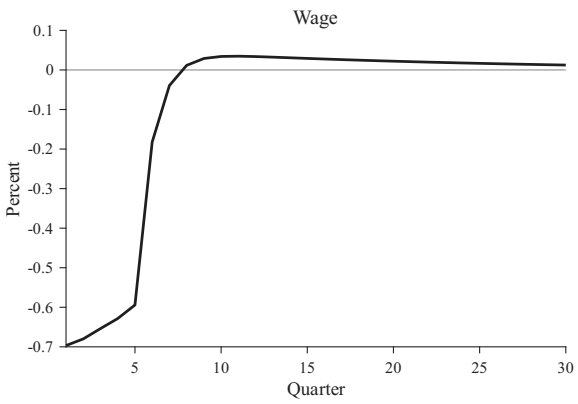



(b) Sectoral capital response

Figure 59: Sectoral responses

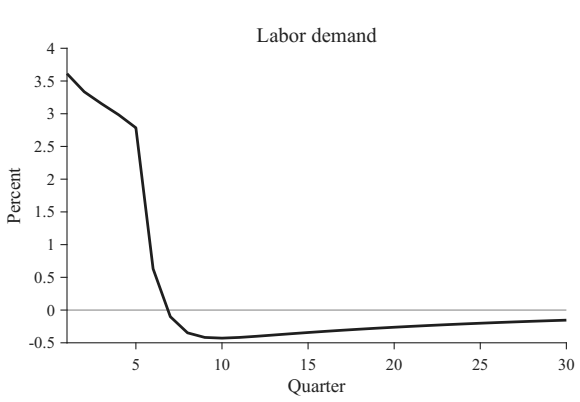



(a) Investment response

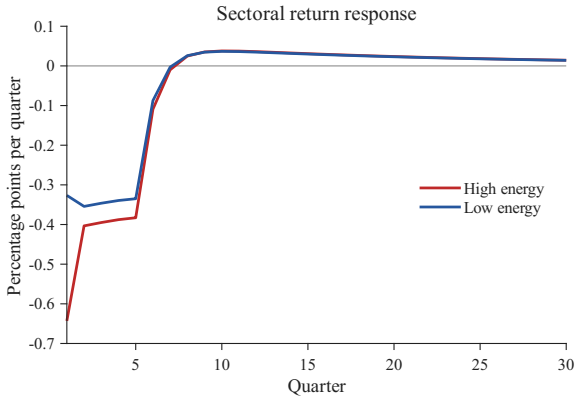



(b) Wage response

Figure 60: Investment and wage




(a) Labor response




(b) Return response

Figure 61: Labor and returns

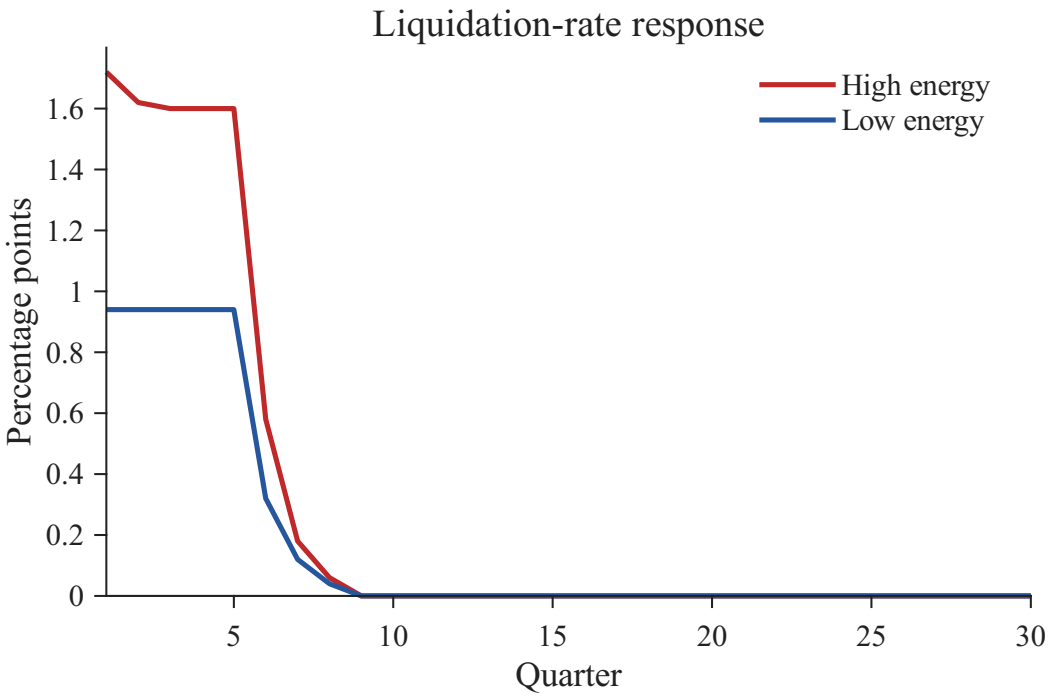



Figure 62: Liquidation-rate response

## B.5 MIT6: Variance shock with high duration

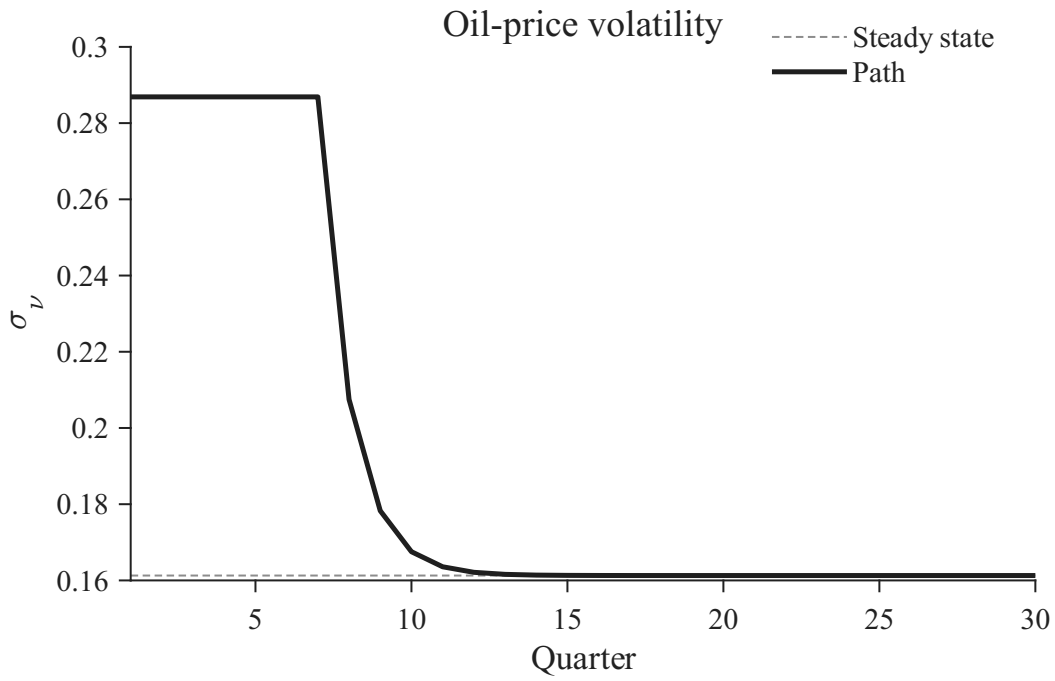
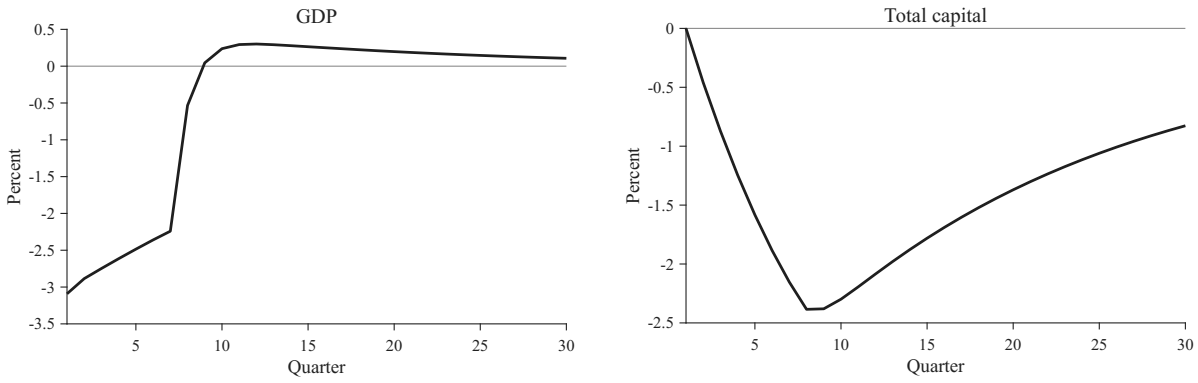


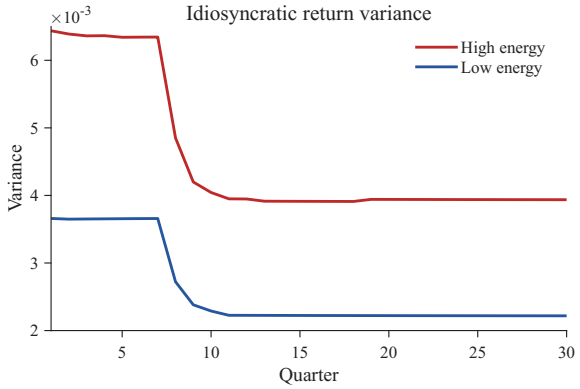

Figure 63: Shock path



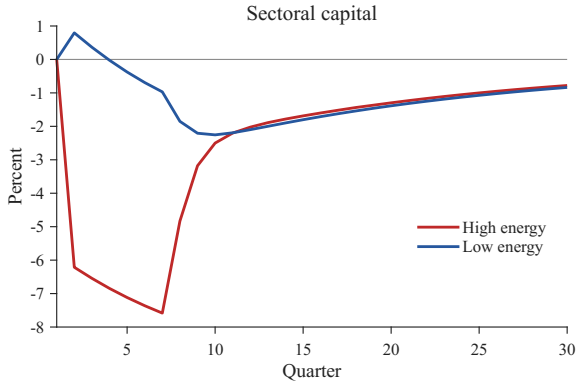

(a) GDP response

(b) Capital response

Figure 64: Aggregate responses

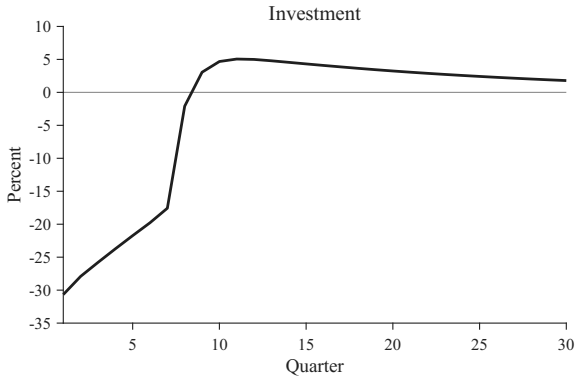



(a) Variance response




(b) Sectoral capital response

Figure 65: Sectoral responses

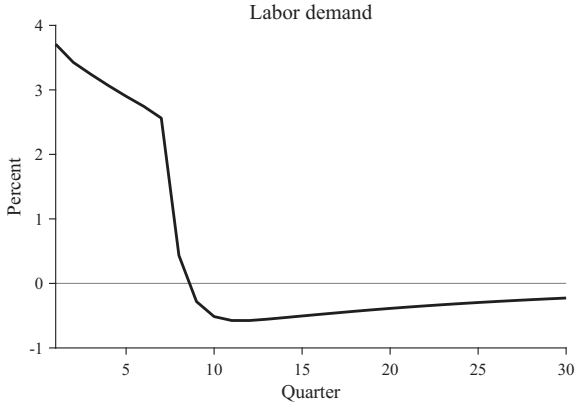



(a) Investment response

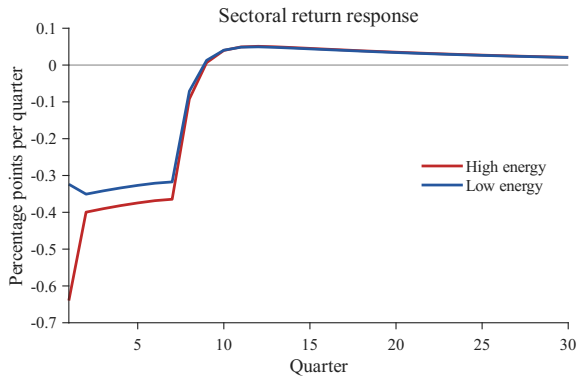



(b) Wage response

Figure 66: Investment and wage




(a) Labor response




(b) Return response

Figure 67: Labor and returns

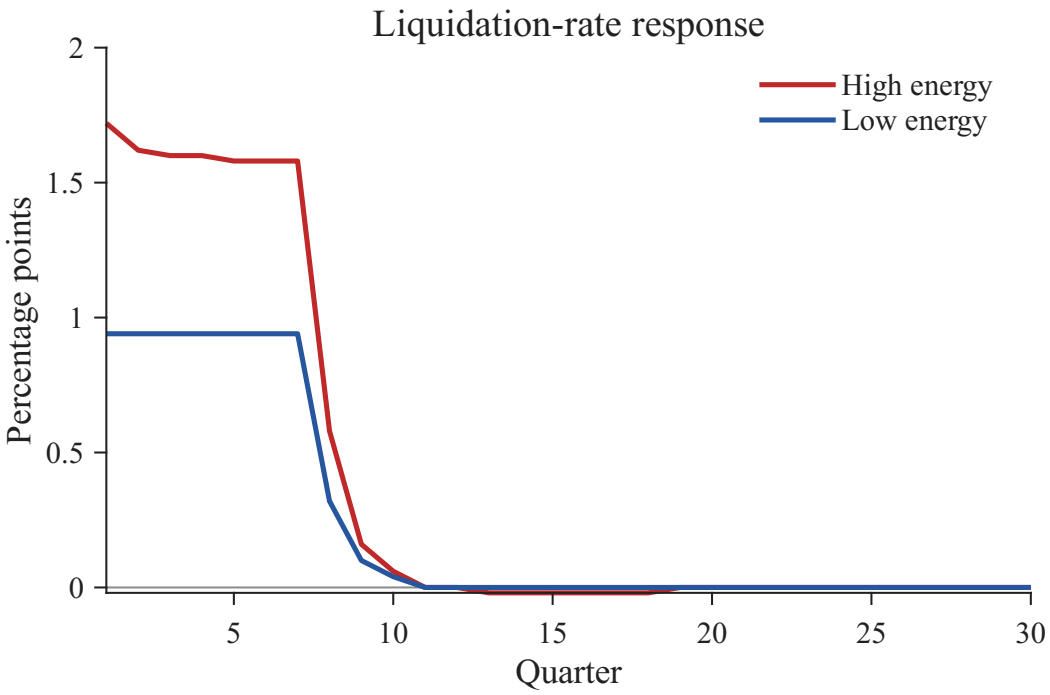



Figure 68: Liquidation-rate response

## B.6 MIT7: Positive price shock with medium duration

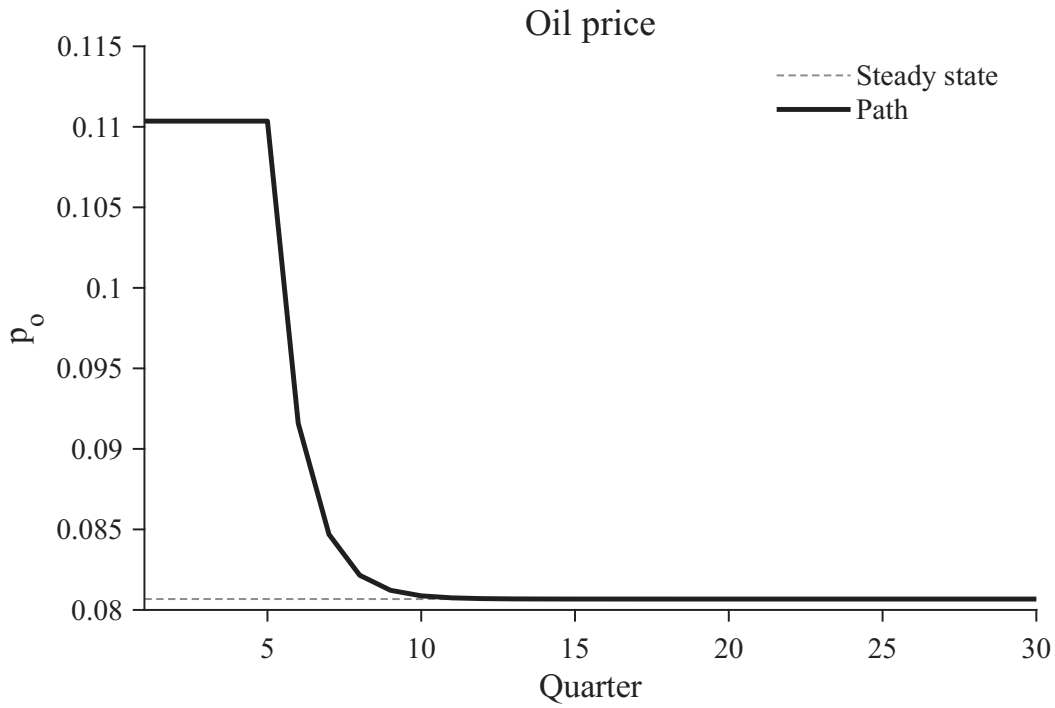
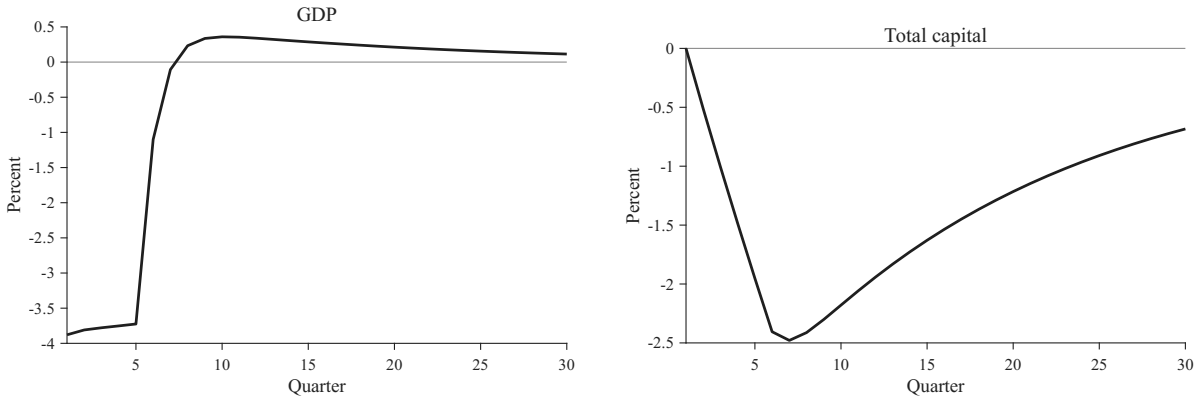


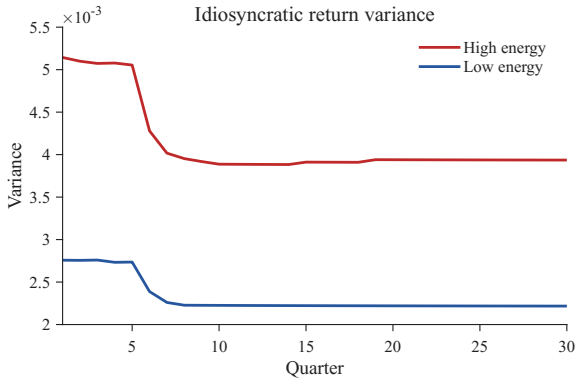

Figure 69: Shock path



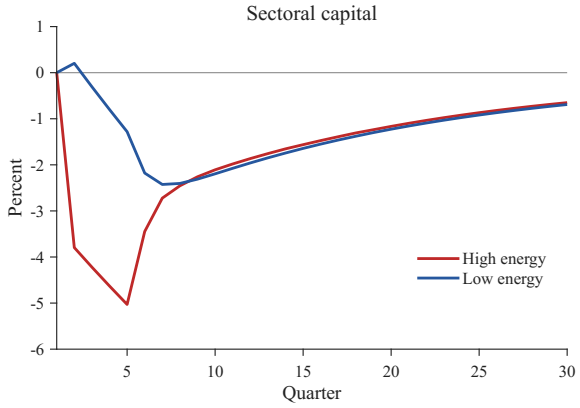

(a) GDP response

(b) Capital response

Figure 70: Aggregate responses

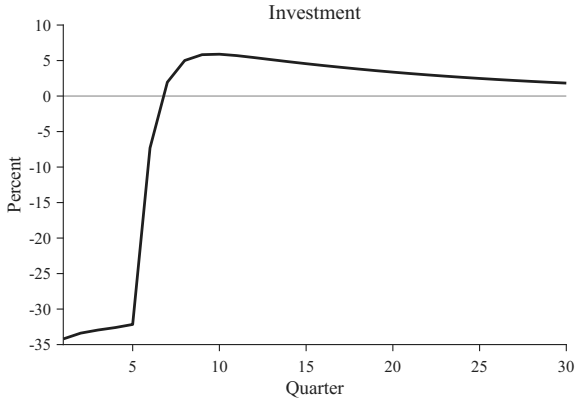



(a) Variance response

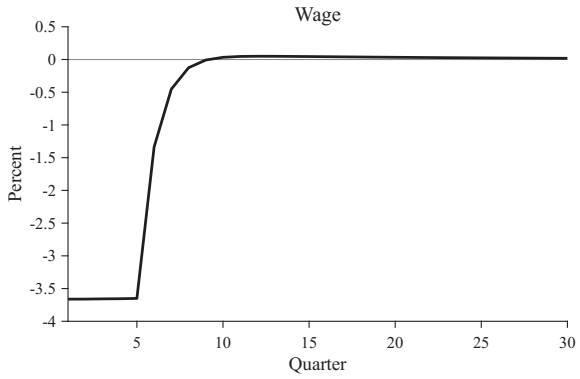



(b) Sectoral capital response

Figure 71: Sectoral responses

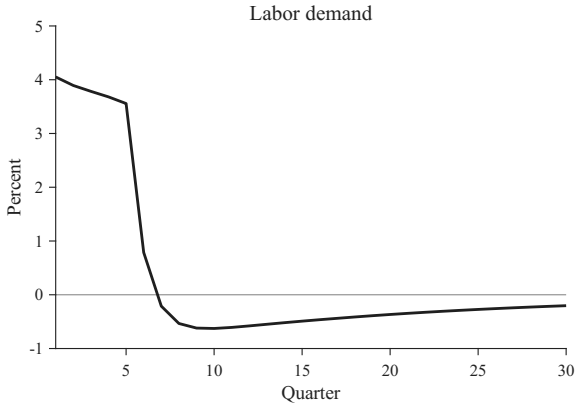



(a) Investment response

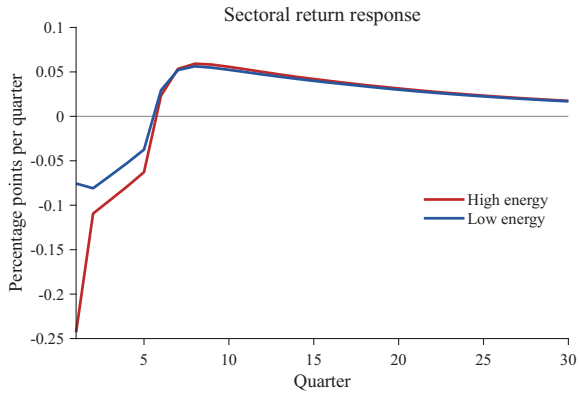



(b) Wage response

Figure 72: Investment and wage




(a) Labor response




(b) Return response

Figure 73: Labor and returns

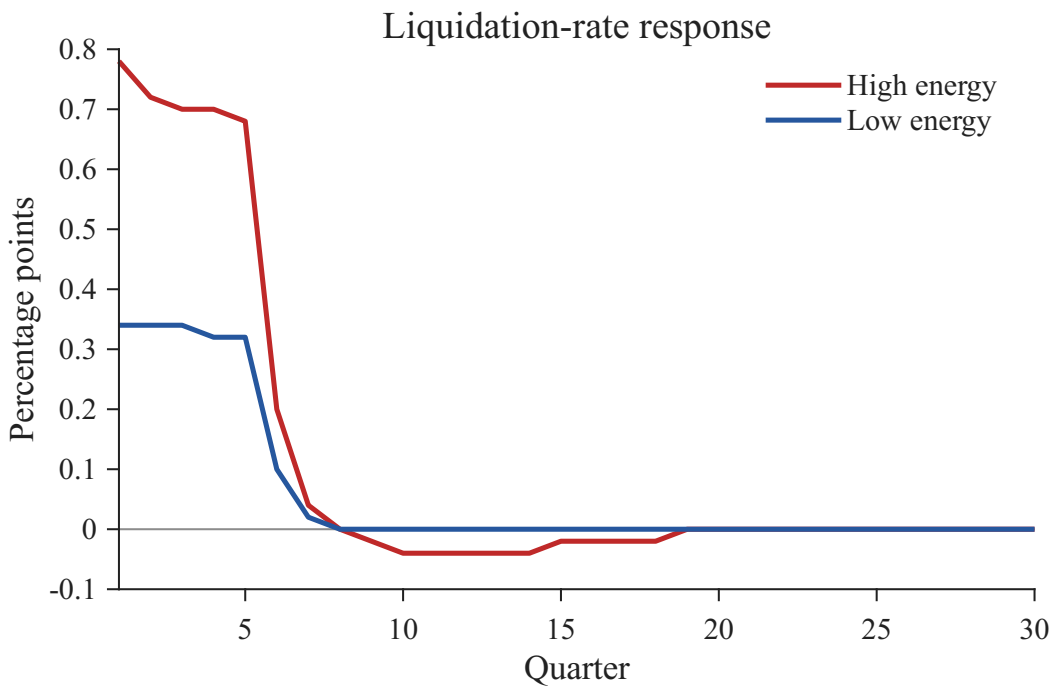



Figure 74: Liquidation-rate response

## B.7 MIT8: Large positive price shock

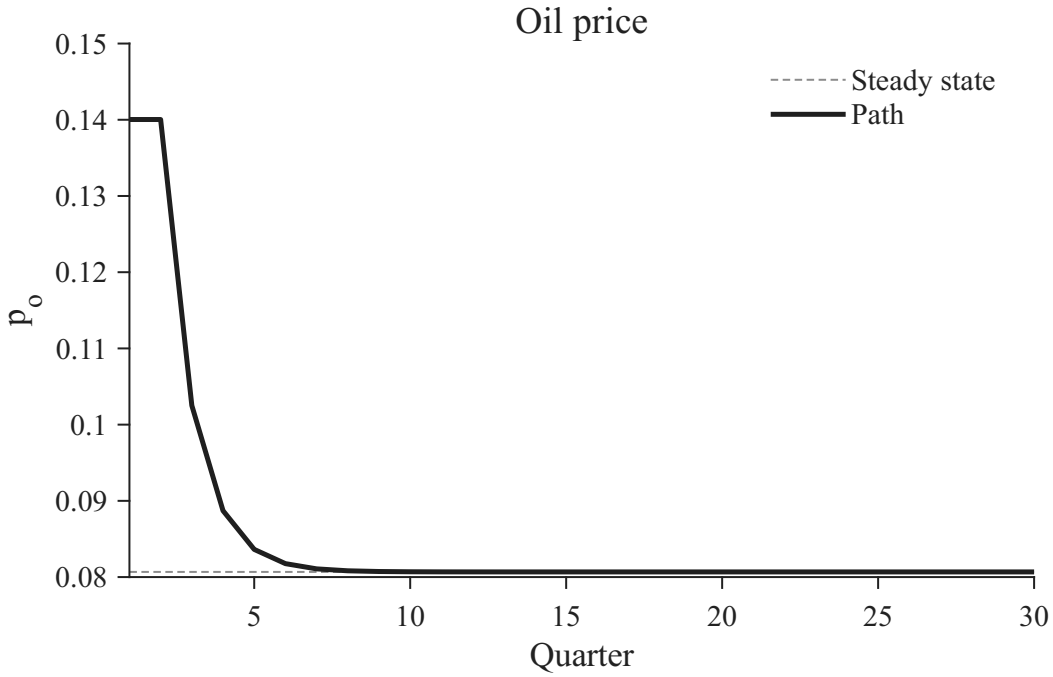
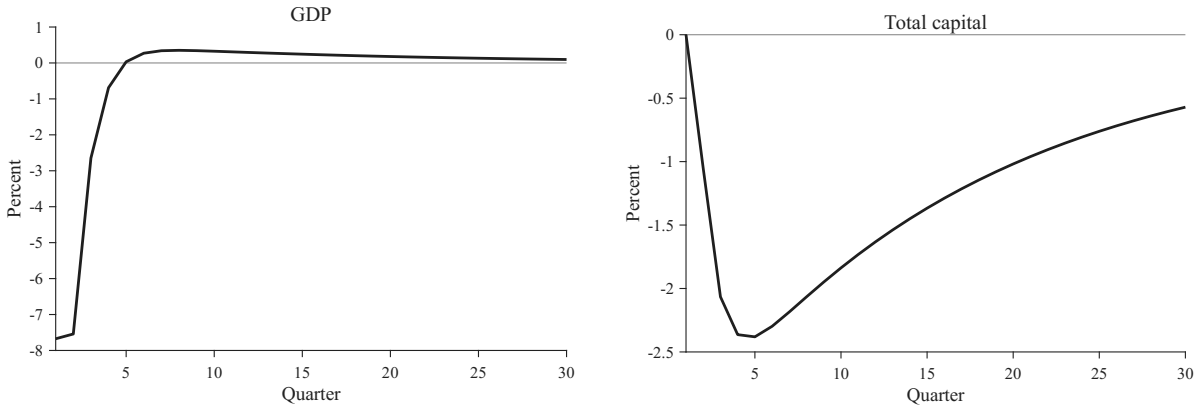


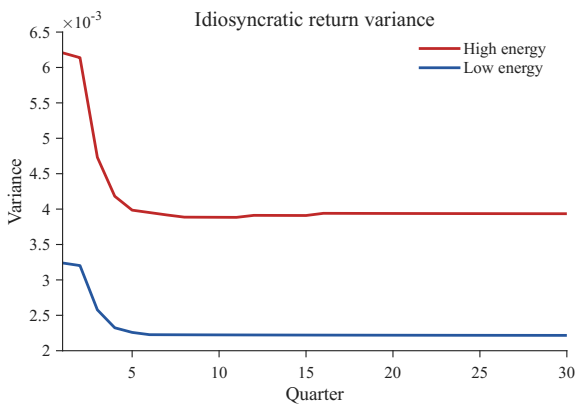

Figure 75: Shock path



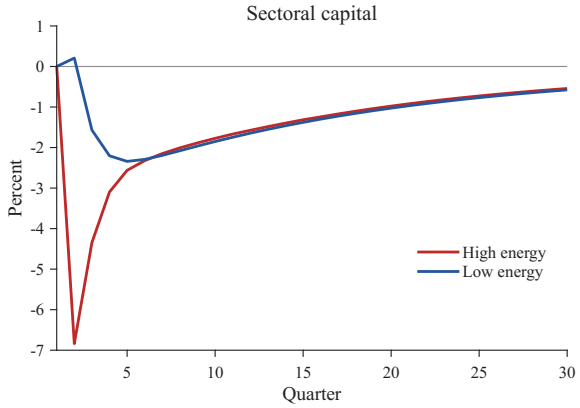

(a) GDP response

(b) Capital response

Figure 76: Aggregate responses

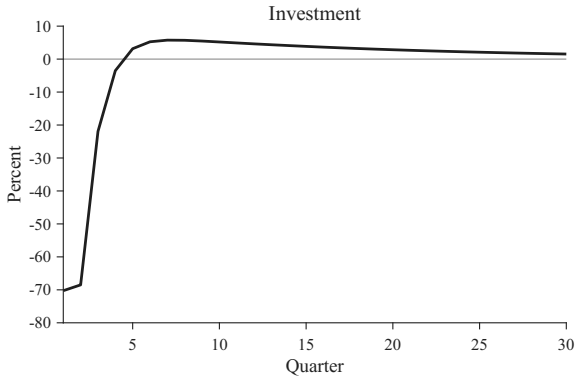



(a) Variance response

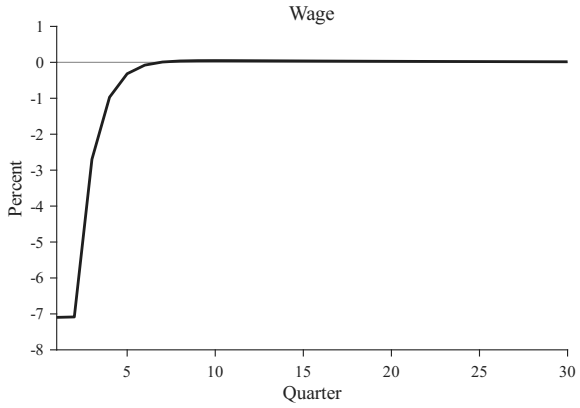



(b) Sectoral capital response

Figure 77: Sectoral responses




(a) Investment response

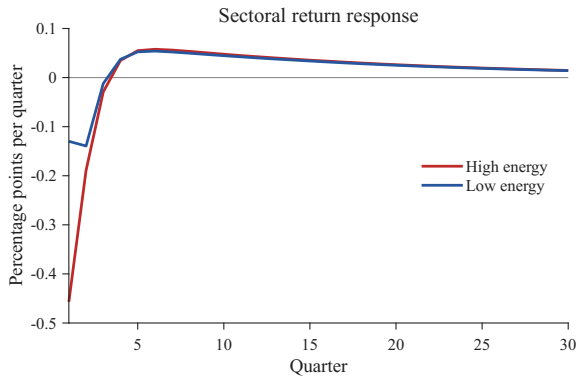



(b) Wage response

Figure 78: Investment and wage




(a) Labor response




(b) Return response

Figure 79: Labor and returns

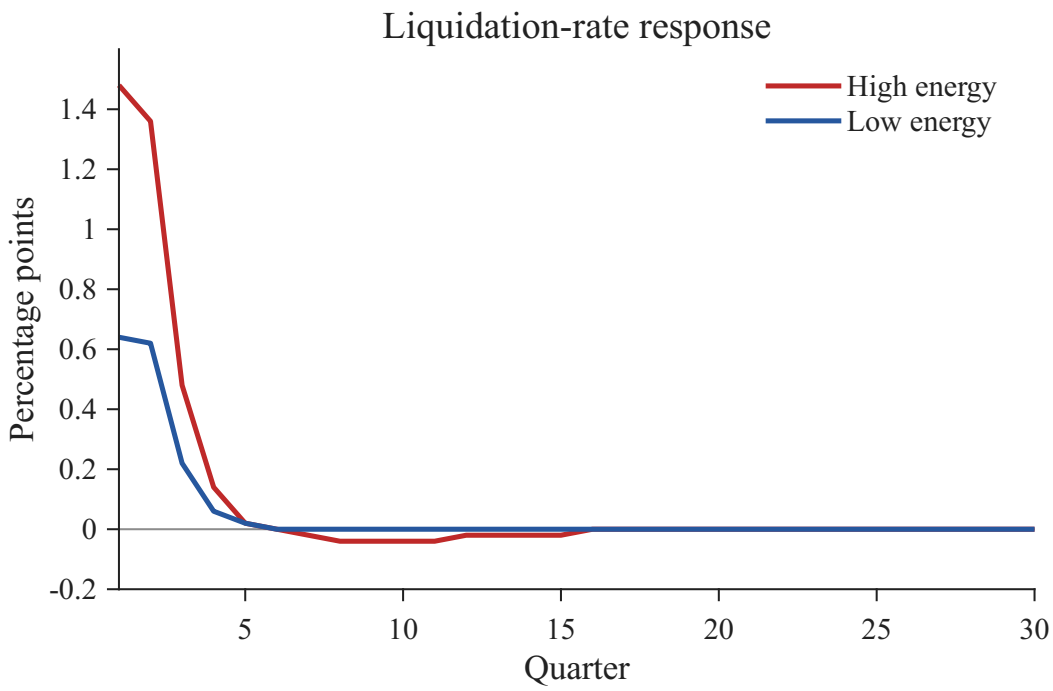



Figure 80: Liquidation-rate response

## B.8 MIT9: Negative price shock with volatility

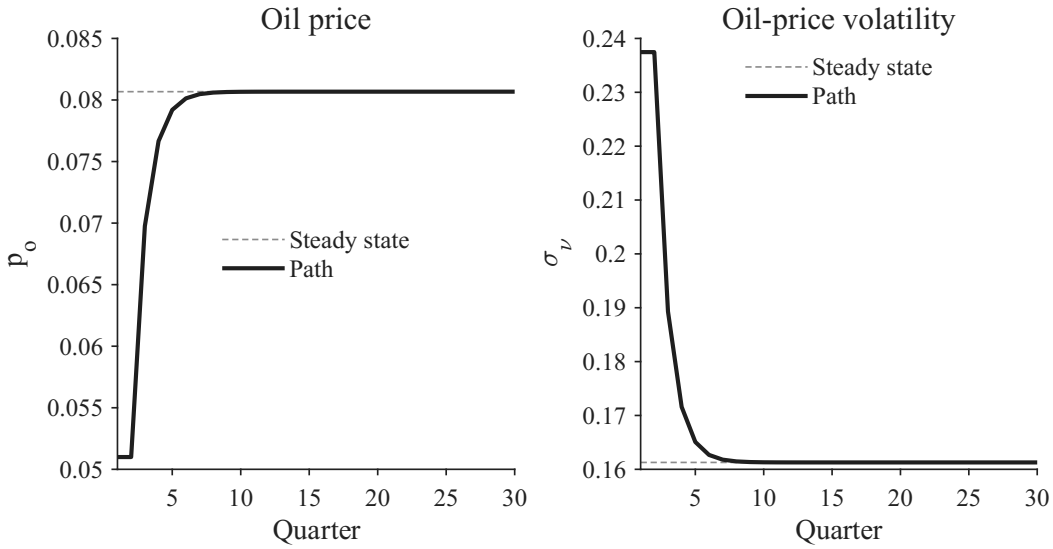
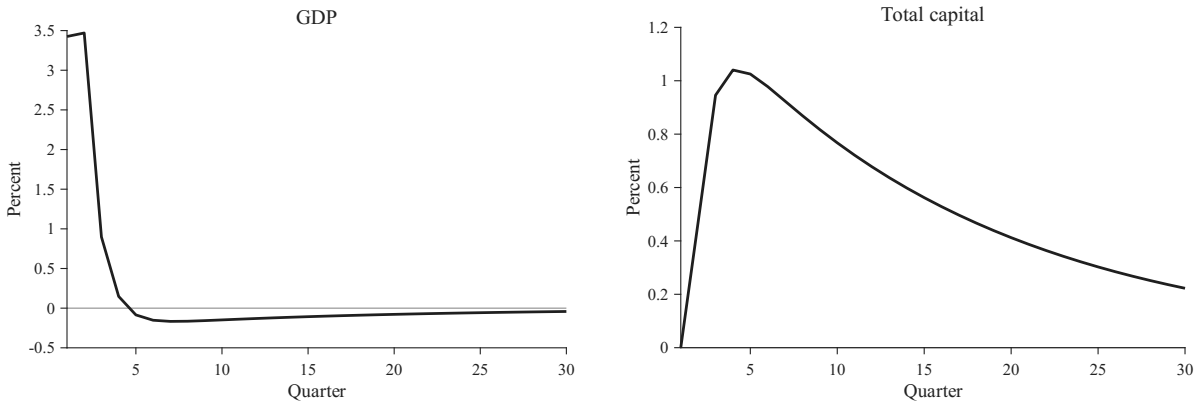


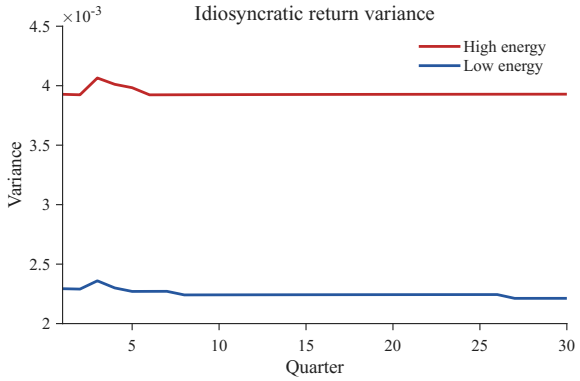

Figure 81: Shock path



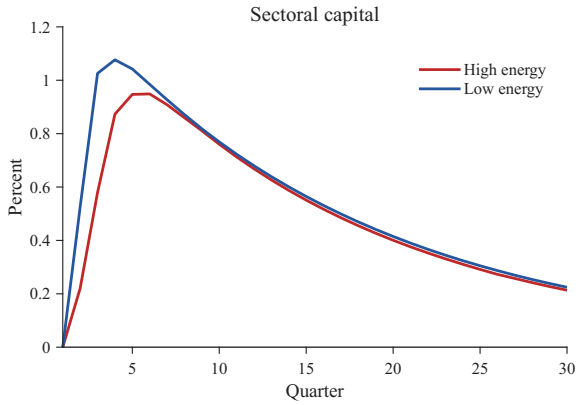

(a) GDP response

(b) Capital response

Figure 82: Aggregate responses

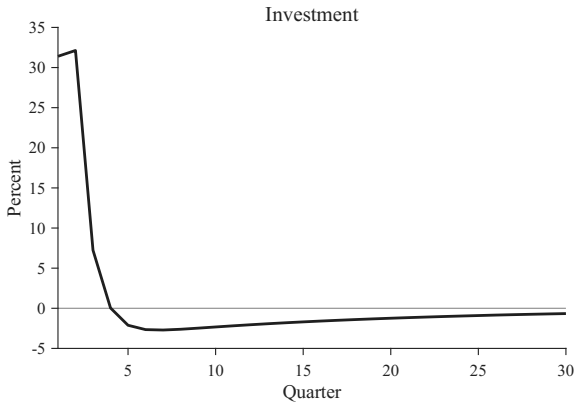



(a) Variance response




(b) Sectoral capital response

Figure 83: Sectoral responses

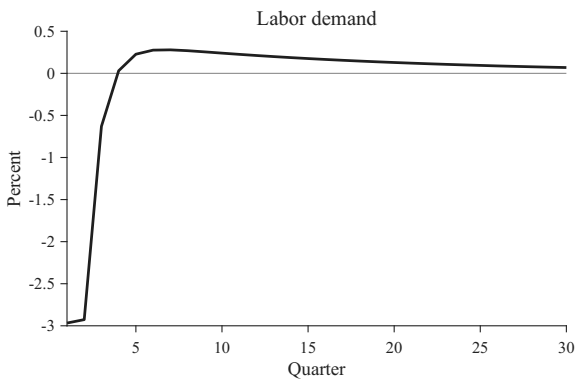



(a) Investment response

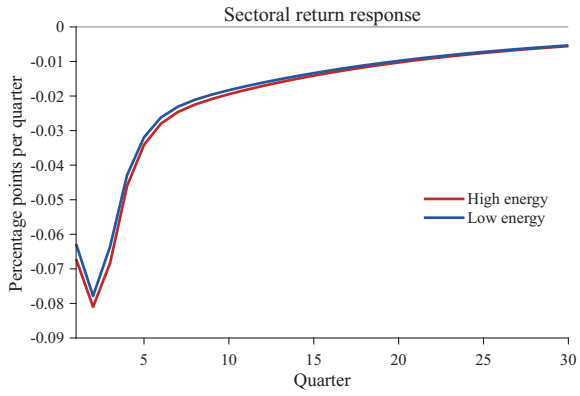



(b) Wage response

Figure 84: Investment and wage




(a) Labor response




(b) Return response

Figure 85: Labor and returns

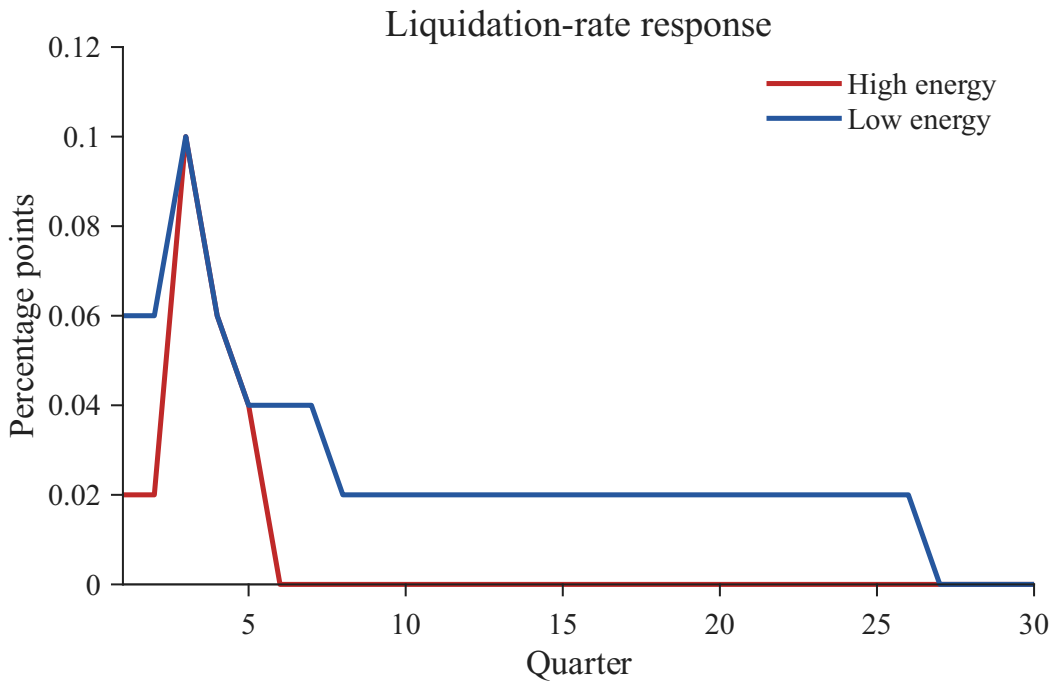



Figure 86: Liquidation-rate response

## B.9 MIT10: Positive price shock with volatility

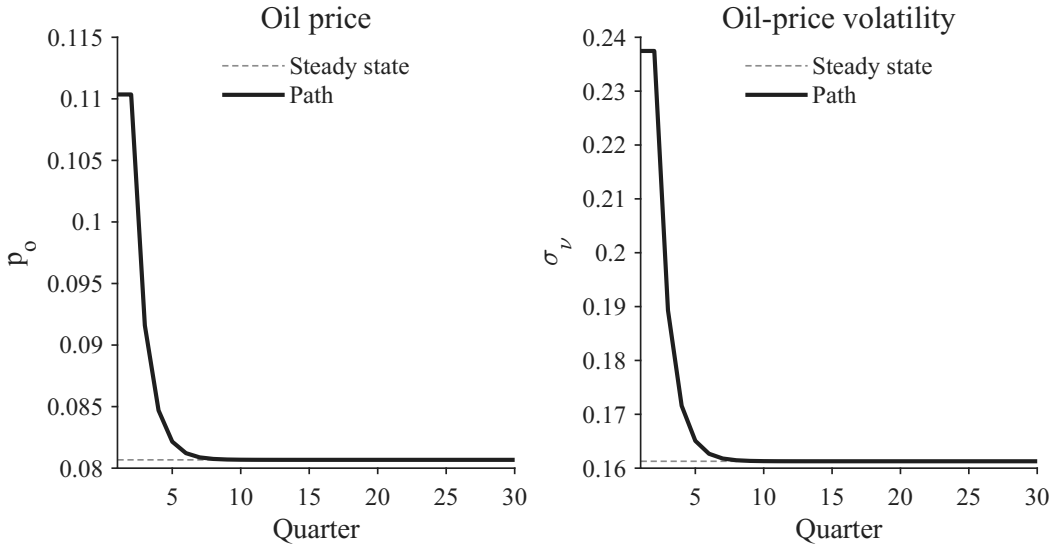
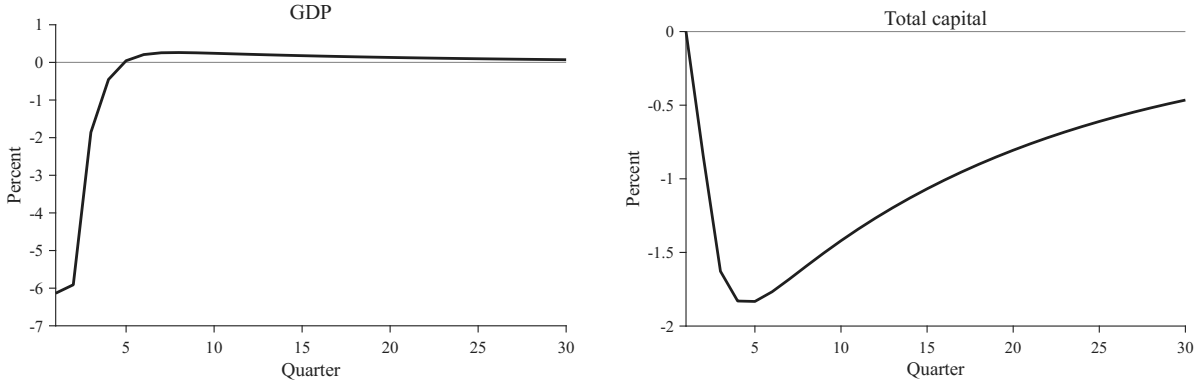


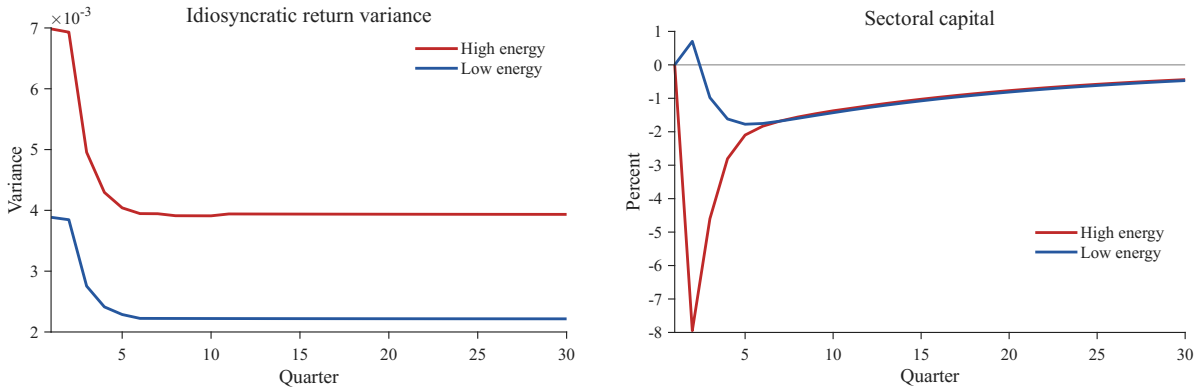

Figure 87: Shock path




(a) GDP response

(b) Capital response

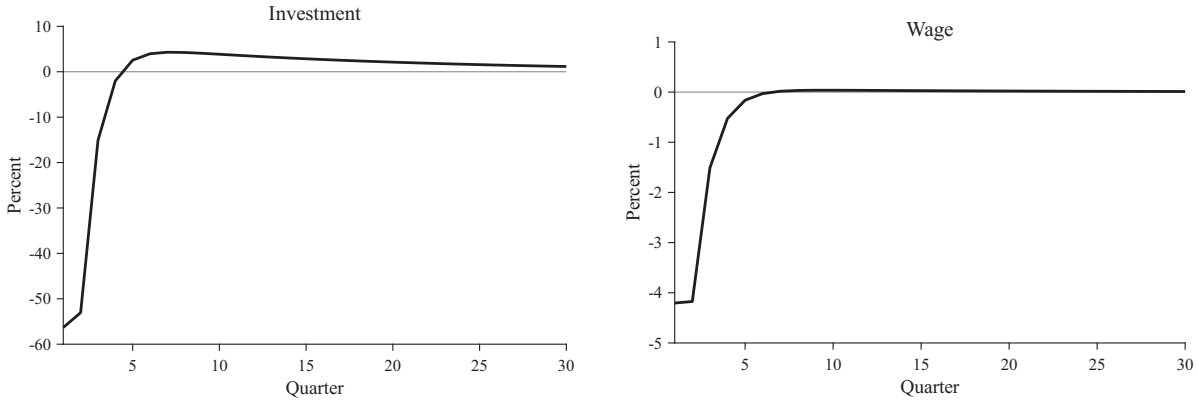
Figure 88: Aggregate responses




(a) Variance response

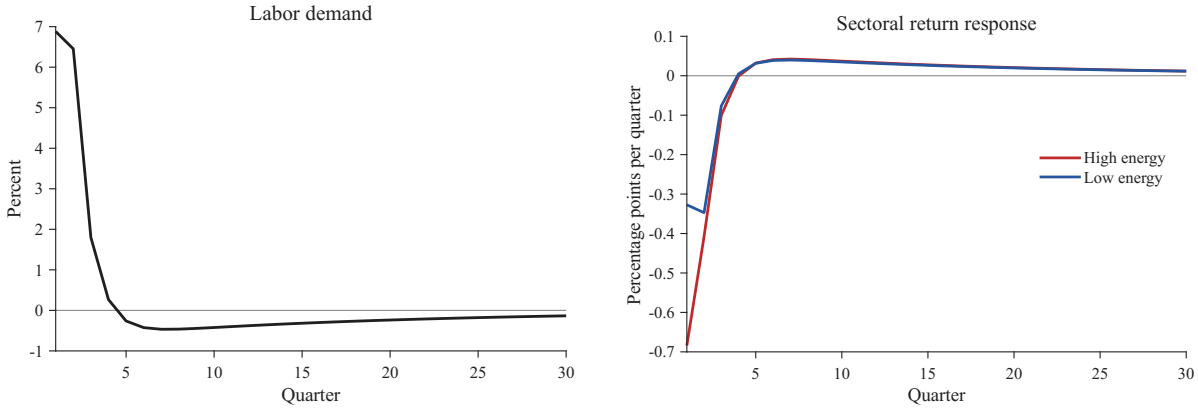
(b) Sectoral capital response

Figure 89: Sectoral responses




(a) Investment response (b) Wage response

Figure 90: Investment and wage




(a) Labor response (b) Return response

Figure 91: Labor and returns

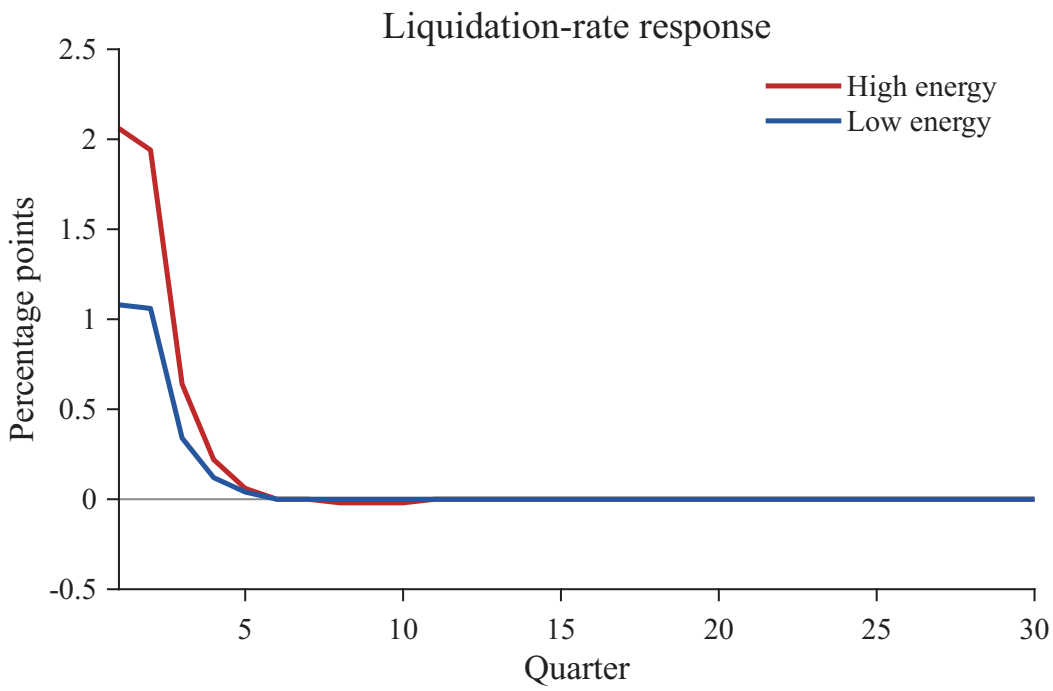



Figure 92: Liquidation-rate response

## B.10 MIT11: Large negative price shock with volatility

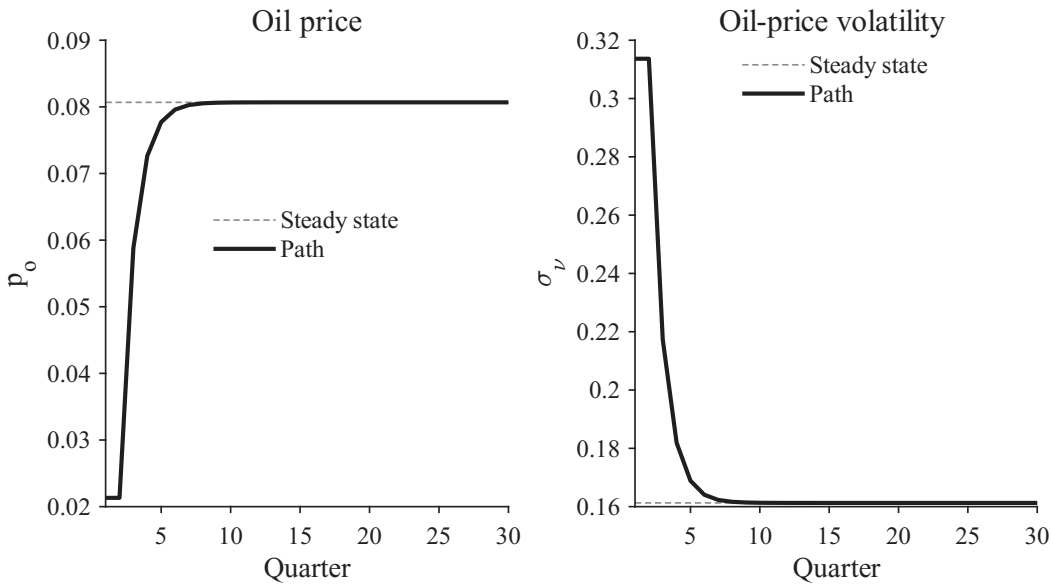
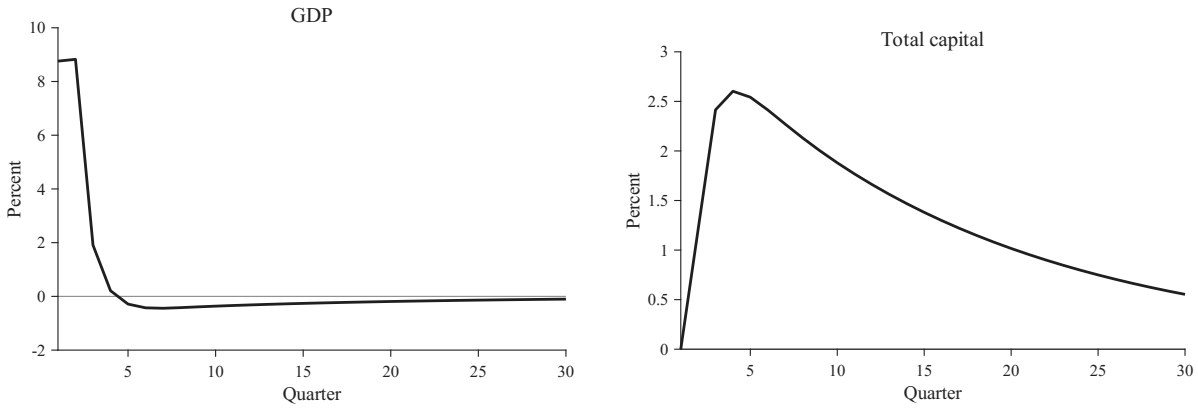


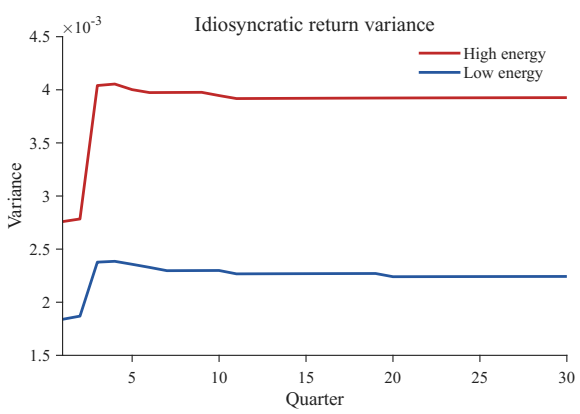

Figure 93: Shock path



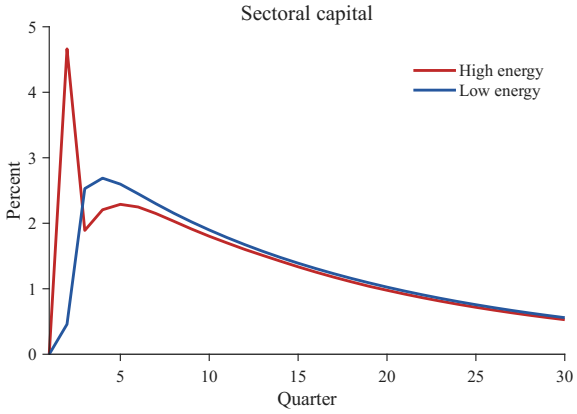

(a) GDP response

(b) Capital response

Figure 94: Aggregate responses

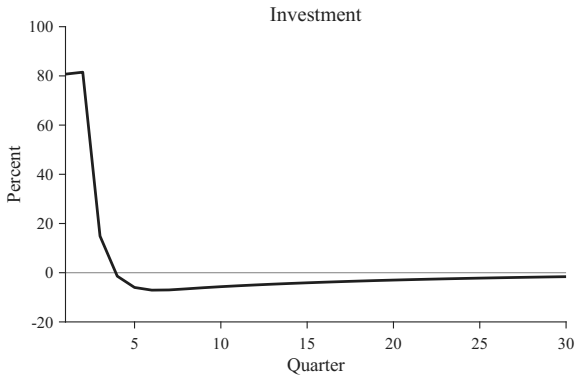



(a) Variance response

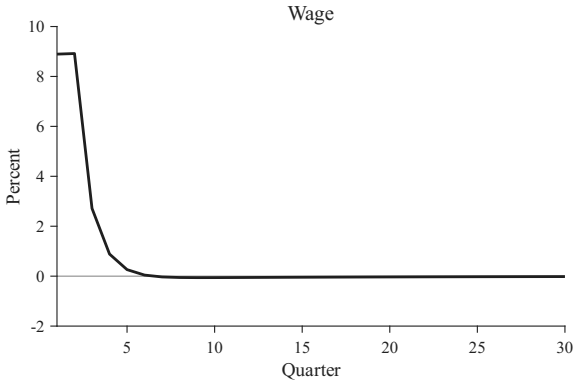



(b) Sectoral capital response

Figure 95: Sectoral responses

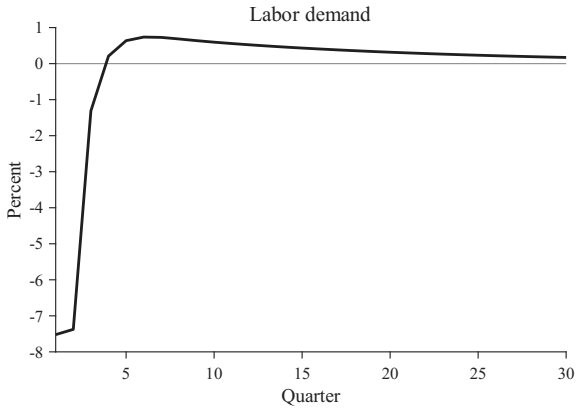



(a) Investment response

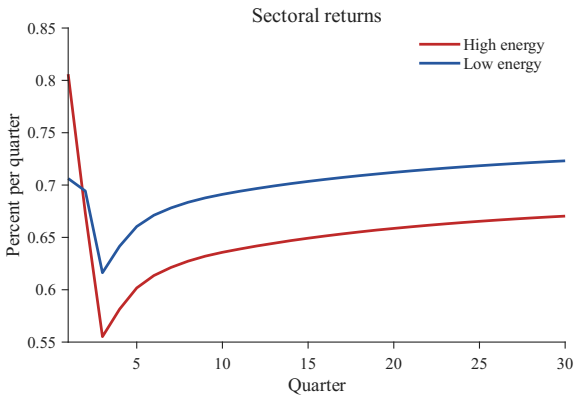



(b) Wage response

Figure 96: Investment and wage




(a) Labor response




(b) Return response

Figure 97: Labor and returns

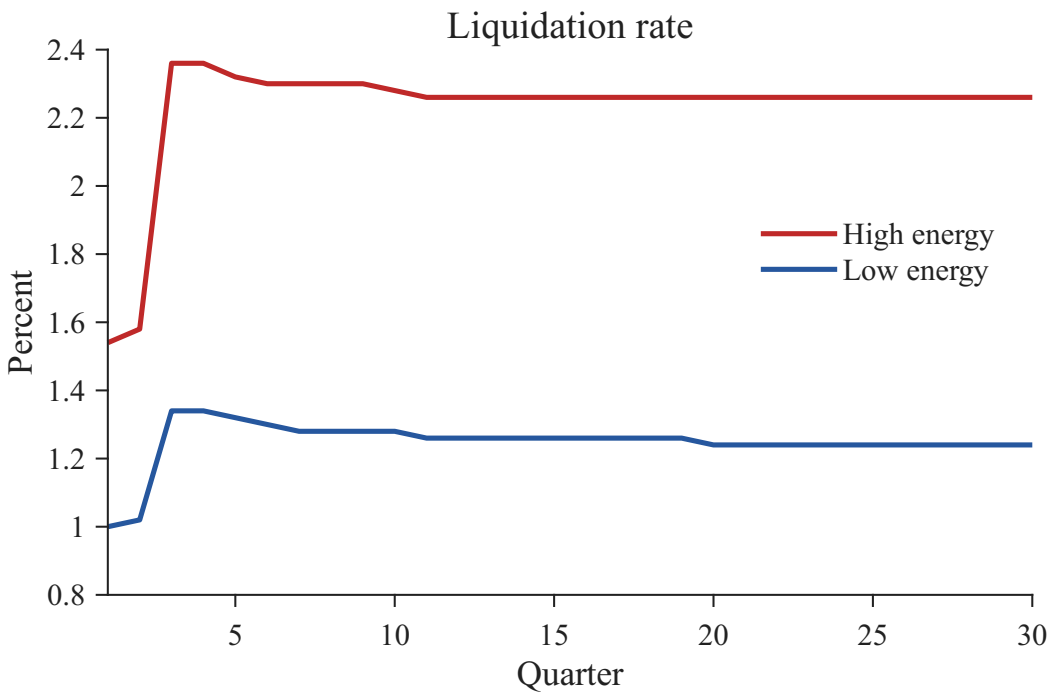



Figure 98: Liquidation-rate response

## B.11 MIT12: Large positive price shock with volatility

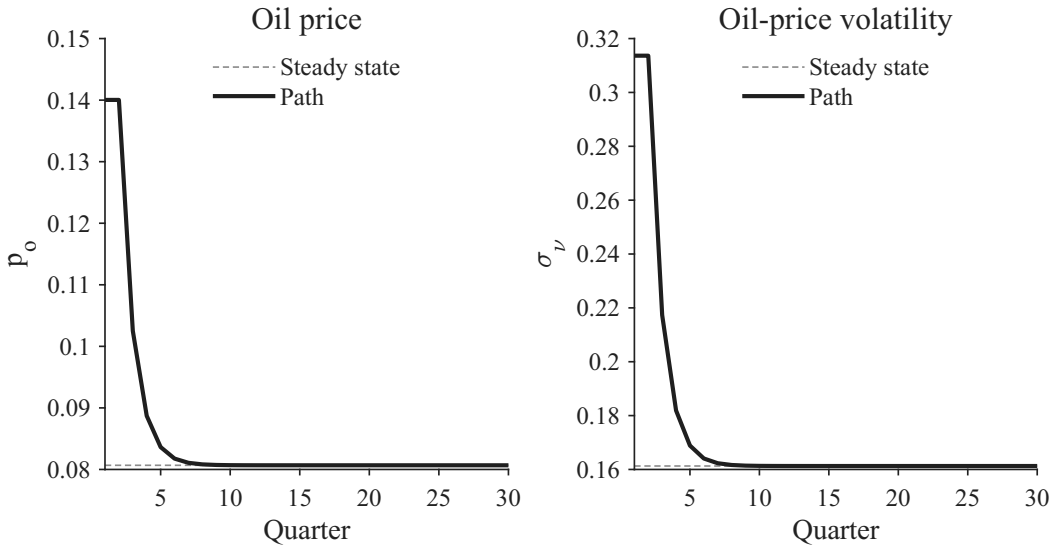
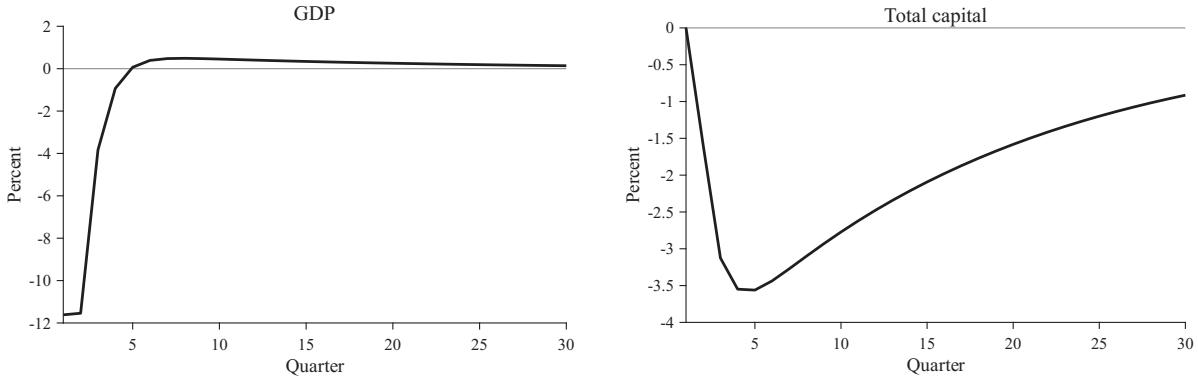


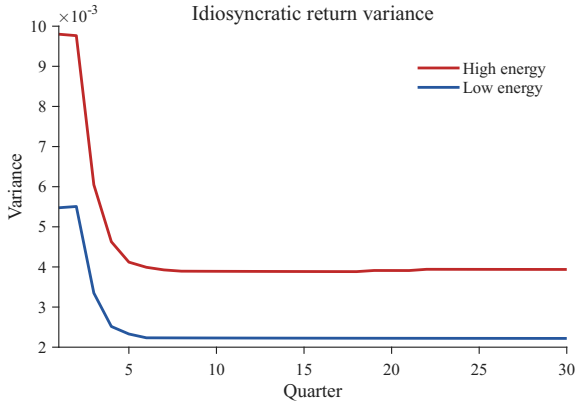

Figure 99: Shock path



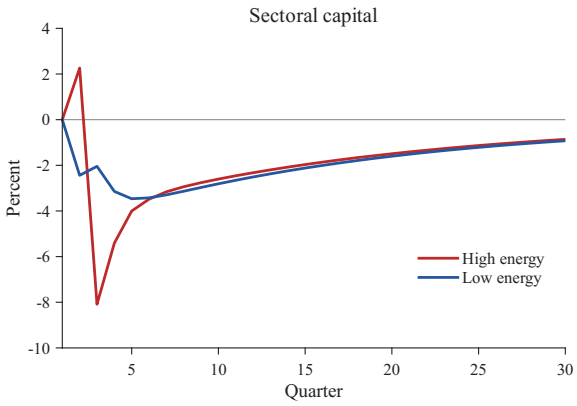

(a) GDP response

(b) Capital response

Figure 100: Aggregate responses

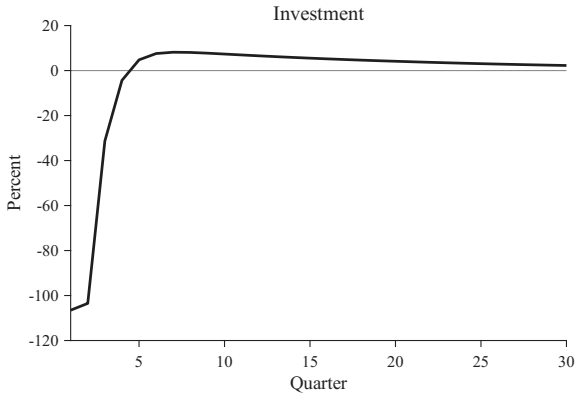



(a) Variance response

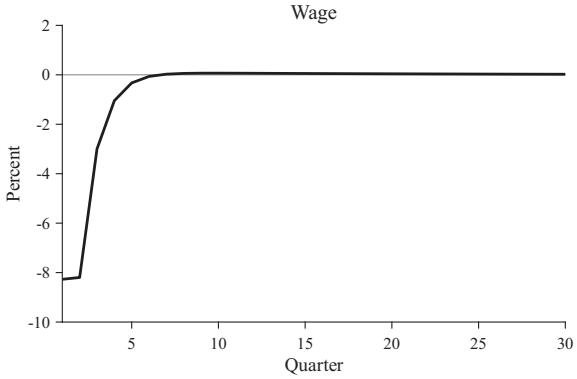



(b) Sectoral capital response

Figure 101: Sectoral responses

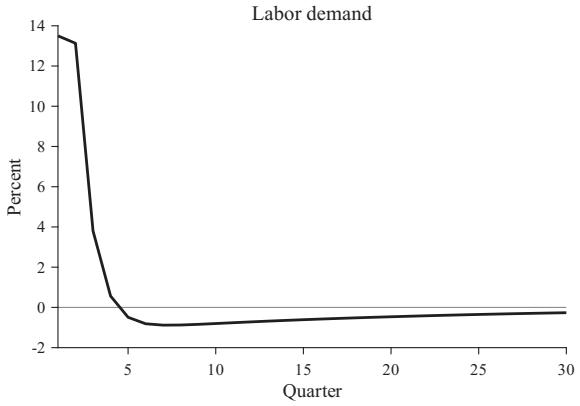



(a) Investment response

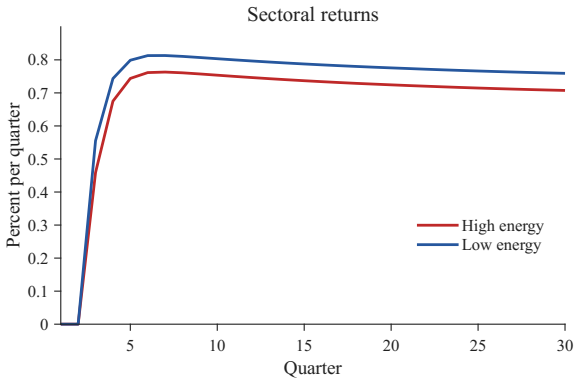



(b) Wage response

Figure 102: Investment and wage




(a) Labor response




(b) Return response

Figure 103: Labor and returns

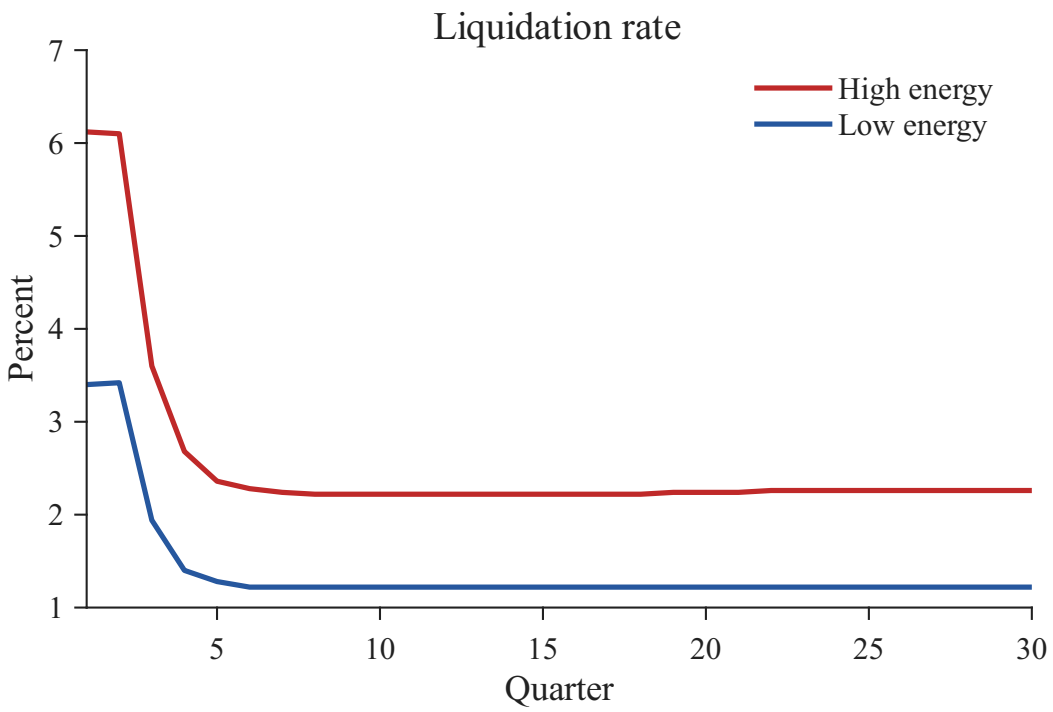



Figure 104: Liquidation-rate response

# C Selected Firm First-Order Conditions

Let $\varrho_f = (\rho_f - 1)/\rho_f$ and define

$$Z_{js} = (1-\phi_s)\alpha_{0,s} O_{js}^{\varrho_f} + (1-\phi_s)\alpha_{LK,s} M_{js}^{\varrho_f} + \phi_s T_s^{\varrho_f}, \qquad Y_{js}^g = Z_{js}^{1/\varrho_f}.$$

For an active oil input and for land, the marginal products are

$$\frac{\partial Y_{js}^g}{\partial O_{js}} = (1-\phi_s)\alpha_{0,s}(Y_{js}^g)^{1/\rho_f} O_{js}^{-1/\rho_f}, \tag{38}$$

$$\frac{\partial Y_{js}^g}{\partial T_s} = \phi_s (Y_{js}^g)^{1/\rho_f} T_s^{-1/\rho_f}. \tag{39}$$

The labor and capital marginal products additionally contain the derivative of the inner capital–labor composite. Expected first-order conditions include the effect of each input on the liquidation threshold. Writing these wedges as $\mu_A$ for revenue and $\mu_B$ for obligations, the land condition is

$$\mathbb{E}_j\left[(1+\mu_{A,js}) p_{js} \frac{\partial Y_{js}^g}{\partial T_s} - (1+\mu_{B,js}) q_T\right] = 0. \tag{40}$$

# D Liquidation Wedges

Let $\zeta_{js}^* = D_{js}/(p_{js} Y_{js}^g)$, with CDF $F_\zeta$ and density $f_\zeta$. Expected liquidation losses are

$$\mathcal{L}_{js} = C_{\text{liq}}(1-\delta) K_s F_\zeta(\zeta_{js}^*). \tag{41}$$

The smooth density implies

$$\mu_{B,js} = \frac{C_{\text{liq}}(1-\delta) K_s \, f_\zeta(\zeta_{js}^*)}{p_{js} Y_{js}^g}, \qquad \mu_{A,js} = \zeta_{js}^* \mu_{B,js}. \tag{42}$$

Further derivation and the treatment of the collateral-coverage condition will be added in the final appendix.

# E Numerical Accuracy

The final paper will report:

(i) steady-state goods, labor, and capital-market residuals;

(ii) transition-path residuals by market and date;
(iii) sensitivity to wealth-grid size and firm-return quadrature;
(iv) comparisons between four-point and full firm-return integration;
(v) sensitivity to transition damping and horizon length;
(vi) terminal-condition errors.